

Supporting information for “A multivariate spatial model for ordinal survey-based data”

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Document purpose

This R Markdown file provides a formatted version of the analysis performed using the original survey data. It contains the code used to obtain the analyses, tables, and figures reported in the manuscript.

Running this file requires access to the original survey data file ESCV2022_UV_Matem.sav. Precomputed posterior samples are loaded to avoid repeating the computationally intensive MCMC fitting step.

Required packages

```
pacman::p_load(foreign, readxl, faraway, spdep, sp, ggplot2, RColorBrewer,
               graphics, ggpubr, leaflet, nimble, ggmcmc, extraDistr,
               parallel, MCMCvis, gridExtra, corrplot, ggcorrplot, readr,
               lattice, sf, patchwork, install = FALSE)
```

Health survey data loading: GHQ-12 items

```
HSRV2022 <- read.spss(file.path("../data", "ESCV2022_UV_Matem.sav"),
                     use.value.labels = TRUE, to.data.frame = TRUE)
# Sample size
NResp <- nrow(HSRV2022)
# P8_1 Concentrate
levels(HSRV2022$P8_1)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"    "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_1)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              470              8195
##      Menos que lo habitual Mucho menos que lo habitual
##              919              194
##              NS/NC
##              19
```

```
HSRV2022$P8_1[HSRV2022$P8_1 == "NS/NC"] <- NA
HSRV2022$P8_1 <- factor(HSRV2022$P8_1, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                    "Menos que lo habitual", "Mucho menos que lo
                                                    ↪ habitual"))
levels(HSRV2022$P8_1) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_1)
```

```
##
##      More      Same      Less Much less
##      470      8195      919      194
```

```
P8_1 <- as.numeric(HSRV2022$P8_1)

# P8_2 Lose sleep over worries
levels(HSRV2022$P8_2)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_2)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              1706              6490
##      Menos que lo habitual Mucho menos que lo habitual
##              1094              488
##              NS/NC
##              19
```

```
HSRV2022$P8_2[HSRV2022$P8_2 == "NS/NC"] <- NA
HSRV2022$P8_2 <- factor(HSRV2022$P8_2, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                    "Menos que lo habitual", "Mucho menos que lo
                                                    ↪ habitual"))
levels(HSRV2022$P8_2) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_2)
```

```
##
##      Not at all      No more      Rather more
##              1706      6490      1094
## Much more than usual
##              488
```

```
P8_2 <- as.numeric(HSRV2022$P8_2)
```

```
# P8_3 Play a useful role
levels(HSRV2022$P8_3)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"    "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_3)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              823              8329
##      Menos que lo habitual Mucho menos que lo habitual
##              483              137
##              NS/NC
##              25
```

```
HSRV2022$P8_3[HSRV2022$P8_3 == "NS/NC"] <- NA
HSRV2022$P8_3 <- factor(HSRV2022$P8_3, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                    "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_3) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_3)
```

```
##
##      More      Same      Less Much less
##      823      8329      483      137
```

```
P8_3 <- as.numeric(HSRV2022$P8_3)
```

```
# P8_4 Make decisions
levels(HSRV2022$P8_4)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"    "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_4)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              619              8572
##      Menos que lo habitual Mucho menos que lo habitual
##              446              140
##              NS/NC
##              20
```

```

HSRV2022$P8_4[HSRV2022$P8_4 == "NS/NC"] <- NA
HSRV2022$P8_4 <- factor(HSRV2022$P8_4, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_4) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_4)

```

```

##
##      More      Same      Less Much less
##      619      8572      446      140

```

```

P8_4 <- as.numeric(HSRV2022$P8_4)

# P8_5 Constantly under strain
levels(HSRV2022$P8_5)

```

```

## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"

```

```

table(HSRV2022$P8_5)

```

```

##
##      Más que lo habitual      Igual que lo habitual
##              1621              5385
##      Menos que lo habitual Mucho menos que lo habitual
##              2175              593
##              NS/NC
##              23

```

```

HSRV2022$P8_5[HSRV2022$P8_5 == "NS/NC"] <- NA
HSRV2022$P8_5 <- factor(HSRV2022$P8_5, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_5) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_5)

```

```

##
##      Not at all      No more      Rather more
##              1621              5385              2175
## Much more than usual
##              593

```

```

P8_5 <- as.numeric(HSRV2022$P8_5)

# P8_6 Unable to overcome difficulties
levels(HSRV2022$P8_6)

```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"     "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_6)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              1173              5748
##      Menos que lo habitual Mucho menos que lo habitual
##              2258              593
##              NS/NC
##              25
```

```
HSRV2022$P8_6[HSRV2022$P8_6 == "NS/NC"] <- NA
HSRV2022$P8_6 <- factor(HSRV2022$P8_6, levels = c("Más que lo habitual", "Igual que lo habitual",
↪      "Menos que lo habitual", "Mucho menos que lo
↪      habitual"))
levels(HSRV2022$P8_6) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_6)
```

```
##
##      Not at all      No more      Rather more
##              1173      5748      2258
## Much more than usual
##              593
```

```
P8_6 <- as.numeric(HSRV2022$P8_6)

# P8_7 Enjoy activities
levels(HSRV2022$P8_7)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"     "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_7)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              624              8176
##      Menos que lo habitual Mucho menos que lo habitual
##              798              175
##              NS/NC
##              24
```

```

HSRV2022$P8_7[HSRV2022$P8_7 == "NS/NC"] <- NA
HSRV2022$P8_7 <- factor(HSRV2022$P8_7, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_7) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_7)

```

```

##
##      More      Same      Less Much less
##      624      8176      798      175

```

```

P8_7 <- as.numeric(HSRV2022$P8_7)

# P8_8 Face up to problems
levels(HSRV2022$P8_8)

```

```

## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"

```

```

table(HSRV2022$P8_8)

```

```

##
##      Más que lo habitual      Igual que lo habitual
##              508              8459
##      Menos que lo habitual Mucho menos que lo habitual
##              624              185
##              NS/NC
##              21

```

```

HSRV2022$P8_8[HSRV2022$P8_8 == "NS/NC"] <- NA
HSRV2022$P8_8 <- factor(HSRV2022$P8_8, levels = c("Más que lo habitual", "Igual que lo habitual",
↪
                                                "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_8) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_8)

```

```

##
##      More      Same      Less Much less
##      508      8459      624      185

```

```

P8_8 <- as.numeric(HSRV2022$P8_8)

# P8_9 Feel depressed
levels(HSRV2022$P8_9)

```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_9)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              1535              4863
##      Menos que lo habitual Mucho menos que lo habitual
##              2271              1102
##              NS/NC
##              26
```

```
HSRV2022$P8_9[HSRV2022$P8_9 == "NS/NC"] <- NA
HSRV2022$P8_9 <- factor(HSRV2022$P8_9, levels = c("Más que lo habitual", "Igual que lo habitual",
↪      "Menos que lo habitual", "Mucho menos que lo
↪      habitual"))
levels(HSRV2022$P8_9) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_9)
```

```
##
##      Not at all      No more      Rather more
##              1535      4863      2271
## Much more than usual
##              1102
```

```
P8_9 <- as.numeric(HSRV2022$P8_9)
# P8_10 Lose confidence
levels(HSRV2022$P8_10)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_10)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              1355              4984
##      Menos que lo habitual Mucho menos que lo habitual
##              2287              1145
##              NS/NC
##              26
```

```

HSRV2022$P8_10[HSRV2022$P8_10 == "NS/NC"] <- NA
HSRV2022$P8_10 <- factor(HSRV2022$P8_10, levels = c("Más que lo habitual", "Igual que lo
↪ habitual",
                                                    "Menos que lo habitual", "Mucho menos que lo
                                                    ↪ habitual"))
levels(HSRV2022$P8_10) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_10)

```

```

##
##          Not at all          No more          Rather more
##              1355              4984              2287
## Much more than usual
##              1145

```

```

P8_10 <- as.numeric(HSRV2022$P8_10)

# P8_11 Feel worthless
levels(HSRV2022$P8_11)

```

```

## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"      "Mucho menos que lo habitual"
## [5] "NS/NC"

```

```

table(HSRV2022$P8_11)

```

```

##
##          Más que lo habitual      Igual que lo habitual
##              1151              5865
##          Menos que lo habitual Mucho menos que lo habitual
##              1513              1238
##              NS/NC
##              30

```

```

HSRV2022$P8_11[HSRV2022$P8_11 == "NS/NC"] <- NA
HSRV2022$P8_11 <- factor(HSRV2022$P8_11, levels = c("Más que lo habitual", "Igual que lo
↪ habitual",
                                                    "Menos que lo habitual", "Mucho menos que lo
                                                    ↪ habitual"))
levels(HSRV2022$P8_11) <- c("Not at all", "No more", "Rather more", "Much more than usual")
table(HSRV2022$P8_11)

```

```

##
##          Not at all          No more          Rather more
##              1151              5865              1513
## Much more than usual
##              1238

```



```
P8_11 <- as.numeric(HSRV2022$P8_11)
```

```
# P8_12 Feel reasonably happy
levels(HSRV2022$P8_12)
```

```
## [1] "Más que lo habitual"      "Igual que lo habitual"
## [3] "Menos que lo habitual"    "Mucho menos que lo habitual"
## [5] "NS/NC"
```

```
table(HSRV2022$P8_12)
```

```
##
##      Más que lo habitual      Igual que lo habitual
##              754              8347
##      Menos que lo habitual Mucho menos que lo habitual
##              566              101
##              NS/NC
##              29
```

```
HSRV2022$P8_12[HSRV2022$P8_12 == "NS/NC"] <- NA
HSRV2022$P8_12 <- factor(HSRV2022$P8_12, levels = c("Más que lo habitual", "Igual que lo
↪ habitual",
                                                    "Menos que lo habitual", "Mucho menos que lo
↪ habitual"))
levels(HSRV2022$P8_12) <- c("More", "Same", "Less", "Much less")
table(HSRV2022$P8_12)
```

```
##
##      More      Same      Less Much less
##      754      8347      566      101
```

```
P8_12 <- as.numeric(HSRV2022$P8_12)

y <- data.frame(P8_1, P8_2, P8_3, P8_4, P8_5, P8_6,
               P8_7, P8_8, P8_9, P8_10, P8_11, P8_12)

# Number of response variables
NVars <- ncol(y)
# Number of levels
NCats <- unique(apply(y, 2, function(x) {length(table(x))}))
# Vector of ones
ones <- rep(1, NCats)

rm(list = c("P8_1", "P8_2", "P8_3", "P8_4", "P8_5", "P8_6",
            "P8_7", "P8_8", "P8_9", "P8_10", "P8_11", "P8_12"))

# Non-response percentages
apply(y, 2, function(x) {sum(is.na(x))}/NResp * 100
```

```
##      P8_1      P8_2      P8_3      P8_4      P8_5      P8_6      P8_7      P8_8
```

```
## 0.1939369 0.1939369 0.2551802 0.2041441 0.2347657 0.2551802 0.2449730 0.2143513
##      P8_9      P8_10      P8_11      P8_12
## 0.2653874 0.2653874 0.3062162 0.2960090
```

```
# Covariate sex: 1 = Male; 2 = Female
sexC <- HSRV2022$sexo
levels(sexC) <- c("Male", "Female")
sex <- as.numeric(sexC)

# Covariate age group: 1 = [15,25); 2 = [25,35); 3 = [35,45); 4 = [45,55); 5 = [55,65);
# 6 = [65,70); 7 = [70,75); 8 = [75,...)
ageC <- cut(HSRV2022$Edad, breaks = c(15, 25, 35, 45, 55, 65, 70, 75, 103),
            include.lowest = TRUE, right = FALSE)
levels(ageC)[length(table(ageC))] <- "[75,...)"
age <- as.numeric(ageC)

# Number of respondents by sex and age group
table(sexC, ageC)
```

```
##      ageC
## sexC    [15,25) [25,35) [35,45) [45,55) [55,65) [65,70) [70,75) [75,...)
## Male      467    504    671    815    655    507    477    830
## Female    458    524    683    771    702    437    414    882
```

```
# Number of levels of each (categorical) covariate
NSex <- length(table(sexC))
NAges <- length(table(ageC))
```

Spatial neighbourhood structure

```
# Cartography of the Region of Valencia
load(file.path("../data", "CartoCV.Rdata"))
# Cartography is sorted by municipality code
order(carto_muni$INE_MUN)-1:542
```

```
## [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [112] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [149] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [186] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [223] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [260] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [297] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [334] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [371] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [408] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [445] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [482] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [519] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```

# Neighbourhood structure by contiguity
cv.nb <- poly2nb(carto_muni)

# Some extra neighbourhoods are added for Rincón de Ademuz comarca
cv.nb[[277]] <- as.integer(sort(c(cv.nb[[277]], 312, 317, 517, 523, 508)))
cv.nb[[363]] <- as.integer(sort(c(cv.nb[[363]], 312, 317, 517, 523, 508)))
cv.nb[[364]] <- as.integer(sort(c(cv.nb[[364]], 312, 317, 517, 523, 508)))
cv.nb[[477]] <- as.integer(sort(c(cv.nb[[477]], 312, 317, 517, 523, 508)))

cv.nb[[312]] <- as.integer(sort(c(cv.nb[[312]], 277, 363, 364, 477)))
cv.nb[[317]] <- as.integer(sort(c(cv.nb[[317]], 277, 363, 364, 477)))
cv.nb[[517]] <- as.integer(sort(c(cv.nb[[517]], 277, 363, 364, 477)))
cv.nb[[523]] <- as.integer(sort(c(cv.nb[[523]], 277, 363, 364, 477)))
cv.nb[[508]] <- as.integer(sort(c(cv.nb[[508]], 277, 363, 364, 477)))

# Municipality codes
INE_MUN <- as.numeric(as.character(carto_muni@data$INE_MUN))

# Municipality of each respondent
muni <- as.numeric(HSRV2022$LOCALIDAD)
# Number of (distinct) municipalities (542)
NMuni <- length(INE_MUN); rm(INE_MUN)

# Number of neighbours of each municipality
nadj <- card(cv.nb)
# Neighbours of each municipality
map <- unlist(cv.nb)
# Sum of all the neighbour numbers of all municipalities
nadj.tot <- length(map)
# Cumulative sums of the number of neighbours of each municipality
index <- c(0, cumsum(nadj))

### Leroux CAR distribution in NIMBLE ###

# Diagonal matrix with the number of neighbours of each area
D <- diag(nadj)
# Adjacency matrix
W <- nb2mat(cv.nb, style = "B", zero.policy = TRUE)
# Eigenvalues of D-W
Lambda <- eigen(D - W)$values
# Identity matrix
I <- diag(rep(1, NMuni))

# All the neighbourhoods j ~ i where i < j
from.to <- cbind(rep(1:NMuni, times = nadj), map); colnames(from.to) <- c("from", "to")
from.to <- from.to[which(from.to[, 1] < from.to[, 2]), ]
NDist <- nrow(from.to)

dcar_leroux <- nimbleFunction(
  name = 'dcar_leroux',
  run = function(x = double(1),          # Spatial random effect (vector)
                 rho = double(0),         # Amount of spatial dependence (scalar)
                 sd.theta = double(0),    # Standard deviation (scalar)
                 Lambda = double(1),      # Eigenvalues of matrix D - W
                 from.to = double(2),     # Matrix of distinct pairs of neighbours from.to[, 1] <
                 ↪ from.to[, 2]
                 log = integer(0, default = 0)) {
    returnType(double(0))
  }

```

```

# Number of small areas
NMuni <- dim(x)[1]
# Number of distinct pairs of neighbours
NDist <- dim(from.to)[1]
# Required vectors
x.from <- nimNumeric(NDist)
x.to <- nimNumeric(NDist)
for (Dist in 1:NDist) {
  x.from[Dist] <- x[from.to[Dist, 1]]
  x.to[Dist] <- x[from.to[Dist, 2]]
}

# Log-density
logDens <- sum(dnorm(x[1:NMuni], mean = 0, sd = sd.theta * pow(1 - rho, -1/2), log = TRUE)) -
  NMuni/2 * log(1 - rho) + 1/2 * sum(log(rho * (Lambda[1:NMuni] - 1) + 1)) -
  1/2 * pow(sd.theta, -2) * rho * sum(pow(x.from[1:NDist] - x.to[1:NDist], 2))
if(log) return(logDens)
else return(exp(logDens))
}
)

```

Multivariate models

Independent Model: Model-Indep

```

# n.chains <- 5
# this_cluster <- makeCluster(n.chains)

# Model code

modelCode <- nimbleCode(
{
  for(Var in 1:NVars) {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp, Var] ~ dcat(prlevels[Resp, Var, 1:NCats])

      # Definition of the probabilities of each category as a function of the
      # cumulative probabilities
      prlevels[Resp, Var, 1] <- p.gamma[Resp, Var, 1]
      for (Cat in 2:(NCats-1)) {
        prlevels[Resp, Var, Cat] <- p.gamma[Resp, Var, Cat] - p.gamma[Resp, Var, Cat-1]
      }
      prlevels[Resp, Var, NCats] <- 1 - p.gamma[Resp, Var, NCats-1]

      # Linear predictor
      for (Cat in 1:(NCats-1)) {
        logit(p.gamma[Resp, Var, Cat]) <- kappa[sex[Resp], age[Resp], Cat, Var] +
          sd.theta[Var] * theta[muni[Resp], Var]
      }
    }
  }
}

# Prior distributions

```

```

# kappa[1:NSEX, 1:NAges, 1:(NCats-1), 1:NVars] cut points
# Monotonic transformation
for (Var in 1:NVars) {
  for (SexGroup in 1:NSEX) {
    for (AgeGroup in 1:NAges) {
      for (Cat in 1:(NCats-1)) {
        kappa[SexGroup, AgeGroup, Cat, Var] <- logit(sum(delta[SexGroup, AgeGroup, Var,
→ 1:Cat]))
      }
      # delta[1:NSEX, 1:NAges, 1:NVars, 1:NCats] Dirichlet prior
      delta[SexGroup, AgeGroup, Var, 1:NCats] ~ ddirch(ones[1:NCats])
    }
  }
}

# theta[1:NMuni, 1:NVars] spatial random effects
for (Var in 1:NVars) {
  # theta[1:NMuni, 1:NVars] spatial random effects
  # LCAR distribution
  theta[1:NMuni, Var] ~ dcar_leroux(rho = rho[Var],
                                   sd.theta = 1,
                                   Lambda = Lambda[1:NMuni],
                                   from.to = from.to[1:NDist, 1:2])
}

# Hyperparameters of the spatial random effects
for (Var in 1:NVars) {
  rho[Var] ~ dunif(0, 1)
  sd.theta[Var] ~ dhalfflat()
}

# Stochastic restrictions
# Required vectors
for (Var in 1:NVars) {
  for (Resp in 1:NResp) {
    theta.Resp[Resp, Var] <- theta[muni[Resp], Var]
  }

  # Zero-mean constraint for theta.Resp
  zero.theta.resp[Var] ~ dnorm(mean.thetas.resp[Var], 10000)
  mean.thetas.resp[Var] <- mean(theta.Resp[1:NResp, Var])
}
}
)

# Model data
modelData <- list(y = as.matrix(y), zero.theta.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars,
                      sex = sex, age = age, muni = muni, NSEX = NSEX,
                      NAges = NAges, NMuni = NMuni, ones = ones,
                      NDist = NDist, Lambda = Lambda, from.to = from.to)

# Parameters to monitor
modelParameters <- c("kappa", "theta", "sd.theta", "rho", "delta")

```

```

# Create a function to run the MCMC
run_MCMC_allcode <- function(X, code, constants, data, monitors) {

  pacman::p_load(nimble, extraDistr, install = FALSE)

  dcar_leroux <- nimbleFunction(
    name = 'dcar_leroux',
    run = function(x = double(1),          # Spatial random effect (vector)
                  rho = double(0),          # Amount of spatial dependence (scalar)
                  sd.theta = double(0),    # Standard deviation (scalar)
                  Lambda = double(1),      # Eigenvalues of matrix D - W
                  from.to = double(2),     # Matrix of distinct pairs of neighbours from.to[, 1] <
                  ↪ from.to[, 2]
                  log = integer(0, default = 0)) {
      returnType(double(0))

      # Number of small areas
      NMuni <- dim(x)[1]
      # Number of distinct pairs of neighbours
      NDist <- dim(from.to)[1]
      # Required vectors
      x.from <- nimNumeric(NDist)
      x.to <- nimNumeric(NDist)
      for (Dist in 1:NDist) {
        x.from[Dist] <- x[from.to[Dist, 1]]
        x.to[Dist] <- x[from.to[Dist, 2]]
      }

      logDens <- sum(dnorm(x[1:NMuni], mean = 0, sd = sd.theta * pow(1 - rho, -1/2), log = TRUE))
      ↪ -
        NMuni/2 * log(1 - rho) + 1/2 * sum(log(rho * (Lambda[1:NMuni] - 1) + 1)) -
        1/2 * pow(sd.theta, -2) * rho * sum(pow(x.from[1:NDist] - x.to[1:NDist], 2))
      if(log) return(logDens)
      else return(exp(logDens))
    }
  )

  rcar_leroux <- nimbleFunction(
    name = 'rcar_leroux',
    run = function(n = integer(0),
                  rho = double(0),
                  sd.theta = double(0),
                  Lambda = double(1),
                  from.to = double(2)) {
      returnType(double(1))

      nimStop("user-defined distribution dcar_leroux provided without random generation
      ↪ function.")
      x <- nimNumeric(542)
      return(x)
    }
  )

  assign('dcar_leroux', dcar_leroux, envir = .GlobalEnv)
  assign('rcar_leroux', rcar_leroux, envir = .GlobalEnv)

  NSex <- constants$NSex
  NAges <- constants$NAges
  NCats <- constants$NCats

```

```

ones <- constants$ones
NResp <- constants$NResp
NMuni <- constants$NMuni
NVars <- constants$NVars

# Create the NIMBLE model and initialize its nodes
model <- nimbleModel(code = code,
  constants = constants,
  data = data,
  inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
    dim = c(NSex, NAges, NVars, NCats)),
    rho = runif(NVars),
    theta = matrix(rnorm(NMuni * NVars, sd = 0.1), nrow = NMuni,
      ↪ ncol = NVars),
    sd.theta = runif(NVars)),
  calculate = FALSE)

# Compile the model, which means generating C++ code, compiling that code, and loading it back
↪ into R
Cmodel <- compileNimble(model)

# model$getParents(model$getNodeNames(dataOnly = TRUE), stochOnly = TRUE)

# Configuration
modelMCMCconfiguration <- configureMCMC(model, useConjugacy = FALSE,
  enableWAIC = TRUE)

# Remove desired samplers
modelMCMCconfiguration$removeSamplers(c("theta", "rho", "sd.theta"))

# Add RW-MH theta[1:NMuni, 1:NVars] samplers
thetas <- matrix(nrow = NMuni, ncol = NVars)
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    thetas[Muni, Var] <- paste0("theta[", Muni, ",", Var, "]")
  }
}

smuni <- sort(unique(constants$smuni))
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    ifelse(Muni %in% smuni,
      modelMCMCconfiguration$addSampler(target = thetas[Muni, Var], type = "RW"), # also
↪ slice
      modelMCMCconfiguration$addSampler(target = thetas[Muni, Var], type = "RW"))
  }
}

# Add slice rho sampler
rhos <- character(NVars)
for (Var in 1:NVars) {
  rhos[Var] <- paste0("rho[", Var, "]")
}

for (Var in 1:NVars) {
  modelMCMCconfiguration$addSampler(target = rhos[Var], type = "slice")
}

# Add slice sd.theta sampler

```

```

sd.thetas <- character(NVars)
for (Var in 1:NVars) {
  sd.thetas[Var] <- paste0("sd.theta[",Var,"]")
}

for (Var in 1:NVars) {
  modelMCMCconfiguration$addSampler(target = sd.thetas[Var], type = "slice")
}

# Add new monitors
modelMCMCconfiguration$monitors <- c()
modelMCMCconfiguration$addMonitors(monitors)
# Build MCMC object
modelMCMC <- buildMCMC(modelMCMCconfiguration)
# Need to reset the nimbleFunctions in order to add the new MCMC
CmodelMCMC <- compileNimble(modelMCMC, project = model, resetFunctions = TRUE)
# Results
results <- runMCMC(CmodelMCMC, niter = 8000, nburnin = 2000, thin = 30, setSeed = X)

return(results)
}

# system.time(salnimble1 <- parLapply(cl = this_cluster, X = 1:n.chains,
#                                     fun = run_MCMC_allcode,
#                                     code = modelCode,
#                                     constants = modelConstants,
#                                     data = modelData,
#                                     monitors = modelParameters))
#
# # Close the cluster after the parallel computation
# stopCluster(this_cluster)

# 1.84h with: niter = 8000, nburnin = 2000, thin = 30
# saveRDS(salnimble1, file = file.path("../", "results",
# → "multi-2022-nimble-MH-indep-8k-2k-30-WAIC.rds"))

```

Correlated Model: Model-Corr

```

# n.chains <- 5
# this_cluster <- makeCluster(n.chains)

# Model code

modelCode <- nimbleCode(
{
  for(Var in 1:NVars) {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp, Var] ~ dcat(prlevels[Resp, Var, 1:NCats])

      # Definition of the probabilities of each category as a function of the
      # cumulative probabilities
      prlevels[Resp, Var, 1] <- p.gamma[Resp, Var, 1]
      for (Cat in 2:(NCats-1)) {
        prlevels[Resp, Var, Cat] <- p.gamma[Resp, Var, Cat] - p.gamma[Resp, Var, Cat-1]
      }
    }
  }
}

```



```

prlevels[Resp, Var, NCats] <- 1 - p.gamma[Resp, Var, NCats-1]

# Linear predictor
for (Cat in 1:(NCats-1)) {
  logit(p.gamma[Resp, Var, Cat]) <- kappa[sex[Resp], age[Resp], Cat, Var] +
    theta[muni[Resp], Var]
}
}

# Prior distributions

# kappa[1:NSEX, 1:NAGES, 1:(NCats-1), 1:NVars] cut points
# Monotonic transformation
for (Var in 1:NVars) {
  for (SexGroup in 1:NSEX) {
    for (AgeGroup in 1:NAGES) {
      for (Cat in 1:(NCats-1)) {
        kappa[SexGroup, AgeGroup, Cat, Var] <- logit(sum(delta[SexGroup, AgeGroup, Var,
↪ 1:Cat]))
      }
      # delta[1:NSEX, 1:NAGES, 1:NVars, 1:NCats] Dirichlet prior
      delta[SexGroup, AgeGroup, Var, 1:NCats] ~ ddirch(ones[1:NCats])
    }
  }
}

# theta[1:NMuni, 1:NVars] spatial random effects
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    theta[Muni, Var] <- inprod(sub.Muni[Muni, ], M.Muni[, Var])
  }
  # sub.Muni[1:NMuni, 1:NVars] underlying spatial REs
  # LCAR distribution
  sub.Muni[1:NMuni, Var] ~ dcar_leroux(rho = rho[Var],
    sd.theta = 1,
    Lambda = Lambda[1:NMuni],
    from.to = from.to[1:NDist, 1:2])
}

# Hyperparameter of the spatial random effects
for (Var in 1:NVars) {
  rho[Var] ~ dunif(0, 1)
}

# M.Resp[1:NVars, 1:NVars] and M.Muni[1:NVars, 1:NVars] M-matrices
for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    M.Muni[Var1, Var2] ~ dnorm(0, tau.M.Muni)
  }
}

# Prior for precisions of M.Muni
tau.M.Muni <- pow(sd.M.Muni, -2)
sd.M.Muni ~ dhalfflat()

# Stochastic restrictions
# Required vectors
for (Var in 1:NVars) {

```

```

    for (Resp in 1:NResp) {
      sub.Muni.Resp[Resp, Var] <- sub.Muni[muni[Resp], Var]
    }

    # Zero-mean constraint for sub.Muni.Resp
    zero.sub.Muni.resp[Var] ~ dnorm(mean.sub.Munis.resp[Var], 10000)
    mean.sub.Munis.resp[Var] <- mean(sub.Muni.Resp[1:NResp, Var])
  }
}
)

# Model data

modelData <- list(y = as.matrix(y), zero.sub.Muni.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars,
  sex = sex, age = age, muni = muni, NSex = NSex,
  NAges = NAges, NMuni = NMuni, ones = ones,
  NDist = NDist, Lambda = Lambda, from.to = from.to)

# Parameters to monitor

modelParameters <- c("kappa", "theta", "M.Muni", "sd.M.Muni", "rho",
  "delta", "sub.Muni")

# Create a function to run the MCMC
run_MCMC_allcode <- function(X, code, constants, data, monitors) {

  pacman::p_load(nimble, extraDistr, install = FALSE)

  dcar_leroux <- nimbleFunction(
    name = 'dcar_leroux',
    run = function(x = double(1),          # Spatial random effect (vector)
      rho = double(0),          # Amount of spatial dependence (scalar)
      sd.theta = double(0),     # Standard deviation (scalar)
      Lambda = double(1),      # Eigenvalues of matrix D - W
      from.to = double(2),     # Matrix of distinct pairs of neighbours from.to[, 1] <
        ↪ from.to[, 2]
      log = integer(0, default = 0)) {
      returnType(double(0))

      # Number of small areas
      NMuni <- dim(x)[1]
      # Number of distinct pairs of neighbours
      NDist <- dim(from.to)[1]
      # Required vectors
      x.from <- nimNumeric(NDist)
      x.to <- nimNumeric(NDist)
      for (Dist in 1:NDist) {
        x.from[Dist] <- x[from.to[Dist, 1]]
        x.to[Dist] <- x[from.to[Dist, 2]]
      }

      logDens <- sum(dnorm(x[1:NMuni], mean = 0, sd = sd.theta * pow(1 - rho, -1/2), log = TRUE))
      ↪ -
        NMuni/2 * log(1 - rho) + 1/2 * sum(log(rho * (Lambda[1:NMuni] - 1) + 1)) -
        1/2 * pow(sd.theta, -2) * rho * sum(pow(x.from[1:NDist] - x.to[1:NDist], 2))
      if(log) return(logDens)
    }
  )
}

```

```

    else return(exp(logDens))
  }
)

rcar_leroux <- nimbleFunction(
  name = 'rcar_leroux',
  run = function(n = integer(0),
                 rho = double(0),
                 sd.theta = double(0),
                 Lambda = double(1),
                 from.to = double(2)) {
    returnType(double(1))

    nimStop("user-defined distribution dcar_leroux provided without random generation
    ↪ function.")
    x <- nimNumeric(542)
    return(x)
  }
)

assign('dcar_leroux', dcar_leroux, envir = .GlobalEnv)
assign('rcar_leroux', rcar_leroux, envir = .GlobalEnv)

NSex <- constants$NSex
NAges <- constants$NAges
NCats <- constants$NCats
ones <- constants$ones
NResp <- constants$NResp
NMuni <- constants$NMuni
NVars <- constants$NVars

# Create the NIMBLE model and initialize its nodes
model <- nimbleModel(code = code,
                     constants = constants,
                     data = data,
                     inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
                                                dim = c(NSex, NAges, NVars, NCats)),
                                   rho = runif(NVars),
                                   sub.Muni = matrix(rnorm(NMuni * NVars, sd = 0.01), nrow =
                                   ↪ NMuni, ncol = NVars),
                                   M.Muni = matrix(rnorm(NVars * NVars, sd = 0.5), ncol = NVars,
                                   ↪ nrow = NVars),
                                   sd.M.Muni = runif(1, min = 0.2, max = 0.8)),
                     calculate = FALSE)

# Compile the model, which means generating C++ code, compiling that code, and loading it back
↪ into R
Cmodel <- compileNimble(model)

# model$getParents(model$getNodeNames(dataOnly = TRUE), stochOnly = TRUE)

# Configuration
modelMCMCconfiguration <- configureMCMC(model, useConjugacy = FALSE,
                                       enableWAIC = TRUE)

# Remove desired samplers
modelMCMCconfiguration$removeSamplers(c("sub.Muni", "rho", "sd.M.Muni"))

# Add RW-MH sub.Muni[1:NMuni, 1:NVars] samplers

```

```

sub.Munis <- matrix(nrow = NMuni, ncol = NVars)
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    sub.Munis[Muni, Var] <- paste0("sub.Muni[",Muni,",",Var,"]")
  }
}

smuni <- sort(unique(constants$muni))
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    ifelse(Muni %in% smuni,
            modelMCMCconfiguration$addSampler(target = sub.Munis[Muni, Var], type = "RW"), #
→   also slice
            modelMCMCconfiguration$addSampler(target = sub.Munis[Muni, Var], type = "RW"))
  }
}

# Add slice rho sampler
rhos <- character(NVars)
for (Var in 1:NVars) {
  rhos[Var] <- paste0("rho[",Var,"]")
}

for (Var in 1:NVars) {
  modelMCMCconfiguration$addSampler(target = rhos[Var], type = "slice")
}

# Add slice sd.M.Muni sampler
modelMCMCconfiguration$addSampler(target = "sd.M.Muni", type = "slice")

# Add new monitors
modelMCMCconfiguration$monitors <- c()
modelMCMCconfiguration$addMonitors(monitors)
# Build MCMC object
modelMCMC <- buildMCMC(modelMCMCconfiguration)
# Need to reset the nimbleFunctions in order to add the new MCMC
CmodelMCMC <- compileNimble(modelMCMC, project = model, resetFunctions = TRUE)
# Results
results <- runMCMC(CmodelMCMC, niter = 8000, nburnin = 2000, thin = 30, setSeed = X)

return(results)
}

# system.time(salnimble2 <- parLapply(cl = this_cluster, X = 1:n.chains,
#                                     fun = run_MCMC_allcode,
#                                     code = modelCode,
#                                     constants = modelConstants,
#                                     data = modelData,
#                                     monitors = modelParameters))
#
# # Close the cluster after the parallel computation
# stopCluster(this_cluster)

# 6.18h with: niter = 8000, nburnin = 2000, thin = 30
# saveRDS(salnimble2, file = file.path("../", "results",
→   "multi-2022-nimble-MH-corr-8k-2k-30-WAIC.rds"))

```

Correlated with IREs Model: Model-Corr&IRE

```
# n.chains <- 5
# this_cluster <- makeCluster(n.chains)

# Model code

modelCode <- nimbleCode(
{
  for(Var in 1:NVars) {
    # Likelihood
    for (Resp in 1:NResp) {
      y[Resp, Var] ~ dcat(prlevels[Resp, Var, 1:NCats])

      # Definition of the probabilities of each category as a function of the
      # cumulative probabilities
      prlevels[Resp, Var, 1] <- p.gamma[Resp, Var, 1]
      for (Cat in 2:(NCats-1)) {
        prlevels[Resp, Var, Cat] <- p.gamma[Resp, Var, Cat] - p.gamma[Resp, Var, Cat-1]
      }
      prlevels[Resp, Var, NCats] <- 1 - p.gamma[Resp, Var, NCats-1]

      # Linear predictor
      for (Cat in 1:(NCats-1)) {
        logit(p.gamma[Resp, Var, Cat]) <- kappa[sex[Resp], age[Resp], Cat, Var] +
          theta[muni[Resp], Var] + psi[Resp, Var]
      }
    }
  }

  # Prior distributions

  # kappa[1:NSEX, 1:NAges, 1:(NCats-1), 1:NVars] cut points
  # Monotonic transformation
  for (Var in 1:NVars) {
    for (SexGroup in 1:NSEX) {
      for (AgeGroup in 1:NAges) {
        for (Cat in 1:(NCats-1)) {
          kappa[SexGroup, AgeGroup, Cat, Var] <- logit(sum(delta[SexGroup, AgeGroup, Var,
↪ 1:Cat]))
        }
        # delta[1:NSEX, 1:NAges, 1:NVars, 1:NCats] Dirichlet prior
        delta[SexGroup, AgeGroup, Var, 1:NCats] ~ ddirch(ones[1:NCats])
      }
    }
  }

  # psi[1:NResp, 1:NVars] individual random effects
  for(Var in 1:NVars) {
    for (Resp in 1:NResp) {
      psi[Resp, Var] <- inprod(sub.Resp[Resp, ], M.Resp[, Var])
      # sub.Resp[1:NResp, 1:NVars] underlying individual REs
      sub.Resp[Resp, Var] ~ dnorm(0, 1)
    }
  }

  # theta[1:NMuni, 1:NVars] spatial random effects
  for (Var in 1:NVars) {
```

```

    for (Muni in 1:NMuni) {
      theta[Muni, Var] <- inprod(sub.Muni[Muni, ], M.Muni[, Var])
    }
    # sub.Muni[1:NMuni, 1:NVars] underlying spatial REs
    # LCAR distribution
    sub.Muni[1:NMuni, Var] ~ dcar_leroux(rho = rho[Var],
                                         sd.theta = 1,
                                         Lambda = Lambda[1:NMuni],
                                         from.to = from.to[1:NDist, 1:2])
  }

  # Hyperparameter of the spatial random effects
  for (Var in 1:NVars) {
    rho[Var] ~ dunif(0, 1)
  }

  # M.Resp[1:NVars, 1:NVars] and M.Muni[1:NVars, 1:NVars] M-matrices
  for (Var1 in 1:NVars) {
    for (Var2 in 1:NVars) {
      M.Resp[Var1, Var2] ~ dnorm(0, tau.M.Resp)
      M.Muni[Var1, Var2] ~ dnorm(0, tau.M.Muni)
    }
  }

  # Prior for precisions of M.Resp and M.Muni
  tau.M.Resp <- pow(sd.M.Resp, -2)
  sd.M.Resp ~ dhalfflat()

  tau.M.Muni <- pow(sd.M.Muni, -2)
  sd.M.Muni ~ dhalfflat()

  # Stochastic restrictions
  # Required vectors
  for (Var in 1:NVars) {
    for (Resp in 1:NResp) {
      sub.Muni.Resp[Resp, Var] <- sub.Muni[muni[Resp], Var]
    }

    # Zero-mean constraint for sub.Muni.Resp
    zero.sub.Muni.resp[Var] ~ dnorm(mean.sub.Munis.resp[Var], 10000)
    mean.sub.Munis.resp[Var] <- mean(sub.Muni.Resp[1:NResp, Var])
  }
}

# Model data
modelData <- list(y = as.matrix(y), zero.sub.Muni.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars, sex = sex,
                      age = age, muni = muni, NSex = NSex, NAges = NAges,
                      NMuni = NMuni, ones = ones, NDist = NDist, Lambda = Lambda,
                      from.to = from.to)

# Parameters to monitor
modelParameters <- c("kappa", "theta", "M.Muni", "rho",
                    "sd.M.Muni", "psi", "M.Resp", "sd.M.Resp",

```

```

      "delta", "sub.Muni", "sub.Resp")

# Create a function to run the MCMC
run_MCMC_allcode <- function(X, code, constants, data, monitors) {

  pacman::p_load(nimble, extraDistr, install = FALSE)

  dcar_leroux <- nimbleFunction(
    name = 'dcar_leroux',
    run = function(x = double(1),      # Spatial random effect (vector)
                  rho = double(0),      # Amount of spatial dependence (scalar)
                  sd.theta = double(0), # Standard deviation (scalar)
                  Lambda = double(1),   # Eigenvalues of matrix D - W
                  from.to = double(2),  # Matrix of distinct pairs of neighbours from.to[, 1] <
                    ↪ from.to[, 2]
                  log = integer(0, default = 0)) {
      returnType(double(0))

      # Number of small areas
      NMuni <- dim(x)[1]
      # Number of distinct pairs of neighbours
      NDist <- dim(from.to)[1]
      # Required vectors
      x.from <- nimNumeric(NDist)
      x.to <- nimNumeric(NDist)
      for (Dist in 1:NDist) {
        x.from[Dist] <- x[from.to[Dist, 1]]
        x.to[Dist] <- x[from.to[Dist, 2]]
      }

      logDens <- sum(dnorm(x[1:NMuni], mean = 0, sd = sd.theta * pow(1 - rho, -1/2), log = TRUE))
      ↪ -
        NMuni/2 * log(1 - rho) + 1/2 * sum(log(rho * (Lambda[1:NMuni] - 1) + 1)) -
        1/2 * pow(sd.theta, -2) * rho * sum(pow(x.from[1:NDist] - x.to[1:NDist], 2))
      if(log) return(logDens)
      else return(exp(logDens))
    }
  )

  rcar_leroux <- nimbleFunction(
    name = 'rcar_leroux',
    run = function(n = integer(0),
                  rho = double(0),
                  sd.theta = double(0),
                  Lambda = double(1),
                  from.to = double(2)) {
      returnType(double(1))

      nimStop("user-defined distribution dcar_leroux provided without random generation
      ↪ function.")
      x <- nimNumeric(542)
      return(x)
    }
  )

  assign('dcar_leroux', dcar_leroux, envir = .GlobalEnv)
  assign('rcar_leroux', rcar_leroux, envir = .GlobalEnv)

  NSex <- constants$NSex

```

```

NAges <- constants$NAges
NCats <- constants$NCats
ones <- constants$ones
NResp <- constants$NResp
NMuni <- constants$NMuni
NVars <- constants$NVars

# Create the NIMBLE model and initialize its nodes
model <- nimbleModel(code = code,
  constants = constants,
  data = data,
  inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
    dim = c(NSex, NAges, NVars, NCats)),
    rho = runif(NVars),
    sub.Resp = matrix(rnorm(NResp * NVars, sd = 0.01), nrow =
      ↪ NResp, ncol = NVars),
    M.Resp = matrix(rnorm(NVars * NVars, sd = 0.5), ncol = NVars,
      ↪ nrow = NVars),
    sd.M.Resp = runif(1, min = 0.2, max = 0.8),
    sub.Muni = matrix(rnorm(NMuni * NVars, sd = 0.01), nrow =
      ↪ NMuni, ncol = NVars),
    M.Muni = matrix(rnorm(NVars * NVars, sd = 0.5), ncol = NVars,
      ↪ nrow = NVars),
    sd.M.Muni = runif(1, min = 0.2, max = 0.8)),
  calculate = FALSE)

# Compile the model, which means generating C++ code, compiling that code, and loading it back
↪ into R
Cmodel <- compileNimble(model)

# model$getParents(model$getNodeNames(dataOnly = TRUE), stochOnly = TRUE)

# Configuration
modelMCMCconfiguration <- configureMCMC(model, useConjugacy = FALSE,
  enableWAIC = TRUE)

# Remove desired samplers
modelMCMCconfiguration$removeSamplers(c("sub.Muni", "rho", "sd.M.Muni",
  "sd.M.Resp"))

# Add RW-MH sub.Muni[1:NMuni, 1:NVars] samplers
sub.Munis <- matrix(nrow = NMuni, ncol = NVars)
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    sub.Munis[Muni, Var] <- paste0("sub.Muni[", Muni, ",", Var, "]")
  }
}

smuni <- sort(unique(constants$smuni))
for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    ifelse(Muni %in% smuni,
      modelMCMCconfiguration$addSampler(target = sub.Munis[Muni, Var], type = "RW"), #
      ↪ also slice
      modelMCMCconfiguration$addSampler(target = sub.Munis[Muni, Var], type = "RW"))
  }
}

# Add slice rho sampler

```



```

rhos <- character(NVars)
for (Var in 1:NVars) {
  rhos[Var] <- paste0("rho[",Var,"]")
}

for (Var in 1:NVars) {
  modelMCMCconfiguration$addSampler(target = rhos[Var], type = "slice")
}

# Add slice sd.M.Muni and sd.M.Resp samplers
modelMCMCconfiguration$addSampler(target = "sd.M.Resp", type = "slice")
modelMCMCconfiguration$addSampler(target = "sd.M.Muni", type = "slice")

# Add new monitors
modelMCMCconfiguration$monitors <- c()
modelMCMCconfiguration$addMonitors(monitors)
# Build MCMC object
modelMCMC <- buildMCMC(modelMCMCconfiguration)
# Need to reset the nimbleFunctions in order to add the new MCMC
CmodelMCMC <- compileNimble(modelMCMC, project = model,
                           resetFunctions = TRUE)

# Results
results <- runMCMC(CmodelMCMC, niter = 8000, nburnin = 2000, thin = 30, setSeed = X)

return(results)
}

# system.time(salnimble3 <- parLapply(cl = this_cluster, X = 1:n.chains,
#                                   fun = run_MCMC_allcode,
#                                   code = modelCode,
#                                   constants = modelConstants,
#                                   data = modelData,
#                                   monitors = modelParameters))
#
# # Close the cluster after the parallel computation
# stopCluster(this_cluster)

# 14.42h with: niter = 8000, nburnin = 2000, thin = 30
# saveRDS(salnimble3, file = file.path("../", "results",
#   ↪ "multi-2022-nimble-MH-corr-ire-8k-2k-30-WAIC.rds"))

```

Loading posterior samples

```

n.chains <- 5
n.sims <- 200 * n.chains
labels <- c("Item 1", "Item 2", "Item 3", "Item 4", "Item 5", "Item 6", "Item 7", "Item 8", "Item
↪ 9", "Item 10", "Item 11", "Item 12")
salnimble1 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-indep-8k-2k-30-WAIC.rds"))
salnimble2 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-8k-2k-30-WAIC.rds"))
salnimble3 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-ire-8k-2k-30-WAIC.rds"))

```

Convert NIMBLE output to WinBUGS-style format: Model-Indep

```

NimToWin <- function(salnimble) {

  n.chains <- length(salnimble)
  n.sims <- n.chains * nrow(salnimble[[1]])

  kappa <- array(dim = c(n.sims, NSex, NAges, NCats - 1, NVars))
  theta <- array(dim = c(n.sims, NMuni, NVars))
  rho <- matrix(nrow = n.sims, ncol = NVars)
  sd.theta <- matrix(nrow = n.sims, ncol = NVars)

  for (Var in 1:NVars) {
    for (Cat in 1:(NCats - 1)) {
      for (Sex in 1:NSex) {
        for (Age in 1:NAges) {
          kappa[, Sex, Age, Cat, Var] <- c(salnimble[[1]][, paste0("kappa[", Sex, ", ", Age, ", ",
→   ", Cat, ", ", Var, "]")],
                                           salnimble[[2]][, paste0("kappa[", Sex, ", ", Age, ", ",
→   ", Cat, ", ", Var, "]")],
                                           salnimble[[3]][, paste0("kappa[", Sex, ", ", Age, ", ",
→   ", Cat, ", ", Var, "]")],
                                           salnimble[[4]][, paste0("kappa[", Sex, ", ", Age, ", ",
→   ", Cat, ", ", Var, "]")],
                                           salnimble[[5]][, paste0("kappa[", Sex, ", ", Age, ", ",
→   ", Cat, ", ", Var, "]")])
        }
      }
    }
  }

  for (Var in 1:NVars) {
    for (Muni in 1:NMuni) {
      theta[, Muni, Var] <- c(salnimble[[1]][, paste0("theta[", Muni, ", ", Var, "]")],
                             salnimble[[2]][, paste0("theta[", Muni, ", ", Var, "]")],
                             salnimble[[3]][, paste0("theta[", Muni, ", ", Var, "]")],
                             salnimble[[4]][, paste0("theta[", Muni, ", ", Var, "]")],
                             salnimble[[5]][, paste0("theta[", Muni, ", ", Var, "]")])
    }
  }

  for (Var in 1:NVars) {
    rho[, Var] <- c(salnimble[[1]][, paste0("rho[", Var, "]")],
                   salnimble[[2]][, paste0("rho[", Var, "]")],
                   salnimble[[3]][, paste0("rho[", Var, "]")],
                   salnimble[[4]][, paste0("rho[", Var, "]")],
                   salnimble[[5]][, paste0("rho[", Var, "]")])
  }

  for (Var in 1:NVars) {
    sd.theta[, Var] <- c(salnimble[[1]][, paste0("sd.theta[", Var, "]")],
                        salnimble[[2]][, paste0("sd.theta[", Var, "]")],
                        salnimble[[3]][, paste0("sd.theta[", Var, "]")],
                        salnimble[[4]][, paste0("sd.theta[", Var, "]")],
                        salnimble[[5]][, paste0("sd.theta[", Var, "]")])
  }

  summary <- MCMCsummary(object = salnimble, round = 4)

```

```

# summary <- "not available"
sims.list <- list("kappa" = kappa, "theta" = theta, "sd.theta" = sd.theta,
                 "rho" = rho)

salwinbugs <- list("summary" = summary, "sims.list" = sims.list,
                 "n.chains" = n.chains, "n.sims" = n.sims)

return(salwinbugs)
}

salwinbugs1 <- NimToWin(salnimble = salnimble1)

```

Convert NIMBLE output to WinBUGS-style format: Model-Corr

```

NimToWin <- function(salnimble) {

  n.chains <- length(salnimble)
  n.sims <- n.chains * nrow(salnimble[[1]])

  kappa <- array(dim = c(n.sims, NSex, NAges, NCats - 1, NVars))
  theta <- array(dim = c(n.sims, NMuni, NVars))
  sd.M.Muni <- numeric(length = n.sims)
  M.Muni <- array(dim = c(n.sims, NVars, NVars))
  rho <- matrix(nrow = n.sims, ncol = NVars)

  for (Var in 1:NVars) {
    for (Cat in 1:(NCats - 1)) {
      for (Sex in 1:NSex) {
        for (Age in 1:NAges) {
          kappa[, Sex, Age, Cat, Var] <- c(salnimble[[1]][, paste0("kappa[", Sex, " ", " ", Age, " ",
→ " ", Cat, " ", " ", Var, ")]"),
          salnimble[[2]][, paste0("kappa[", Sex, " ", " ", Age, " ",
→ " ", Cat, " ", " ", Var, ")]"),
          salnimble[[3]][, paste0("kappa[", Sex, " ", " ", Age, " ",
→ " ", Cat, " ", " ", Var, ")]"),
          salnimble[[4]][, paste0("kappa[", Sex, " ", " ", Age, " ",
→ " ", Cat, " ", " ", Var, ")]"),
          salnimble[[5]][, paste0("kappa[", Sex, " ", " ", Age, " ",
→ " ", Cat, " ", " ", Var, ")]")])
        }
      }
    }
  }

  for (Var in 1:NVars) {
    for (Muni in 1:NMuni) {
      theta[, Muni, Var] <- c(salnimble[[1]][, paste0("theta[", Muni, " ", " ", Var, ")]"),
      salnimble[[2]][, paste0("theta[", Muni, " ", " ", Var, ")]"),
      salnimble[[3]][, paste0("theta[", Muni, " ", " ", Var, ")]"),
      salnimble[[4]][, paste0("theta[", Muni, " ", " ", Var, ")]"),
      salnimble[[5]][, paste0("theta[", Muni, " ", " ", Var, ")]")])
    }
  }

  sd.M.Muni <- c(salnimble[[1]][, "sd.M.Muni"], salnimble[[2]][, "sd.M.Muni"],

```

```

        salnimble[[3]][, "sd.M.Muni"], salnimble[[4]][, "sd.M.Muni"],
        salnimble[[5]][, "sd.M.Muni"])

for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    M.Muni[, Var1, Var2] <- c(salnimble[[1]][, paste0("M.Muni[, Var1, ", Var2, "]")],
                             salnimble[[2]][, paste0("M.Muni[, Var1, ", Var2, "]")],
                             salnimble[[3]][, paste0("M.Muni[, Var1, ", Var2, "]")],
                             salnimble[[4]][, paste0("M.Muni[, Var1, ", Var2, "]")],
                             salnimble[[5]][, paste0("M.Muni[, Var1, ", Var2, "]")])
  }
}

for (Var in 1:NVars) {
  rho[, Var] <- c(salnimble[[1]][, paste0("rho[, Var, "]")],
                 salnimble[[2]][, paste0("rho[, Var, "]")],
                 salnimble[[3]][, paste0("rho[, Var, "]")],
                 salnimble[[4]][, paste0("rho[, Var, "]")],
                 salnimble[[5]][, paste0("rho[, Var, "]")])
}

summary <- MCMCsummary(object = salnimble, round = 4)
# summary <- "not available"
sims.list <- list("kappa" = kappa, "theta" = theta, "sd.M.Muni" = sd.M.Muni,
                 "M.Muni" = M.Muni, "rho" = rho)

salwinbugs <- list("summary" = summary, "sims.list" = sims.list,
                 "n.chains" = n.chains, "n.sims" = n.sims)

return(salwinbugs)
}

salwinbugs2 <- NimToWin(salnimble = salnimble2)

```

Convert NIMBLE output to WinBUGS-style format: Model-Corr&IRE

```

NimToWin <- function(salnimble) {

  n.chains <- length(salnimble)
  n.sims <- n.chains * nrow(salnimble[[1]])

  kappa <- array(dim = c(n.sims, NSex, NAges, NCats - 1, NVars))
  psi <- array(dim = c(n.sims, NResp, NVars))
  sd.M.Resp <- numeric(length = n.sims)
  M.Resp <- array(dim = c(n.sims, NVars, NVars))
  theta <- array(dim = c(n.sims, NMuni, NVars))
  sd.M.Muni <- numeric(length = n.sims)
  M.Muni <- array(dim = c(n.sims, NVars, NVars))
  rho <- matrix(nrow = n.sims, ncol = NVars)

  for (Var in 1:NVars) {
    for (Cat in 1:(NCats - 1)) {
      for (Sex in 1:NSex) {

```

```

    for (Age in 1:Nages) {
      kappa[, Sex, Age, Cat, Var] <- c(salnimble[[1]][, paste0("kappa[", Sex, ", ", Age, ", 
→ ", Cat, ", ", Var, "]")],
                                     salnimble[[2]][, paste0("kappa[", Sex, ", ", Age, ", 
→ ", Cat, ", ", Var, "]")],
                                     salnimble[[3]][, paste0("kappa[", Sex, ", ", Age, ", 
→ ", Cat, ", ", Var, "]")],
                                     salnimble[[4]][, paste0("kappa[", Sex, ", ", Age, ", 
→ ", Cat, ", ", Var, "]")],
                                     salnimble[[5]][, paste0("kappa[", Sex, ", ", Age, ", 
→ ", Cat, ", ", Var, "]")])
    }
  }
}

for (Var in 1:NVars) {
  for (Resp in 1:NResp) {
    psi[, Resp, Var] <- c(salnimble[[1]][, paste0("psi[", Resp, ", ", Var, "]")],
                         salnimble[[2]][, paste0("psi[", Resp, ", ", Var, "]")],
                         salnimble[[3]][, paste0("psi[", Resp, ", ", Var, "]")],
                         salnimble[[4]][, paste0("psi[", Resp, ", ", Var, "]")],
                         salnimble[[5]][, paste0("psi[", Resp, ", ", Var, "]")])
  }
}

sd.M.Resp <- c(salnimble[[1]][, "sd.M.Resp"], salnimble[[2]][, "sd.M.Resp"],
              salnimble[[3]][, "sd.M.Resp"], salnimble[[4]][, "sd.M.Resp"],
              salnimble[[5]][, "sd.M.Resp"])

for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    M.Resp[, Var1, Var2] <- c(salnimble[[1]][, paste0("M.Resp[", Var1, ", ", Var2, "]")],
                             salnimble[[2]][, paste0("M.Resp[", Var1, ", ", Var2, "]")],
                             salnimble[[3]][, paste0("M.Resp[", Var1, ", ", Var2, "]")],
                             salnimble[[4]][, paste0("M.Resp[", Var1, ", ", Var2, "]")],
                             salnimble[[5]][, paste0("M.Resp[", Var1, ", ", Var2, "]")])
  }
}

for (Var in 1:NVars) {
  for (Muni in 1:NMuni) {
    theta[, Muni, Var] <- c(salnimble[[1]][, paste0("theta[", Muni, ", ", Var, "]")],
                           salnimble[[2]][, paste0("theta[", Muni, ", ", Var, "]")],
                           salnimble[[3]][, paste0("theta[", Muni, ", ", Var, "]")],
                           salnimble[[4]][, paste0("theta[", Muni, ", ", Var, "]")],
                           salnimble[[5]][, paste0("theta[", Muni, ", ", Var, "]")])
  }
}

sd.M.Muni <- c(salnimble[[1]][, "sd.M.Muni"], salnimble[[2]][, "sd.M.Muni"],
              salnimble[[3]][, "sd.M.Muni"], salnimble[[4]][, "sd.M.Muni"],
              salnimble[[5]][, "sd.M.Muni"])

for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    M.Muni[, Var1, Var2] <- c(salnimble[[1]][, paste0("M.Muni[", Var1, ", ", Var2, "]")],
                             salnimble[[2]][, paste0("M.Muni[", Var1, ", ", Var2, "]")],
                             salnimble[[3]][, paste0("M.Muni[", Var1, ", ", Var2, "]")],

```

```

        salnimble[[4]][, paste0("M.Muni[, Var1, ", ", Var2, "]")],
        salnimble[[5]][, paste0("M.Muni[, Var1, ", ", Var2, "]")])
    }
}

for (Var in 1:NVars) {
  rho[, Var] <- c(salnimble[[1]][, paste0("rho[, Var, "]")],
    salnimble[[2]][, paste0("rho[, Var, "]")],
    salnimble[[3]][, paste0("rho[, Var, "]")],
    salnimble[[4]][, paste0("rho[, Var, "]")],
    salnimble[[5]][, paste0("rho[, Var, "]")])
}

summary <- MCMCsummary(object = salnimble, round = 4)
# summary <- "not available"
sims.list <- list("kappa" = kappa, "theta" = theta, "M.Muni" = M.Muni,
  "rho" = rho, "sd.M.Muni" = sd.M.Muni, "psi" = psi,
  "sd.M.Resp" = sd.M.Resp, "M.Resp" = M.Resp)

salwinbugs <- list("summary" = summary, "sims.list" = sims.list,
  "n.chains" = n.chains, "n.sims" = n.sims)

return(salwinbugs)
}

salwinbugs3 <- NimToWin(salnimble = salnimble3)

```

Figure 1: Posterior means of the spatial effects

```

# Cartography of the Region of Valencia
load(file.path("../data", "CartoCV.Rdata"))

NMods <- 3
thetasim <- array(dim = c(n.sims, NMuni, NVars, NMods))
thetasim[, , , 1] <- salwinbugs1$sims.list$theta
thetasim[, , , 2] <- salwinbugs2$sims.list$theta
thetasim[, , , 3] <- salwinbugs3$sims.list$theta

selection <- 1:4
NSel <- length(selection)
thetasim <- thetasim[, , selection, ]

# Fifteen equal-probability intervals
breaks <- c(min(apply(thetasim, 2:4, mean)) - 0.001, quantile(apply(thetasim, 2:4, mean), probs =
  ↪ seq(1/15, 14/15, length.out = 14)), max(apply(thetasim, 2:4, mean)))
breaks <- c(-2.20, -0.5, -0.35, -0.25, -0.20, -0.15, -0.10, -0.05,
  0.05, 0.10, 0.15, 0.20, 0.25, 0.35, 0.5, 2.20)
break_labels <- c("\u2264 -0.50", "(-0.50, -0.35]", "(-0.35, -0.25]",
  "(-0.25, -0.20]", "(-0.20, -0.15]", "(-0.15, -0.10]",
  "(-0.10, -0.05]", "(-0.05, 0.05]", "(0.05, 0.10]",
  "(0.10, 0.15]", "(0.15, 0.20]", "(0.20, 0.25]",
  "(0.25, 0.35]", "(0.35, 0.50]", "> 0.50")

carto_muni_sf <- st_as_sf(carto_muni)
mod_labels <- c("Model-Indep", "Model-Corr", "Model-Corr&IRE")

```

```

res_list <- list()
for (Sel in 1:NSel) {
  for (Mod in 1:NMods) {
    tmp <- carto_muni_sf
    tmp$thetamean <- apply(thetasim[, , Sel, Mod], 2, mean)
    tmp$thetacat <- cut(tmp$thetamean, breaks = breaks, include.lowest = FALSE, right = TRUE,
    ↪ labels = break_labels)
    tmp$Item <- paste("Item", Sel)
    tmp$Model <- mod_labels[Mod]
    res_list[[length(res_list) + 1]] <- tmp
  }
}

plot_sf <- do.call(rbind, res_list)
plot_sf$Model <- factor(plot_sf$Model, levels = c("Model-Indep", "Model-Corr", "Model-Corr&IRE"))
plot_sf$Item <- factor(plot_sf$Item, levels = paste("Item", 1:NSel))
plot_sf$thetacat <- factor(plot_sf$thetacat, levels = break_labels)

```

```

ggplot(plot_sf) +
  geom_sf(aes(fill = thetacat), colour = "black", linewidth = 0.01) +
  facet_grid(Model ~ Item, switch = "y") +
  scale_fill_manual(values = colorRampPalette(brewer.pal(7, "BrBG"))(15),
    drop = FALSE, name = NULL, guide = guide_legend(reverse = TRUE)) +
  theme_void() +
  theme(strip.background = element_blank(),
    strip.text.x = element_text(size = 12, face = "bold"),
    strip.text.y.left = element_text(size = 12, face = "bold", angle = 90,
      margin = margin(r = 8, l = 8)),
    strip.placement = "outside", panel.spacing = unit(0.05, "lines"),
    legend.text = element_text(size = 9), legend.position = "right",
    legend.key.height = unit(0.9, "cm"), legend.key.width = unit(0.7, "cm"),
    legend.spacing.y = unit(0.08, "cm"))

```

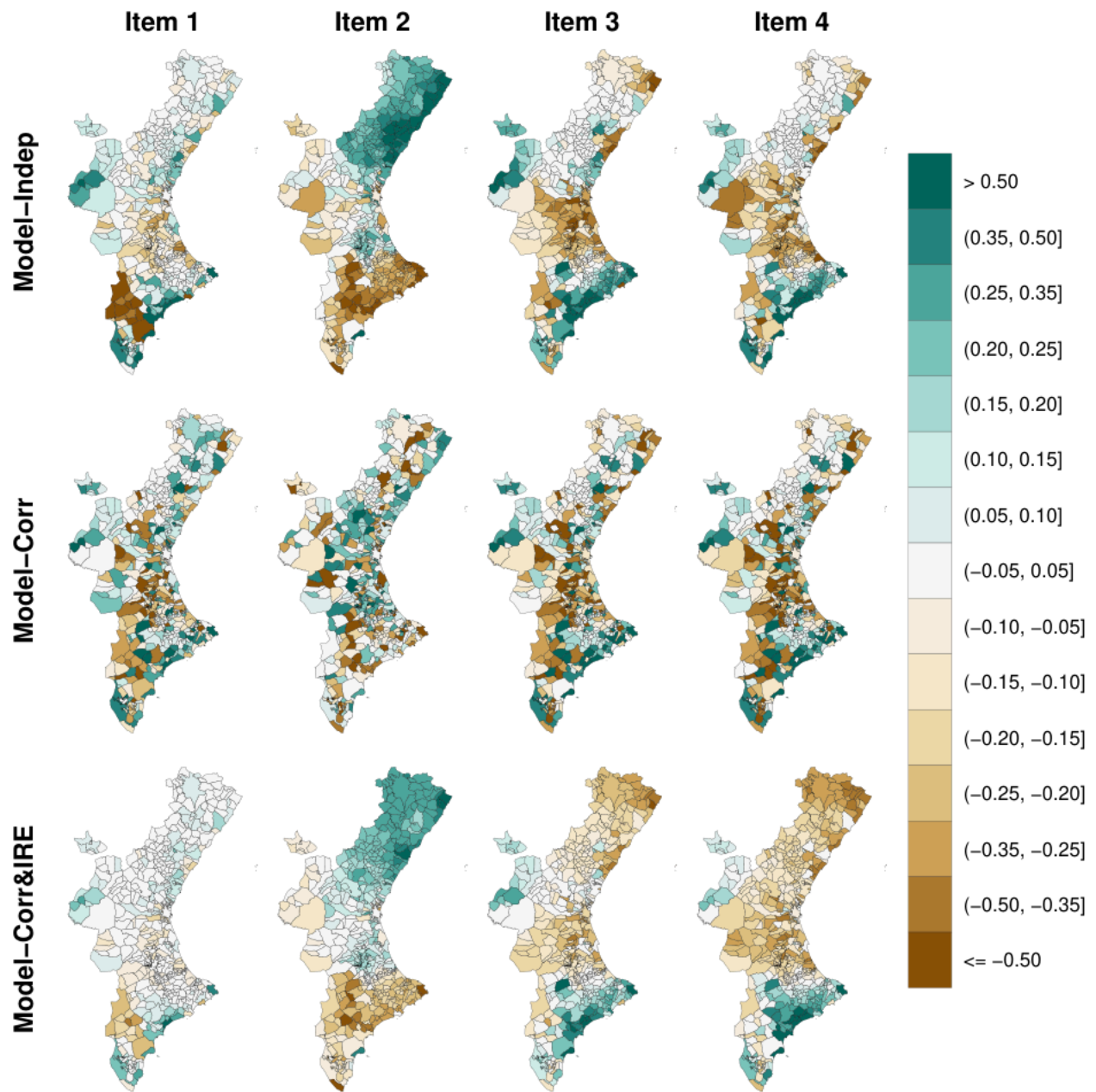


Figure S13: Posterior means of the spatial effects

```
# Cartography of the Region of Valencia
load(file.path("../data", "CartoCV.Rdata"))

NMods <- 3
thetasim <- array(dim = c(n.sims, NMuni, NVars, NMods))
thetasim[, , , 1] <- salwinbugs1$sims.list$theta
thetasim[, , , 2] <- salwinbugs2$sims.list$theta
thetasim[, , , 3] <- salwinbugs3$sims.list$theta
```



```

selection <- 1:NVars
NSel <- length(selection)
thetasim <- thetasim[, , selection, ]

# Fifteen equal-probability intervals
breaks <- c(min(apply(thetasim, 2:4, mean)) - 0.001, quantile(apply(thetasim, 2:4, mean), probs =
  ↪ seq(1/15, 14/15, length.out = 14)), max(apply(thetasim, 2:4, mean)))
breaks <- c(-3.75, -0.75, -0.25, -0.20, -0.15, -0.10, -0.05, -0.02,
  0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.75, 3.75)
break_labels <- c("\u2264 -0.75", "(-0.75, -0.25]", "(-0.25, -0.20]",
  "(-0.20, -0.15]", "(-0.15, -0.10]", "(-0.10, -0.05]",
  "(-0.05, -0.02]", "(-0.02, 0.02]", "(0.02, 0.05]",
  "(0.05, 0.10]", "(0.10, 0.15]", "(0.15, 0.20]",
  "(0.20, 0.25]", "(0.25, 0.75]", "> 0.75")

carto_muni_sf <- st_as_sf(carto_muni)
mod_labels <- c("Model-Indep", "Model-Corr", "Model-Corr&IRE")

res_list <- list()
for (Sel in 1:NSel) {
  for (Mod in 1:NMods) {
    tmp <- carto_muni_sf
    tmp$thetamean <- apply(thetasim[, , Sel, Mod], 2, mean)
    tmp$thetacat <- cut(tmp$thetamean, breaks = breaks, include.lowest = FALSE, right = TRUE,
  ↪ labels = break_labels)
    tmp$Item <- paste("Item", Sel)
    tmp$Model <- mod_labels[Mod]
    res_list[[length(res_list) + 1]] <- tmp
  }
}

plot_sf <- do.call(rbind, res_list)
plot_sf$Model <- factor(plot_sf$Model, levels = c("Model-Indep", "Model-Corr", "Model-Corr&IRE"))
plot_sf$Item <- factor(plot_sf$Item, levels = paste("Item", 1:NSel))
plot_sf$thetacat <- factor(plot_sf$thetacat, levels = break_labels)

```

```

ggplot(plot_sf) +
  geom_sf(aes(fill = thetacat), colour = "black", linewidth = 0.01) +
  facet_grid(Model ~ Item, switch = "y") +
  scale_fill_manual(values = colorRampPalette(brewer.pal(7, "BrBG"))(15),
    drop = FALSE, name = NULL, guide = guide_legend(reverse = TRUE)) +
  theme_void() +
  theme(strip.background = element_blank(),
    strip.text.x = element_text(size = 12, face = "bold"),
    strip.text.y.left = element_text(size = 12, face = "bold", angle = 90,
      margin = margin(r = 8, l = 8)),
    strip.placement = "outside", panel.spacing = unit(0.05, "lines"),
    legend.text = element_text(size = 9), legend.position = "right",
    legend.key.height = unit(0.9, "cm"), legend.key.width = unit(0.7, "cm"),
    legend.spacing.y = unit(0.08, "cm"))

```

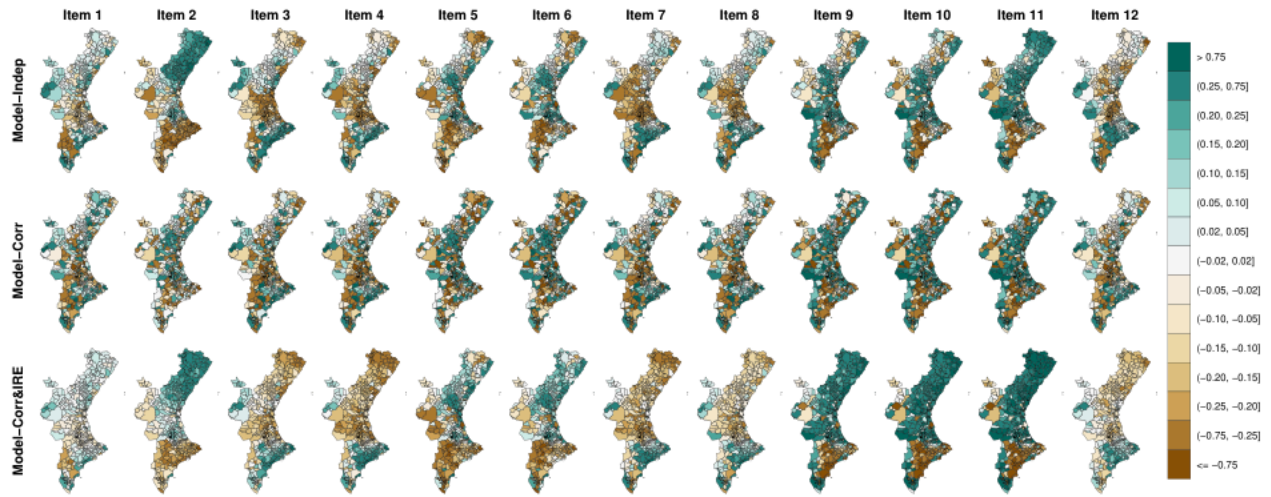


Figure 2: Posterior probabilities of the spatial effects

```
# Cartography of the Region of Valencia
load(file.path("../data", "CartoCV.Rdata"))

NMods <- 3
mod_labels <- c("Indep", "Corr", "CorrIRE")
thetasim <- array(dim = c(n.sims, NMuni, NVars, NMods))
thetasim[, , , 1] <- salwinbugs1$sims.list$theta
thetasim[, , , 2] <- salwinbugs2$sims.list$theta
thetasim[, , , 3] <- salwinbugs3$sims.list$theta

# Checking when theta is LESS than zero
stepsim <- 1 * (thetasim < 0)

selection <- 1:4
NSel <- length(selection)
stepsim <- stepsim[, , selection, ]

carto_muni_sf <- st_as_sf(carto_muni)
mod_labels <- c("Model-Indep", "Model-Corr", "Model-Corr&IRE")

res_list <- list()
for (Sel in 1:NSel) {
  for (Mod in 1:NMods) {
    tmp <- carto_muni_sf
    tmp$probmean <- apply(stepsim[, , Sel, Mod], 2, mean)
    tmp$Item <- paste("Item", Sel)
    tmp$Model <- mod_labels[Mod]
    res_list[[length(res_list) + 1]] <- tmp
  }
}

plot_sf <- do.call(rbind, res_list)
plot_sf$Model <- factor(plot_sf$Model, levels = c("Model-Indep", "Model-Corr", "Model-Corr&IRE"))
plot_sf$Item <- factor(plot_sf$Item, levels = paste("Item", 1:NSel))
```

```

ggplot(plot_sf) +
  geom_sf(aes(fill = probmean), colour = "black", linewidth = 0.01) +
  facet_grid(Model ~ Item, switch = "y") +
  scale_fill_stepsn(colours = colorRampPalette(brewer.pal(7, "RdYlGn"))(15)[15:1],
    breaks = seq(0, 1, length.out = 16), limits = c(0, 1),
    labels = c("0", "", "", "0.2", "", "", "0.4", "",
      "", "0.6", "", "", "0.8", "", "", "1"),
    guide = guide_colorsteps(title = NULL, even.steps = TRUE,
      show.limits = FALSE, ticks = TRUE,
      barheight = unit(14, "cm"),
      barwidth = unit(0.7, "cm")))) +
  theme_void() +
  theme(strip.background = element_blank(),
    strip.text.x = element_text(size = 12, face = "bold"),
    strip.text.y.left = element_text(size = 12, face = "bold", angle = 90,
      margin = margin(r = 8, l = 8)),
    strip.placement = "outside", panel.spacing = unit(0.05, "lines"),
    legend.text = element_text(size = 9), legend.position = "right")

```

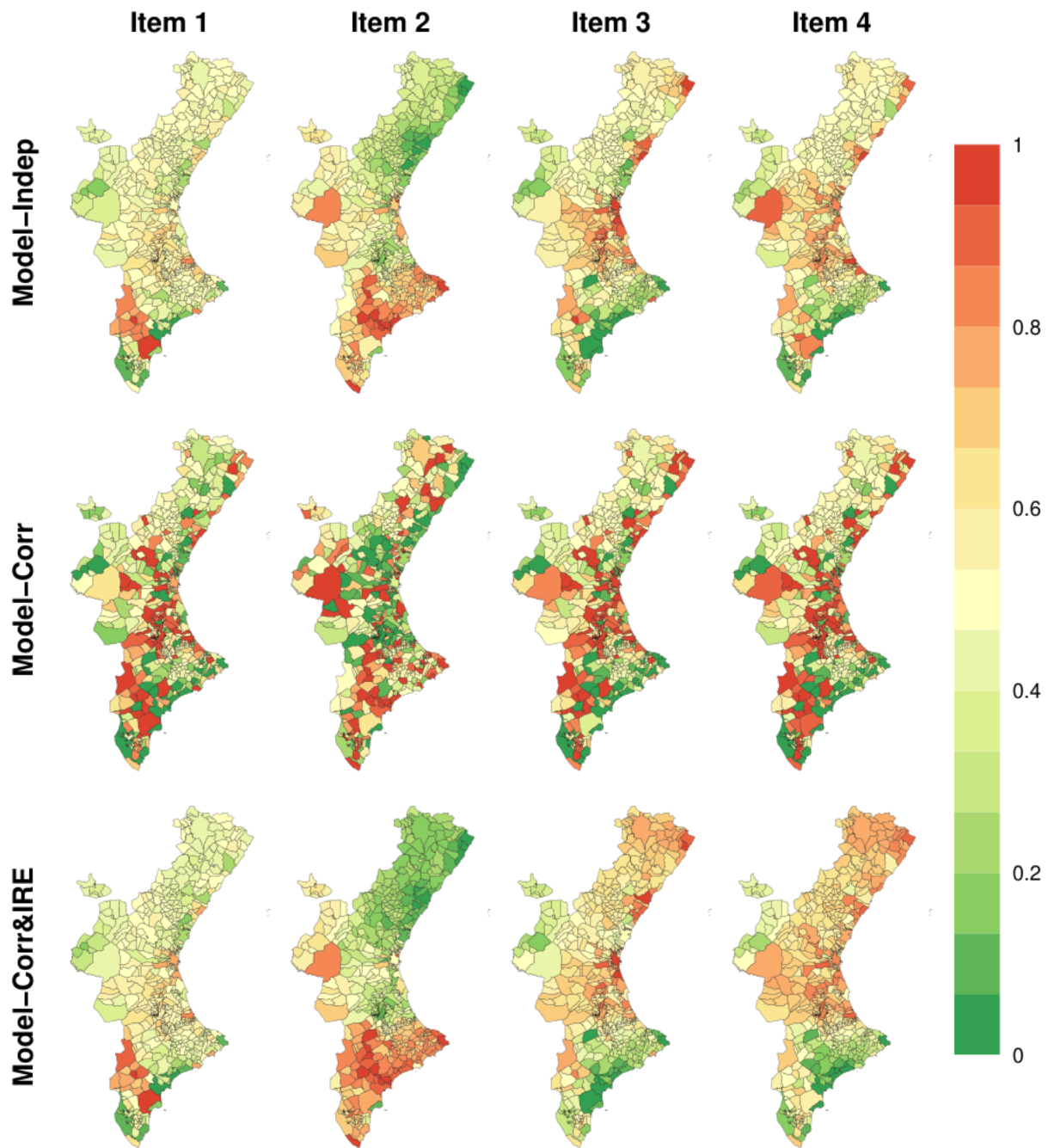


Figure S14: Posterior probabilities of the spatial effects

```
# Cartography of the Region of Valencia
load(file.path("../data", "CartoCV.Rdata"))

NMods <- 3
mod_labels <- c("Indep", "Corr", "CorrIRE")
thetasim <- array(dim = c(n.sims, NMuni, NVars, NMods))
thetasim[, , , 1] <- salwinbugs1$sims.list$theta
```

```

thetasim[, , , 2] <- salwinbugs2$sims.list$theta
thetasim[, , , 3] <- salwinbugs3$sims.list$theta

# Checking when theta is LESS than zero
stepsim <- 1 * (thetasim < 0)

selection <- 1:NVars
NSel <- length(selection)
stepsim <- stepsim[, , selection, ]

carto_muni_sf <- st_as_sf(carto_muni)
mod_labels <- c("Model-Indep", "Model-Corr", "Model-Corr&IRE")

res_list <- list()
for (Sel in 1:NSel) {
  for (Mod in 1:NMods) {
    tmp <- carto_muni_sf
    tmp$probmean <- apply(stepsim[, , Sel, Mod], 2, mean)
    tmp$Item <- paste("Item", Sel)
    tmp$Model <- mod_labels[Mod]
    res_list[[length(res_list) + 1]] <- tmp
  }
}

plot_sf <- do.call(rbind, res_list)
plot_sf$Model <- factor(plot_sf$Model, levels = c("Model-Indep", "Model-Corr", "Model-Corr&IRE"))
plot_sf$Item <- factor(plot_sf$Item, levels = paste("Item", 1:NSel))

```

```

ggplot(plot_sf) +
  geom_sf(aes(fill = probmean), colour = "black", linewidth = 0.01) +
  facet_grid(Model ~ Item, switch = "y") +
  scale_fill_stepsn(colours = colorRampPalette(brewer.pal(7, "RdYlGn"))(15)[15:1],
    breaks = seq(0, 1, length.out = 16), limits = c(0, 1),
    labels = c("0", "", "", "0.2", "", "", "0.4", "",
      "", "0.6", "", "", "0.8", "", "", "1"),
    guide = guide_colorsteps(title = NULL, even.steps = TRUE,
      show.limits = FALSE, ticks = TRUE,
      barheight = unit(14, "cm"),
      barwidth = unit(0.7, "cm"))) +
  theme_void() +
  theme(strip.background = element_blank(),
    strip.text.x = element_text(size = 12, face = "bold"),
    strip.text.y.left = element_text(size = 12, face = "bold", angle = 90,
      margin = margin(r = 8, l = 8)),
    strip.placement = "outside", panel.spacing = unit(0.05, "lines"),
    legend.text = element_text(size = 9), legend.position = "right")

```

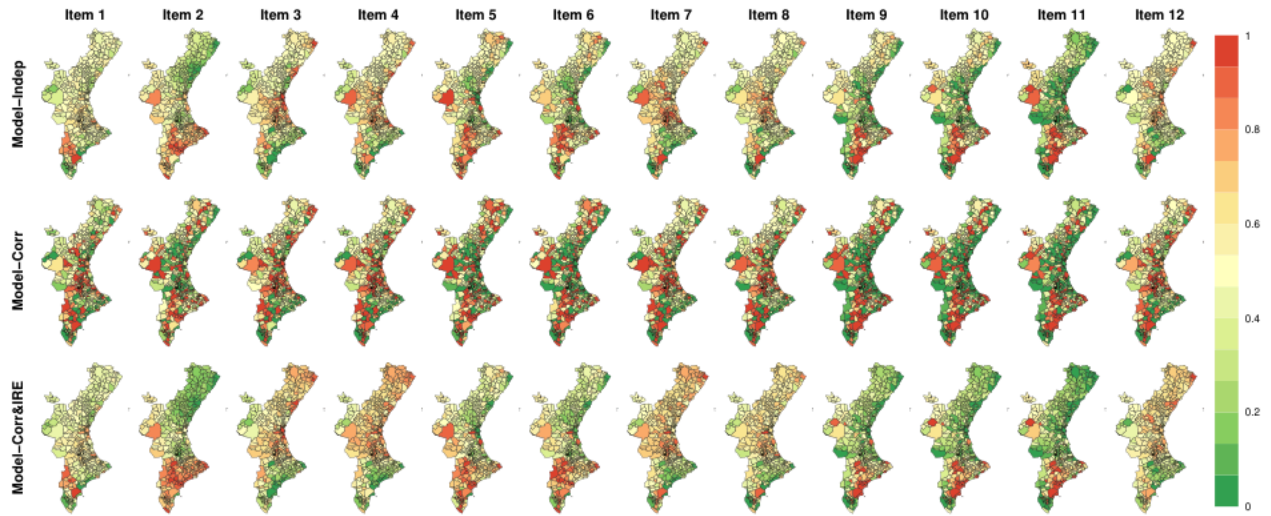


Figure 3: Municipality-level correlation matrix under Model-Corr&IRE

```
SurveyMapping.Sigma.Muni <- function(salwinbugs) {

  n.sims <- salwinbugs$n.sims
  NVars <- dim(salwinbugs$sims.list$theta)[3]
  M.Muni <- salwinbugs$sims.list$M.Muni

  Sigma.Muni <- array(dim = c(n.sims, NVars, NVars))

  for (sim in 1:n.sims) {
    Sigma.Muni[sim, , ] <- t(M.Muni[sim, , ]) %*% M.Muni[sim, , ]
  }
  return(Sigma.Muni)
}

salwinbugs <- salwinbugs3
Sigma.Munisim <- SurveyMapping.Sigma.Muni(salwinbugs = salwinbugs)
Corr <- array(dim = c(n.sims, NVars, NVars))

# Posterior sample of correlation matrices
for (sim in 1:n.sims) {
  Corr[sim, , ] <- diag(diag(Sigma.Munisim[sim, , ])^(-1/2)) %*% Sigma.Munisim[sim, , ] %*%
  → diag(diag(Sigma.Munisim[sim, , ])^(-1/2))
}

Corr.mean <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileL <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileU <- matrix(ncol = NVars, nrow = NVars)
Sigma.Munimean <- matrix(ncol = NVars, nrow = NVars)
for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    Corr.mean[Var1, Var2] <- mean(Corr[, Var1, Var2])
    Corr.quantileL[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.025)
    Corr.quantileU[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.975)
    Sigma.Munimean[Var1, Var2] <- mean(Sigma.Munisim[, Var1, Var2])
  }
}
```



```

    }
  }

  rm(Corr); rm(Sigma.Munisim)

  Corr.mean <- data.frame(Corr.mean); rownames(Corr.mean) <- labels;
  colnames(Corr.mean) <- labels

  orden <- c(3, 4, 1, 8, 7, 12, 11, 9, 10, 2, 5, 6)

  Corr.mean.orden <- as.matrix(Corr.mean)
  Corr.mean.orden <- Corr.mean.orden[orden, orden]

  Corr.quantileL <- data.frame(Corr.quantileL); rownames(Corr.quantileL) <- labels;
  colnames(Corr.quantileL) <- labels

  Corr.quantileL.orden <- as.matrix(Corr.quantileL)
  Corr.quantileL.orden <- Corr.quantileL.orden[orden, orden]

  Corr.quantileU <- data.frame(Corr.quantileU); rownames(Corr.quantileU) <- labels;
  colnames(Corr.quantileU) <- labels

  Corr.quantileU.orden <- as.matrix(Corr.quantileU)
  Corr.quantileU.orden <- Corr.quantileU.orden[orden, orden]

  Significance <- matrix(as.numeric(Corr.quantileL.orden > 0 | Corr.quantileU.orden < 0), ncol =
    ↪ NVars, nrow = NVars, byrow = FALSE)
  colnames(Significance) <- rownames(Significance) <- colnames(Corr.mean.orden)
  Significance <- (Significance - 1) * (-1)
  for (Var in 1:NVars) { Significance[Var, Var] <- 1 }

```

```

# First: ellipses in lower triangular
corrplot(as.matrix(Corr.mean.orden),
  type = "lower", method = "ellipse",
  p.mat = Significance, sig.level = 0.05, insig = "label_sig",
  pch.cex = 1.5, pch.col = "grey20",
  addCoef.col = "black", number.cex = 0.8,
  tl.pos = "d", tl.cex = 0.9, cl.pos = "r")

# Second: CI in upper triangular
corrplot(as.matrix(Corr.mean.orden),
  type = "upper", method = "square",
  diag = FALSE, add = TRUE, cl.pos = "n",
  plotCI = "rect", lowCI = as.matrix(Corr.quantileL.orden),
  uppCI = as.matrix(Corr.quantileU.orden), rect.col = "navy", tl.pos = "n")

```

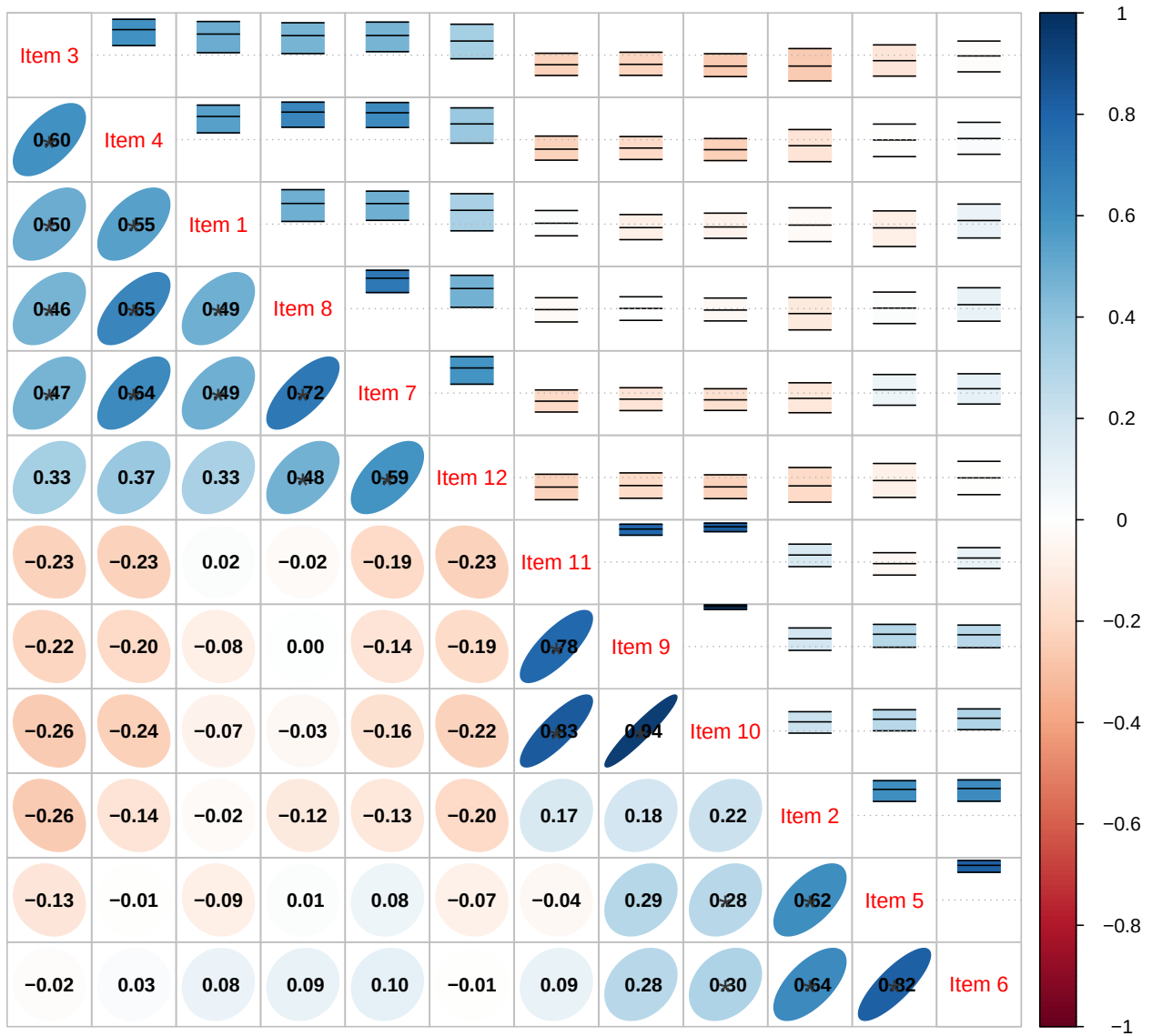


Figure 4: Individual-level correlation matrix under Model-Corr&IRE

```
SurveyMapping.Sigma.Resp <- function(salwinbugs) {

  n.sims <- salwinbugs$n.sims
  NVars <- dim(salwinbugs$sims.list$theta)[3]
  M.Resp <- salwinbugs$sims.list$M.Resp

  Sigma.Resp <- array(dim = c(n.sims, NVars, NVars))

  for (sim in 1:n.sims) {
    Sigma.Resp[sim, , ] <- t(M.Resp[sim, , ]) %*% M.Resp[sim, , ]
  }
  return(Sigma.Resp)
}
```



```

}

salwinbugs <- salwinbugs3

n.sims <- salwinbugs$n.sims
Sigma.Respsim <- SurveyMapping.Sigma.Resp(salwinbugs = salwinbugs)
Corr <- array(dim = c(n.sims, NVars, NVars))

# Posterior sample of correlation matrices
for (sim in 1:n.sims) {
  Corr[sim, , ] <- diag(diag(Sigma.Respsim[sim, , ])^(-1/2)) %*% Sigma.Respsim[sim, , ] %*%
  → diag(diag(Sigma.Respsim[sim, , ])^(-1/2))
}

Corr.mean <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileL <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileU <- matrix(ncol = NVars, nrow = NVars)
Sigma.Respmean <- matrix(ncol = NVars, nrow = NVars)
for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    Corr.mean[Var1, Var2] <- mean(Corr[, Var1, Var2])
    Corr.quantileL[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.025)
    Corr.quantileU[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.975)
    Sigma.Respmean[Var1, Var2] <- mean(Sigma.Respsim[, Var1, Var2])
  }
}

Corr.mean <- data.frame(Corr.mean); rownames(Corr.mean) <- labels; colnames(Corr.mean) <- labels

orden <- corrMatOrder(as.matrix(Corr.mean), order = "hclust", hclust.method = "ward.D2")
# orden <- c(3, 4, 1, 8, 7, 12, 11, 9, 10, 2, 5, 6)

Corr.mean.orden <- as.matrix(Corr.mean)
Corr.mean.orden <- Corr.mean.orden[orden, orden]

Corr.quantileL <- data.frame(Corr.quantileL); rownames(Corr.quantileL) <- labels;
colnames(Corr.quantileL) <- labels

Corr.quantileL.orden <- as.matrix(Corr.quantileL)
Corr.quantileL.orden <- Corr.quantileL.orden[orden, orden]

Corr.quantileU <- data.frame(Corr.quantileU); rownames(Corr.quantileU) <- labels;
colnames(Corr.quantileU) <- labels

Corr.quantileU.orden <- as.matrix(Corr.quantileU)
Corr.quantileU.orden <- Corr.quantileU.orden[orden, orden]

Significance <- matrix(as.numeric(Corr.quantileL.orden > 0 | Corr.quantileU.orden < 0), ncol =
→ NVars, nrow = NVars, byrow = FALSE)
colnames(Significance) <- rownames(Significance) <- colnames(Corr.mean.orden)
Significance <- (Significance - 1) * (-1)
for (Var in 1:NVars) { Significance[Var, Var] <- 1 }

# First: ellipses in lower triangular
corrplot(as.matrix(Corr.mean.orden),
  type = "lower", method = "ellipse",
  p.mat = Significance, sig.level = 0.05, insig = "label_sig",
  pch.cex = 1.5, pch.col = "grey20",

```

```

addCoef.col = "black", number.cex = 0.8,
tl.pos = "d", tl.cex = 0.9, cl.pos = "r")

# Second: CI in upper triangular
corrplot(as.matrix(Corr.mean.orden),
         type = "upper", method = "square",
         diag = FALSE, add = TRUE, cl.pos = "n",
         plotCI = "rect", lowCI = as.matrix(Corr.quantileL.orden),
         uppCI = as.matrix(Corr.quantileU.orden), rect.col = "navy", tl.pos = "n")

```

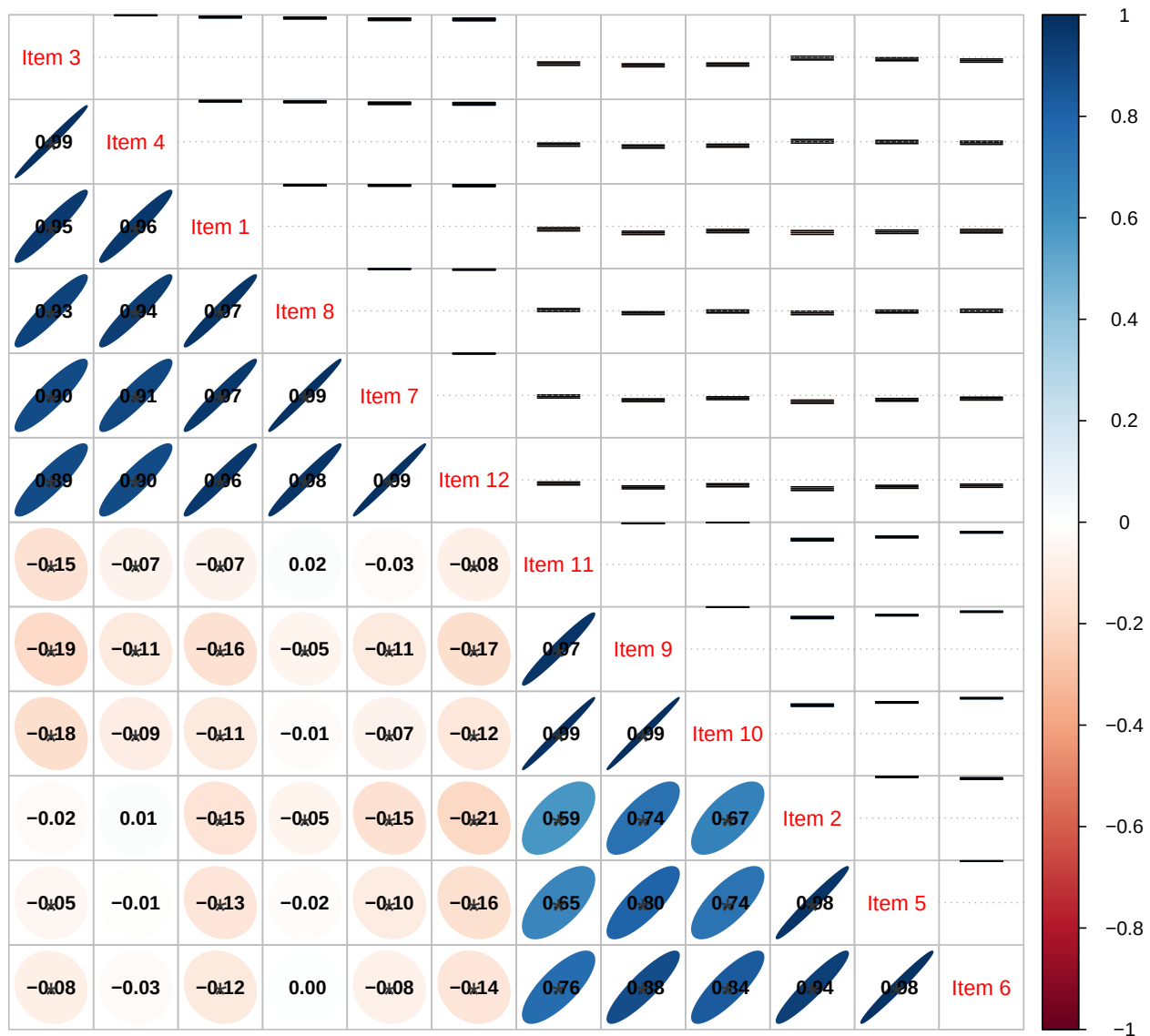


Figure S15: Municipality-level correlation matrix under Model-Corr

```

SurveyMapping.Sigma.Muni <- function(salwinbugs) {

  n.sims <- salwinbugs$n.sims
  NVars <- dim(salwinbugs$sims.list$theta)[3]
  M.Muni <- salwinbugs$sims.list$M.Muni

  Sigma.Muni <- array(dim = c(n.sims, NVars, NVars))

  for (sim in 1:n.sims) {
    Sigma.Muni[sim, , ] <- t(M.Muni[sim, , ]) %*% M.Muni[sim, , ]
  }
  return(Sigma.Muni)
}

salwinbugs <- salwinbugs2
n.sims <- salwinbugs$n.sims
Sigma.Munisim <- SurveyMapping.Sigma.Muni(salwinbugs = salwinbugs)
Corr <- array(dim = c(n.sims, NVars, NVars))

# Posterior sample of correlation matrices
for (sim in 1:n.sims) {
  Corr[sim, , ] <- diag(diag(Sigma.Munisim[sim, , ])^(-1/2)) %*% Sigma.Munisim[sim, , ] %*%
  ↪ diag(diag(Sigma.Munisim[sim, , ])^(-1/2))
}

Corr.mean <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileL <- matrix(ncol = NVars, nrow = NVars)
Corr.quantileU <- matrix(ncol = NVars, nrow = NVars)
Sigma.Munimean <- matrix(ncol = NVars, nrow = NVars)
for (Var1 in 1:NVars) {
  for (Var2 in 1:NVars) {
    Corr.mean[Var1, Var2] <- mean(Corr[, Var1, Var2])
    Corr.quantileL[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.025)
    Corr.quantileU[Var1, Var2] <- quantile(Corr[, Var1, Var2], probs = 0.975)
    Sigma.Munimean[Var1, Var2] <- mean(Sigma.Munisim[, Var1, Var2])
  }
}

rm(Corr); rm(Sigma.Munisim)

Corr.mean <- data.frame(Corr.mean); rownames(Corr.mean) <- labels;
colnames(Corr.mean) <- labels

Corr.mean.orden <- as.matrix(Corr.mean)
Corr.mean.orden <- Corr.mean.orden[orden, orden]

Corr.quantileL <- data.frame(Corr.quantileL); rownames(Corr.quantileL) <- labels;
colnames(Corr.quantileL) <- labels

orden <- c(3, 4, 1, 8, 7, 12, 11, 9, 10, 2, 5, 6)

Corr.quantileL.orden <- as.matrix(Corr.quantileL)
Corr.quantileL.orden <- Corr.quantileL.orden[orden, orden]

Corr.quantileU <- data.frame(Corr.quantileU); rownames(Corr.quantileU) <- labels;
colnames(Corr.quantileU) <- labels

Corr.quantileU.orden <- as.matrix(Corr.quantileU)
Corr.quantileU.orden <- Corr.quantileU.orden[orden, orden]

```

```

Significance <- matrix(as.numeric(Corr.quantileL.orden > 0 | Corr.quantileU.orden < 0), ncol =
  ↪ NVars, nrow = NVars, byrow = FALSE)
colnames(Significance) <- rownames(Significance) <- colnames(Corr.mean.orden)
Significance <- (Significance - 1) * (-1)
for (Var in 1:NVars) { Significance[Var, Var] <- 1 }

```

```

# First: ellipses in lower triangular
corrplot(as.matrix(Corr.mean.orden),
  type = "lower", method = "ellipse",
  p.mat = Significance, sig.level = 0.05, insig = "label_sig",
  pch.cex = 1.5, pch.col = "grey20",
  addCoef.col = "black", number.cex = 0.8,
  tl.pos = "d", tl.cex = 0.9, cl.pos = "r")

# Second: CI in upper triangular
corrplot(as.matrix(Corr.mean.orden),
  type = "upper", method = "square",
  diag = FALSE, add = TRUE, cl.pos = "n",
  plotCI = "rect", lowCI = as.matrix(Corr.quantileL.orden),
  uppCI = as.matrix(Corr.quantileU.orden), rect.col = "navy", tl.pos = "n")

```

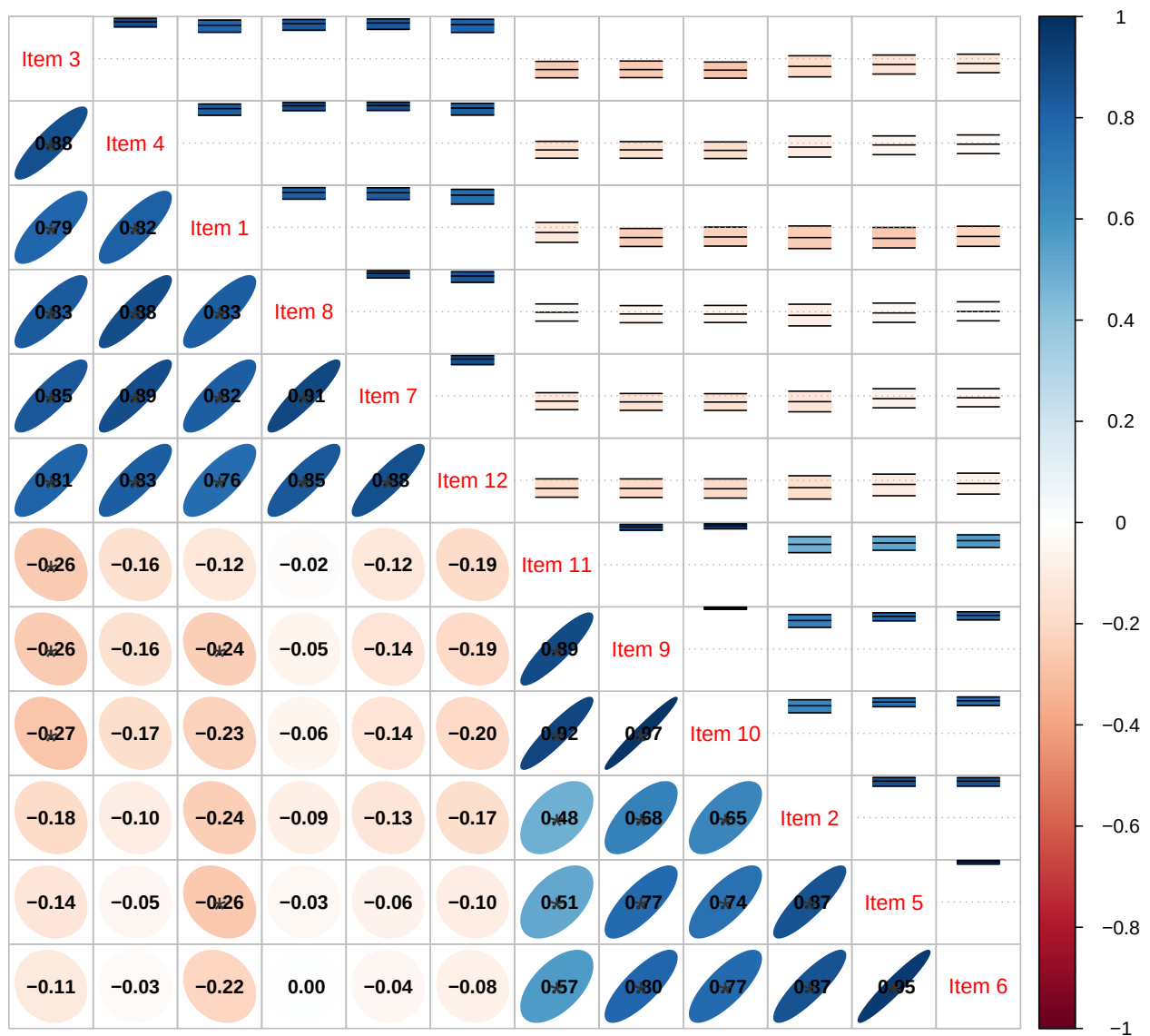


Figure 5: Principal component analysis of the spatial effects

```
salwinbugs <- salwinbugs3
thetasim <- salwinbugs$sims.list$theta
thetamean <- apply(thetasim, 2:3, mean)
pca <- princomp(thetamean, cor = TRUE)
# summary(pca)
pca$loadings[, 1:2]
```

```
##          Comp.1      Comp.2
## [1,]  0.1500401  0.46181484
## [2,] -0.2984278  0.23852458
## [3,]  0.3397215  0.16086642
## [4,]  0.3412119  0.21268818
```

```
## [5,] -0.2052693 0.37894367
## [6,] -0.2249631 0.43823141
## [7,] 0.3246205 0.27712025
## [8,] 0.2802306 0.35851188
## [9,] -0.2963037 0.20552491
## [10,] -0.3065521 0.20412406
## [11,] -0.2994604 0.15756273
## [12,] 0.3293849 0.09494317
```

```
# plot(pca, type = "l")
thetaComp <- pca$scores[, 1:2]
cor(thetaComp)
```

```
##           Comp.1      Comp.2
## Comp.1 1.00000e+00 3.30795e-17
## Comp.2 3.30795e-17 1.00000e+00
```

```
carto_muni_sf <- st_as_sf(carto_muni)
carto_muni_sf$pcal_num <- thetaComp[, 1]

# Fifteen equal-probability intervals
breaks <- c(min(carto_muni_sf$pcal_num) - 0.001, quantile(carto_muni_sf$pcal_num, probs =
  ↪ seq(1/15, 14/15, length.out = 14)), max(carto_muni_sf$pcal_num))
breaks <- c(-17.50, -3.00, -2.50, -2.00, -1.50, -1.00, -0.50, -0.25,
  0.25, 0.50, 1.00, 1.50, 2.00, 2.50, 3.00, 17.50)
labels <- c("Worse S1", "Better S2", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ")

carto_muni_sf$pcal_cat <- cut(carto_muni_sf$pcal_num, breaks = breaks, include.lowest = FALSE,
  ↪ right = TRUE, labels = labels)

p1 <- ggplot(carto_muni_sf) +
  geom_sf(aes(fill = pcal_cat), colour = "black", linewidth = 0.01) +
  scale_fill_manual(values = colorRampPalette(brewer.pal(7, "RdYlBu"))(15),
    drop = FALSE, name = NULL, guide = guide_legend(reverse = TRUE)) +
  labs(title = "(A)") + theme_void() +
  theme(plot.title = element_text(size = 18, face = "bold", hjust = 0),
    legend.text = element_text(size = 14), legend.position = "right",
    legend.key.height = unit(0.9, "cm"), legend.key.width = unit(0.7, "cm"),
    legend.spacing.y = unit(0.08, "cm"))

carto_muni_sf$pcal2_num <- thetaComp[, 2]

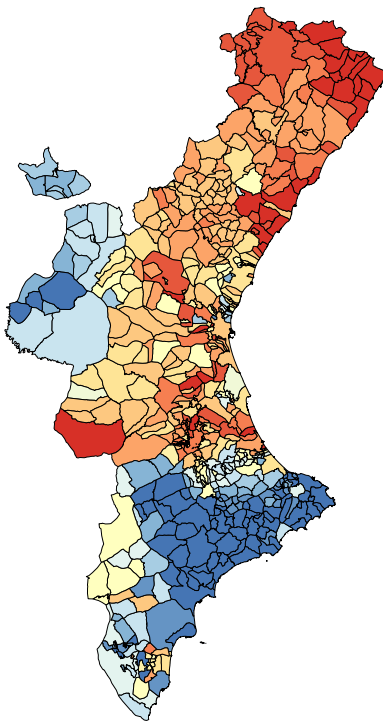
# Fifteen equal-probability intervals
breaks <- c(min(carto_muni_sf$pcal2_num) - 0.001, quantile(carto_muni_sf$pcal2_num, probs =
  ↪ seq(1/15, 14/15, length.out = 14)), max(carto_muni_sf$pcal2_num))
labels <- c("Better MH", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ",
  " ", " ", " ", " ", " ", " ", " ")

carto_muni_sf$pcal2_cat <- cut(carto_muni_sf$pcal2_num, breaks = breaks, include.lowest = FALSE,
  ↪ right = TRUE)
carto_muni_sf$pcal2_cat <- factor(carto_muni_sf$pcal2_cat, levels =
  ↪ rev(levels(carto_muni_sf$pcal2_cat)), labels = labels)
```

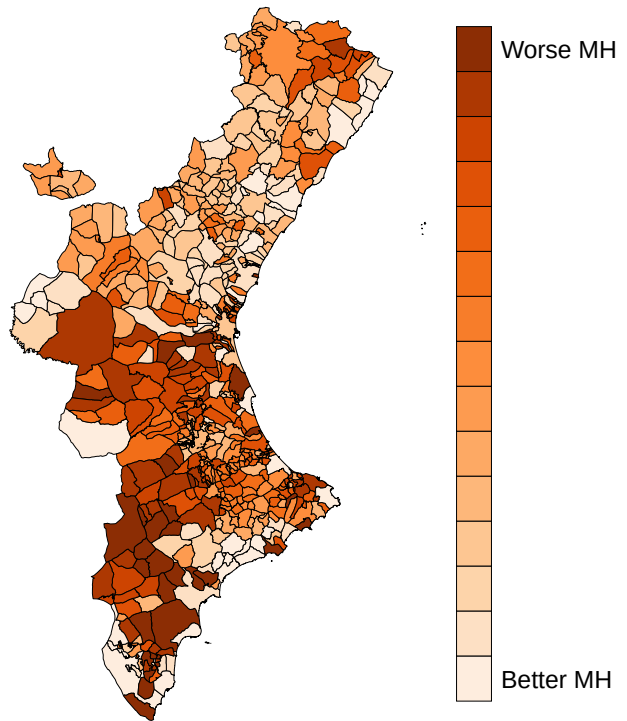
```
p2 <- ggplot(carto_muni_sf) +
  geom_sf(aes(fill = pca2_cat), colour = "black", linewidth = 0.01) +
  scale_fill_manual(values = colorRampPalette(brewer.pal(7, "Oranges"))(15),
    drop = FALSE, name = NULL, guide = guide_legend(reverse = TRUE)) +
  labs(title = "(B)") + theme_void() +
  theme(plot.title = element_text(size = 18, face = "bold", hjust = 0),
    legend.text = element_text(size = 14), legend.position = "right",
    legend.key.height = unit(0.9, "cm"), legend.key.width = unit(0.7, "cm"),
    legend.spacing.y = unit(0.08, "cm"))
```

```
p1 + p2
```

(A)



(B)



Figures S1–S12: Cut points by sex and age group

```
round(salwinbugs3$summary, 4)[startsWith(labels(salwinbugs3$summary)[[1]], "kappa"), ][1:10, ]
```

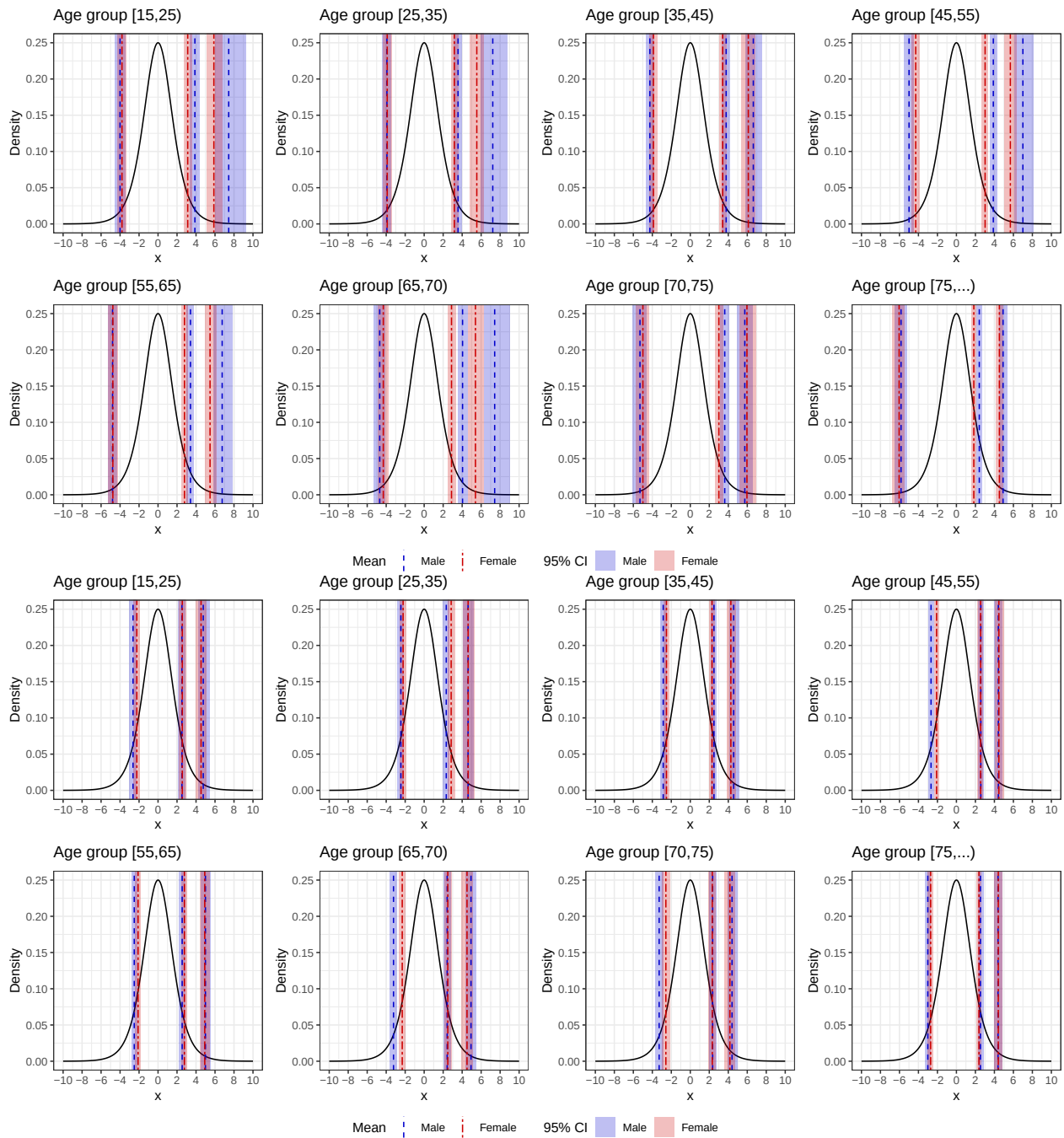
##		mean	sd	2.5%	50%	97.5%	Rhat	n.eff
##	kappa[1, 1, 1, 1]	-4.0030	0.2768	-4.5731	-4.0036	-3.4708	1.00	1003
##	kappa[2, 1, 1, 1]	-3.7967	0.2473	-4.2924	-3.7924	-3.3261	1.00	970
##	kappa[1, 2, 1, 1]	-3.9218	0.2430	-4.4046	-3.9187	-3.4581	1.01	1445
##	kappa[2, 2, 1, 1]	-3.8904	0.2363	-4.3534	-3.8956	-3.4216	1.00	988
##	kappa[1, 3, 1, 1]	-4.2577	0.2305	-4.6876	-4.2617	-3.8036	1.01	1111

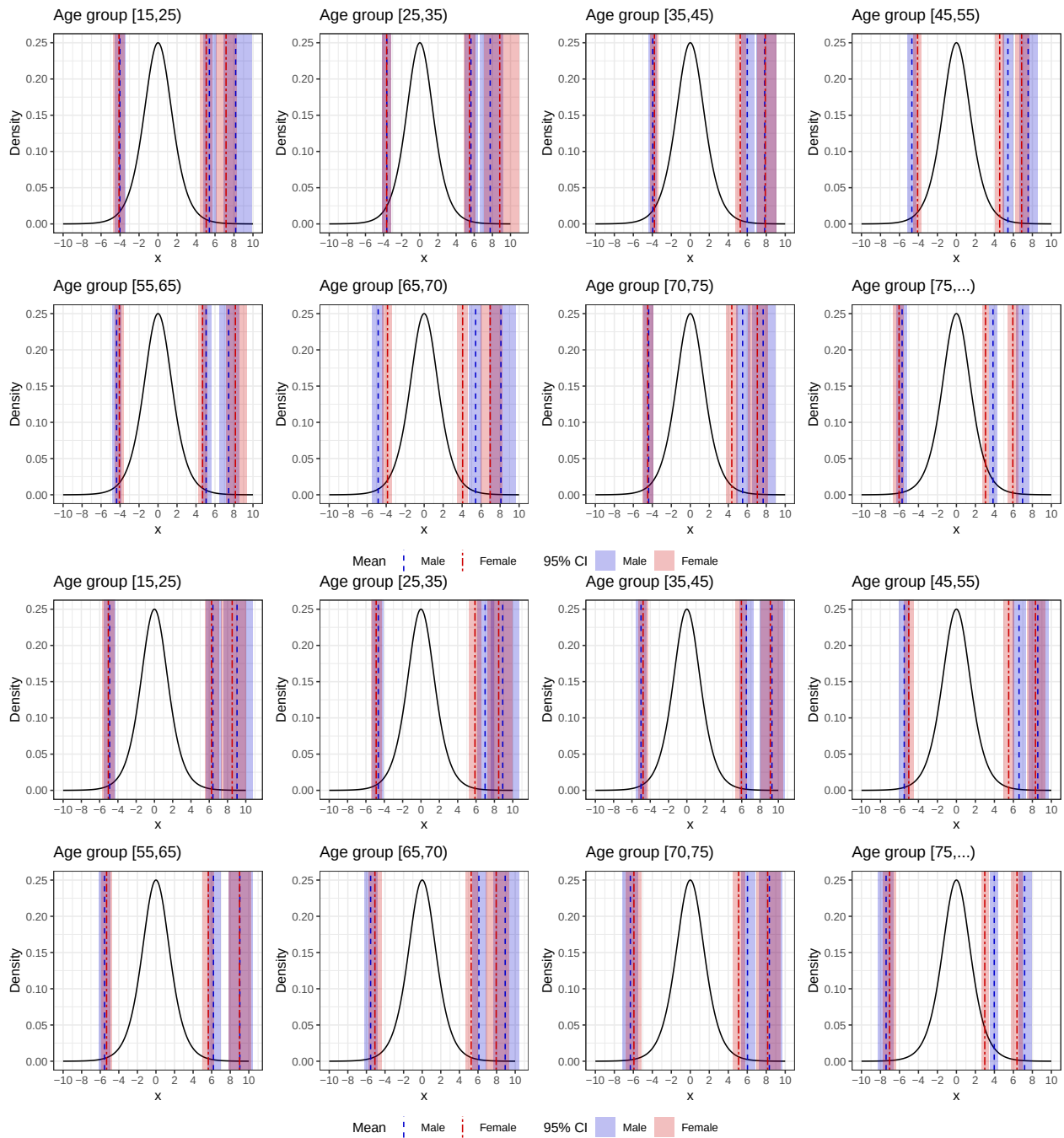
```
## kappa[2, 3, 1, 1] -3.8801 0.2070 -4.2995 -3.8756 -3.4905 1.00 965
## kappa[1, 4, 1, 1] -4.9890 0.2604 -5.5495 -4.9876 -4.5058 1.00 944
## kappa[2, 4, 1, 1] -4.2869 0.2238 -4.7612 -4.2826 -3.8899 1.00 934
## kappa[1, 5, 1, 1] -4.7905 0.2562 -5.2968 -4.7829 -4.3253 1.00 1097
## kappa[2, 5, 1, 1] -4.7581 0.2590 -5.2821 -4.7522 -4.2430 1.00 1000
```

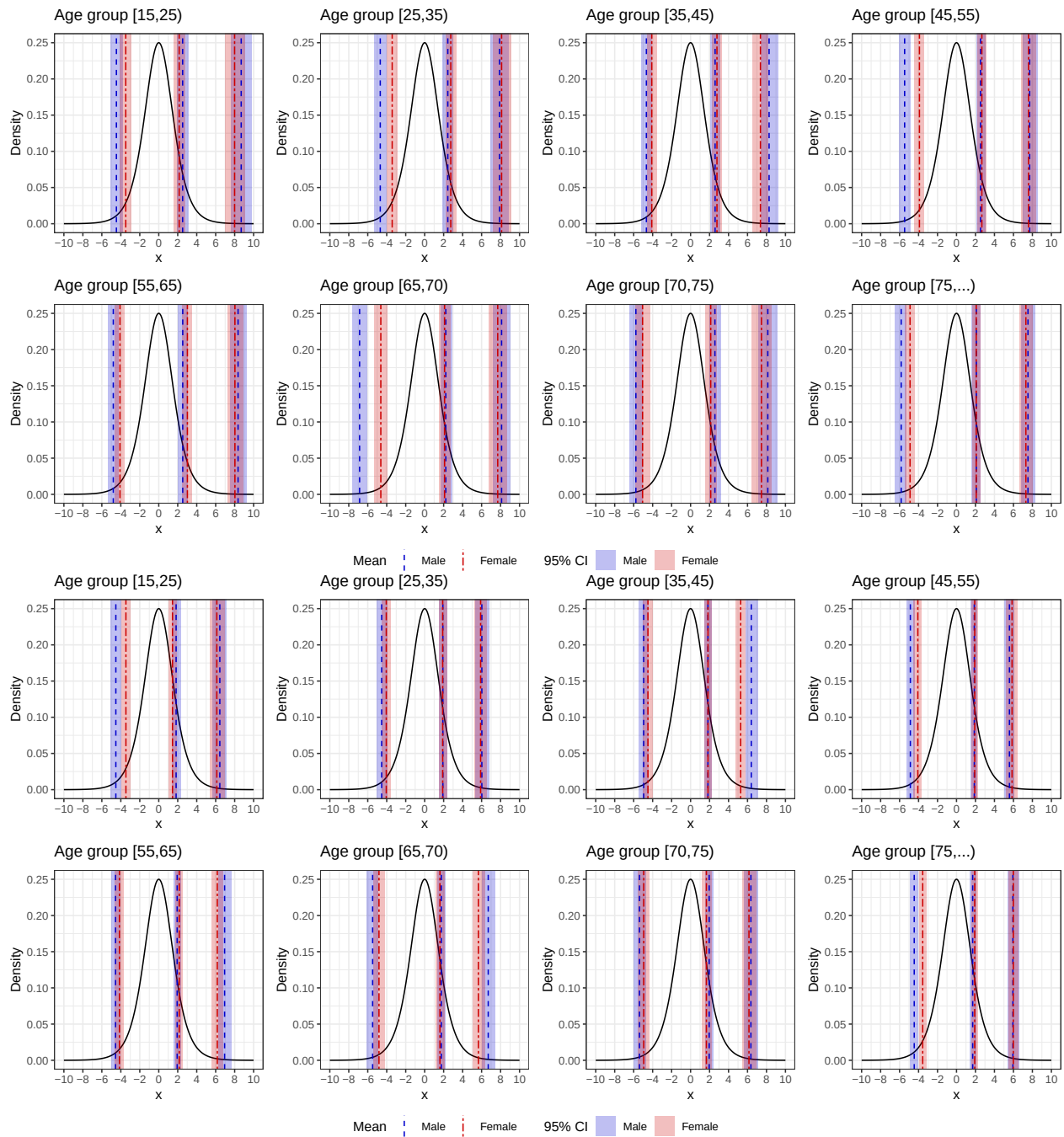
```
kappamean <- apply(salwinbugs3$sims.list$kappa, 2:5, mean)
kappaCI <- apply(salwinbugs3$sims.list$kappa, 2:5, quantile, probs = c(0.025, 0.975))
x <- seq(from = -10, to = 10, length.out = 1000)
```

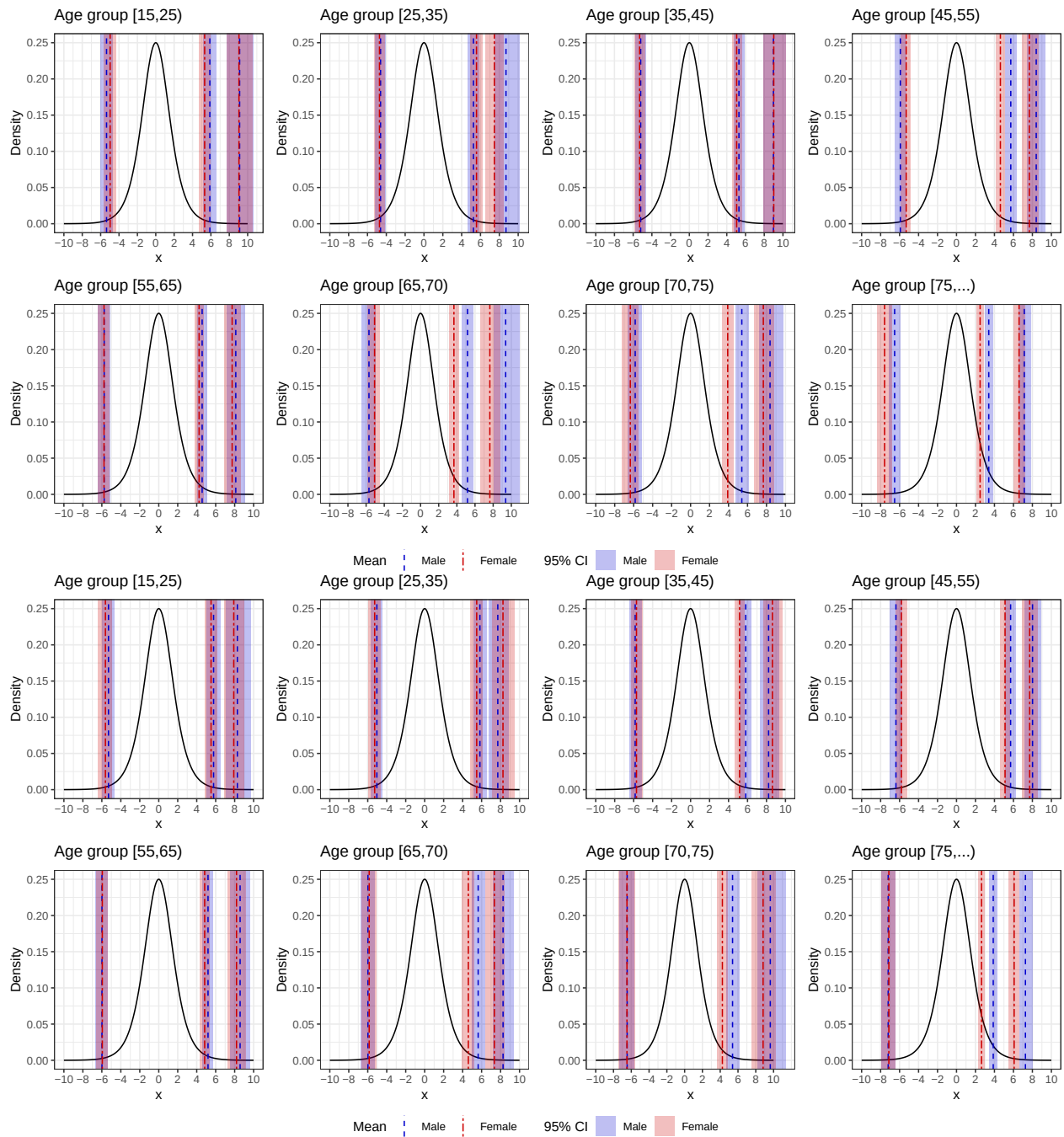
```
for (Var in 1:NVars) {
  p <- vector("list", NAges)
  for (AgeGroup in 1:NAges) {
    AgeLevel <- levels(ageC)[AgeGroup]
    df <- data.frame("x" = x, "y" = dlogis(x))
    lines <- data.frame("intercepts" = as.numeric(kappamean[, AgeGroup, , Var]),
                       "Sex" = rep(c("Male", "Female"), NCats - 1))
    ic <- data.frame("lower" = as.numeric(kappaCI[1, , AgeGroup, , Var]),
                    "upper" = as.numeric(kappaCI[2, , AgeGroup, , Var]),
                    "Sex" = rep(c("Male", "Female"), NCats - 1))
    lines$Sex <- factor(lines$Sex, levels = c("Male", "Female"))
    ic$Sex <- factor(ic$Sex, levels = c("Male", "Female"))
    p[[AgeGroup]] <- ggplot() +
      geom_line(data = df, mapping = aes(x = x, y = y)) +
      geom_rect(data = ic, mapping = aes(xmin = lower, xmax = upper,
                                         ymin = -0.1, ymax = 0.30, fill = Sex),
               alpha = 0.25) +
      geom_vline(data = lines,
                 mapping = aes(xintercept = intercepts, linetype = Sex, color = Sex)) +
      scale_fill_manual(name = "95% CI", breaks = c("Male", "Female"),
                        values = c("Male" = "blue3", "Female" = "red3")) +
      scale_color_manual(name = "Mean", breaks = c("Male", "Female"),
                         values = c("Male" = "blue3", "Female" = "red3")) +
      scale_linetype_manual(name = "Mean", breaks = c("Male", "Female"),
                            values = c("Male" = "dashed", "Female" = "twodash")) +
      labs(title = substitute(paste("Age group ", a, list(a = AgeLevel)), x = "x", y =
        ↪ "Density") +
      coord_cartesian(ylim = c(0, 0.25)) + theme_bw() + scale_x_continuous(breaks = seq(-10, 10,
        ↪ by = 2))
  }

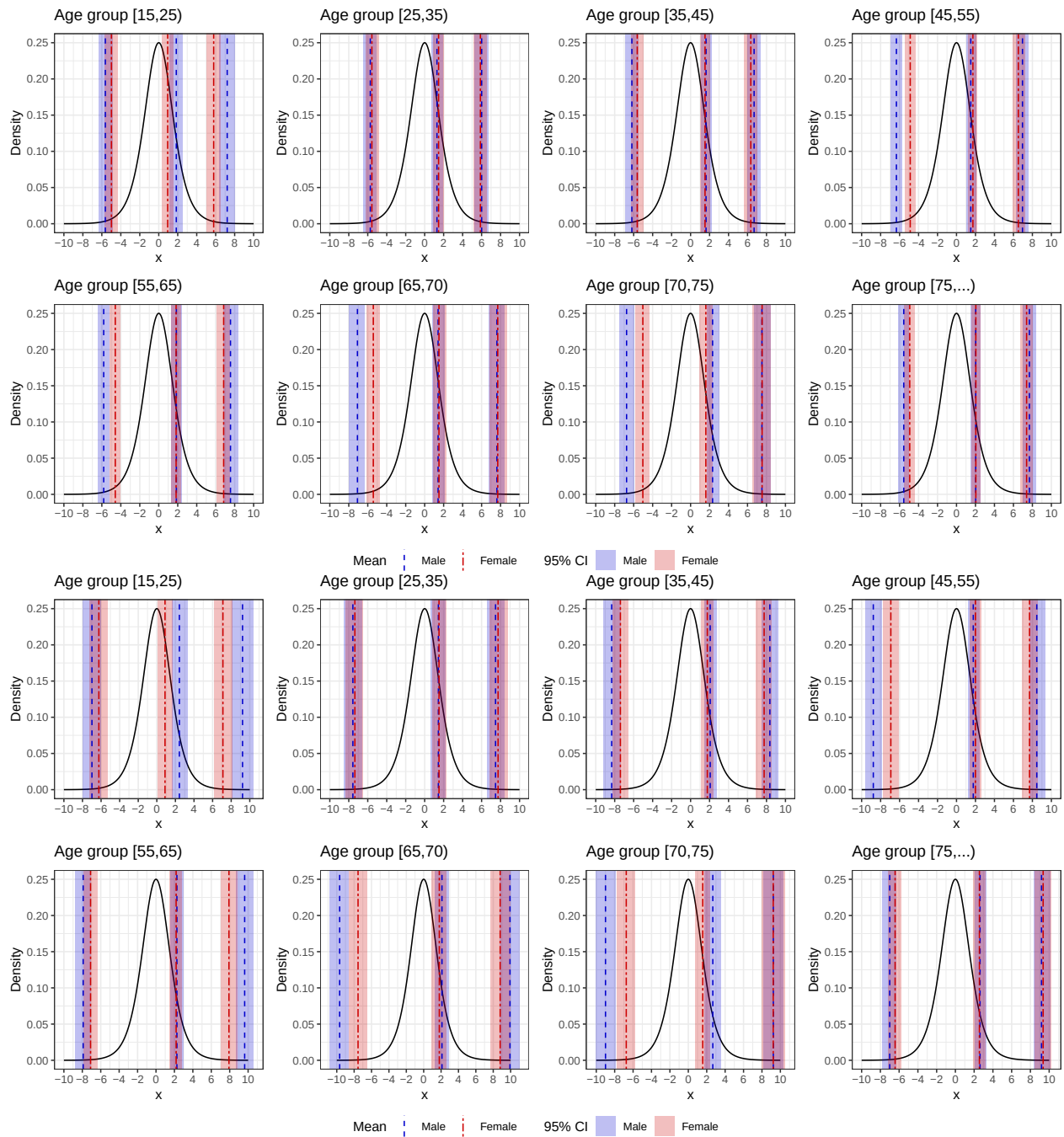
  kappa <- ggarrange(plotlist = p, nrow = 2, ncol = 4, common.legend = TRUE,
                    legend = "bottom")
  print(kappa)
}
```

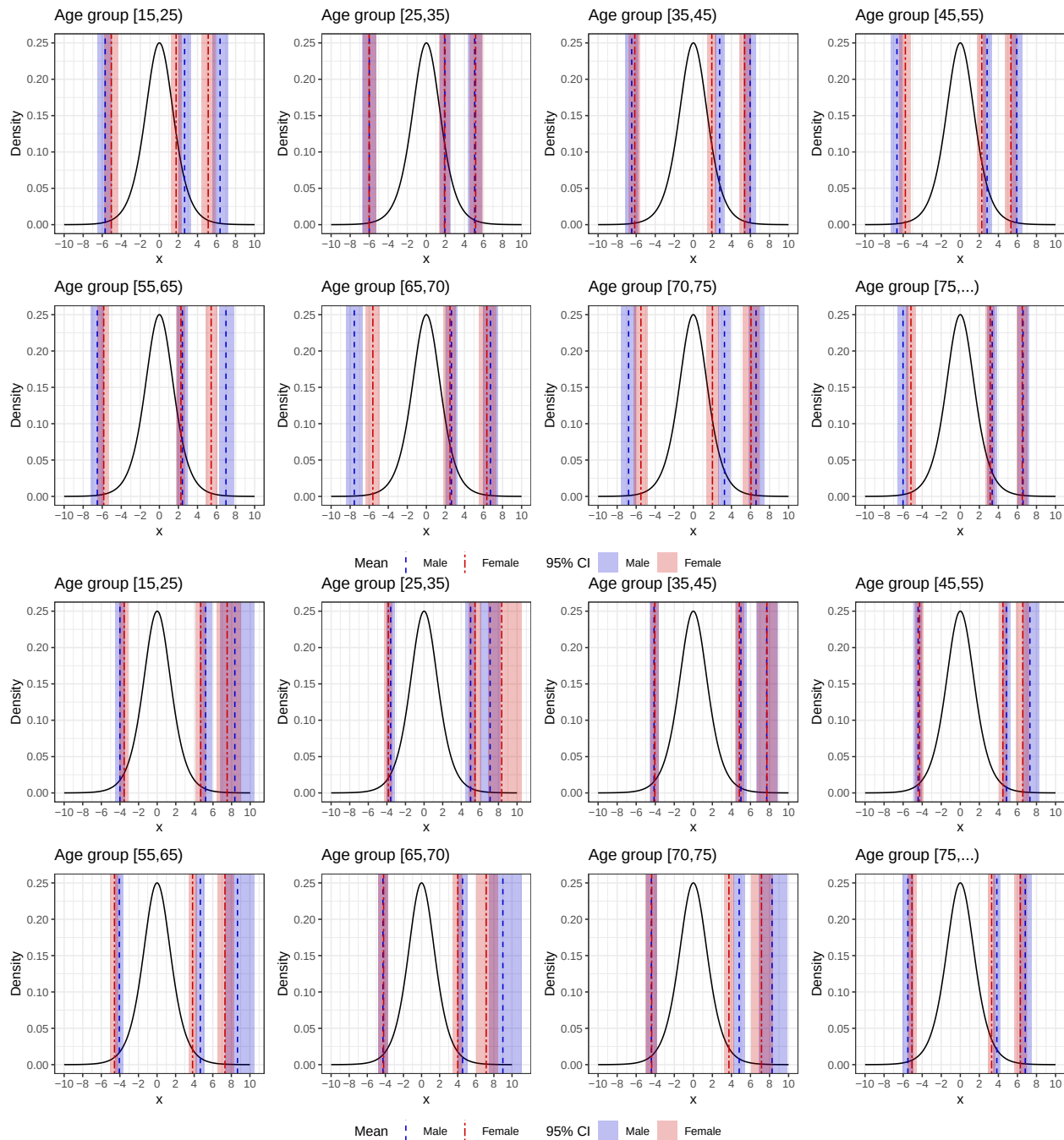













Model assessment

```
### Model-Indep ###

# SurveyMapping.prlevels1:
# - computes the n.sims simulated probabilities for each respondent under Model-Indep

SurveyMapping.prlevels1 <- function(salwinbugs) {
```

```

n.sims <- salwinbugs$n.sims
n.chains <- salwinbugs$n.chains
p.gamma <- array(dim = c(n.sims, NResp, NVars, NCats - 1))
prlevels <- array(dim = c(n.sims, NResp, NVars, NCats))
for (sim in 1:n.sims) {
  for (Resp in 1:NResp) {
    for (Var in 1:NVars) {
      for (Cat in 1:(NCats - 1)) {
        p.gamma[sim, Resp, Var, Cat] <-
          ilogit(salwinbugs$sims.list$kappa[sim, sex[Resp], age[Resp], Cat, Var] +
            salwinbugs$sims.list$sd.theta[sim, Var] * salwinbugs$sims.list$theta[sim,
→ muni[Resp], Var])
      }

      prlevels[sim, Resp, Var, 1] <- p.gamma[sim, Resp, Var, 1]
      prlevels[sim, Resp, Var, NCats] <- 1 - p.gamma[sim, Resp, Var, NCats - 1]

      for (Cat in 2:(NCats - 1)) {
        prlevels[sim, Resp, Var, Cat] <-
          p.gamma[sim, Resp, Var, Cat] - p.gamma[sim, Resp, Var, Cat - 1]
      }
    }
  }

  if (sim %in% c(1, seq(n.sims/n.chains, n.sims, n.sims/n.chains))) {
    cat(sim, "of", n.sims, "simulations", "\n")
  } else {}
}

return(prlevels)
}

prlevels1 <- SurveyMapping.prlevels1(salwinbugs = salwinbugs1)

```

```

## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
## 800 of 1000 simulations
## 1000 of 1000 simulations

```

```

### Model-Corr ###

# SurveyMapping.prlevels2:
# - computes the n.sims simulated probabilities for each respondent under Model-Corr

SurveyMapping.prlevels2 <- function(salwinbugs) {

  n.sims <- salwinbugs$n.sims
  n.chains <- salwinbugs$n.chains
  p.gamma <- array(dim = c(n.sims, NResp, NVars, NCats - 1))
  prlevels <- array(dim = c(n.sims, NResp, NVars, NCats))
  for (sim in 1:n.sims) {
    for (Resp in 1:NResp) {
      for (Var in 1:NVars) {
        for (Cat in 1:(NCats - 1)) {

```

```

    p.gamma[sim, Resp, Var, Cat] <-
      ilogit(salwinbugs$sims.list$kappa[sim, sex[Resp], age[Resp], Cat, Var] +
        salwinbugs$sims.list$theta[sim, muni[Resp], Var])
  }

  prlevels[sim, Resp, Var, 1] <- p.gamma[sim, Resp, Var, 1]
  prlevels[sim, Resp, Var, NCats] <- 1 - p.gamma[sim, Resp, Var, NCats - 1]

  for (Cat in 2:(NCats - 1)) {
    prlevels[sim, Resp, Var, Cat] <-
      p.gamma[sim, Resp, Var, Cat] - p.gamma[sim, Resp, Var, Cat - 1]
  }
}

if (sim %in% c(1, seq(n.sims/n.chains, n.sims, n.sims/n.chains))) {
  cat(sim, "of", n.sims, "simulations", "\n")
} else {}
}

return(prlevels)
}

prlevels2 <- SurveyMapping.prlevels2(salwinbugs = salwinbugs2)

```

```

## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
## 800 of 1000 simulations
## 1000 of 1000 simulations

```

```

### Model-Corr&IRE ###

# SurveyMapping.prlevels3:
# - computes the n.sims simulated probabilities for each respondent under Model-Corr&IRE

SurveyMapping.prlevels3 <- function(salwinbugs) {

  n.sims <- salwinbugs$n.sims
  n.chains <- salwinbugs$n.chains
  p.gamma <- array(dim = c(n.sims, NResp, NVars, NCats - 1))
  prlevels <- array(dim = c(n.sims, NResp, NVars, NCats))
  for (sim in 1:n.sims) {
    for (Resp in 1:NResp) {
      for (Var in 1:NVars) {
        for (Cat in 1:(NCats - 1)) {
          p.gamma[sim, Resp, Var, Cat] <-
            ilogit(salwinbugs$sims.list$kappa[sim, sex[Resp], age[Resp], Cat, Var] +
              salwinbugs$sims.list$theta[sim, muni[Resp], Var] +
              salwinbugs$sims.list$psi[sim, Resp, Var])
        }

        prlevels[sim, Resp, Var, 1] <- p.gamma[sim, Resp, Var, 1]
        prlevels[sim, Resp, Var, NCats] <- 1 - p.gamma[sim, Resp, Var, NCats - 1]

        for (Cat in 2:(NCats - 1)) {

```



```

      prlevels[sim, Resp, Var, Cat] <-
        p.gamma[sim, Resp, Var, Cat] - p.gamma[sim, Resp, Var, Cat - 1]
    }
  }
}

if (sim %in% c(1, seq(n.sims/n.chains, n.sims, n.sims/n.chains))) {
  cat(sim, "of", n.sims, "simulations", "\n")
} else {}
}

return(prlevels)
}

prlevels3 <- SurveyMapping.prlevels3(salwinbugs = salwinbugs3)

```

```

## 1 of 1000 simulations
## 200 of 1000 simulations
## 400 of 1000 simulations
## 600 of 1000 simulations
## 800 of 1000 simulations
## 1000 of 1000 simulations

```

```

### Validation ###

# Sample size by municipality, sex, and age group
sample <- array(dim = c(NMuni, NSex, NAges))
for (Muni in 1:NMuni) {
  for (SexGroup in 1:NSex) {
    for (AgeGroup in 1:NAges) {
      sample[Muni, SexGroup, AgeGroup] <- sum(muni == Muni & sex == SexGroup & age == AgeGroup)
    }
  }
}

SurveyMapping.Validation <- function(prlevels, Muni) {

  NSamp <- sum(sample[Muni, , ])
  index <- which(muni == Muni)
  realvalue <- matrix(nrow = NVars, ncol = NCats)
  predictive <- array(dim = c(n.sims, NSamp, NVars))
  predictive.muni <- array(dim = c(n.sims, NVars, NCats))
  for (sim in 1:n.sims) {
    for (Var in 1:NVars) {
      for (Resp in 1:NSamp) {
        predictive[sim, Resp, Var] <- which(
          rmultinom(n = 1,
                    size = 1,
                    prob = prlevels[sim, index[Resp], Var, ]) == 1)
      }
      predictive.muni[sim, Var, ] <- table(factor(predictive[sim, , Var], levels =
↪ 1:NCats))/NSamp * 100
    }
  }

  if (sim %in% c(1, seq(n.sims/n.chains, n.sims, n.sims/n.chains))) {
    cat(sim, "of", n.sims, "simulations", "\n")
  } else {}
}

```

```

}

posteriormean <- round(apply(predictive.muni, 2:3, mean), 2)
PInterval0.025 <- round(apply(predictive.muni, 2:3, quantile, prob = 0.025), 2)
PInterval0.975 <- round(apply(predictive.muni, 2:3, quantile, prob = 0.975), 2)
for (Var in 1:NVars) {
  realvalue[Var, ] <- round(table(factor(y[muni == Muni, Var], levels = 1:NCats))/NSamp * 100,
↪ 2)
}
return(list("mean" = posteriormean,
           "PI" = list("lower" = PInterval0.025, "upper" = PInterval0.975),
           "real" = realvalue))
return(predictive.muni)
}

# Six municipalities with the largest population in the Region of Valencia
Munis <- order(apply(sample, 1, sum), decreasing = TRUE)[1:6]

# validation1 <- list()
# for (Muni in 1:length(Munis)) {
#   set.seed(9747783)
#   validation1[[Muni]] <- SurveyMapping.Validation(prlevels = prlevels1, Muni = Munis[Muni])
# }

# saveRDS(validation1, file = file.path("../", "results",
↪ "multi-2022-nimble-MH-indep-assessment.rds"))

# validation2 <- list()
# for (Muni in 1:length(Munis)) {
#   set.seed(9747783)
#   validation2[[Muni]] <- SurveyMapping.Validation(prlevels = prlevels2, Muni = Munis[Muni])
# }

# saveRDS(validation2, file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-assessment.rds"))

# validation3 <- list()
# for (Muni in 1:length(Munis)) {
#   set.seed(9747783)
#   validation3[[Muni]] <- SurveyMapping.Validation(prlevels = prlevels3, Muni = Munis[Muni])
# }

# saveRDS(validation3, file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-ire-assessment.rds"))

validation1 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-indep-assessment.rds"))
validation2 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-assessment.rds"))
validation3 <- readRDS(file = file.path("../", "results",
↪ "multi-2022-nimble-MH-corr-ire-assessment.rds"))

### Functions to construct the model assessment tables ###

# Municipalities in the order used in the model assessment objects
table_municipality <- as.character(carto_muni$NOMBRE_MUNI[Munis])
table_municipality <- c("Valencia", "Elche", "Alicante",
                       "Castellón de la Plana", "Orihuela", "Torrevieja")

```

```

# Function to format posterior mean and 95% prediction interval
format_prediction <- function(validation, Muni, Var) {

  mean <- validation[[Muni]]$mean[Var, ]
  lower <- validation[[Muni]]$PI$lower[Var, ]
  upper <- validation[[Muni]]$PI$upper[Var, ]

  sprintf("%.2f (%.2f-%.2f)", mean, lower, upper)
}

# Function to construct a model assessment table for a given item
construct_assessment_table <- function(Var) {

  TableVar <- do.call(rbind, lapply(seq_along(table_municipality), function(Muni) {

    indep <- format_prediction(validation1, Muni, Var)
    corr <- format_prediction(validation2, Muni, Var)
    corrire <- format_prediction(validation3, Muni, Var)
    observed <- sprintf("%.2f", validation3[[Muni]]$real[Var, ])

    data.frame(Municipality = rep(table_municipality[Muni], 4),
               Model = c("Indep", "Corr", "Corr&IRE", "Observed"),
               "1" = c(indep[1], corr[1], corrire[1], observed[1]),
               "2" = c(indep[2], corr[2], corrire[2], observed[2]),
               "3" = c(indep[3], corr[3], corrire[3], observed[3]),
               "4" = c(indep[4], corr[4], corrire[4], observed[4]),
               check.names = FALSE)

  })
  )

  rownames(TableVar) <- NULL
  TableVar
}

```

Table 2: Model assessment

```

Table2 <- construct_assessment_table(Var = 5)

old_width <- getOption("width")
options(width = 200)
print(Table2, row.names = FALSE, right = FALSE)
options(width = old_width)

```

## Municipality	Model	1	2	3	4
## Valencia	Indep	17.81 (15.18-20.55)	56.02 (52.91-59.27)	20.73 (18.00-23.73)	5.43 (4.00-7.00)
## Valencia	Corr	17.18 (14.45-19.91)	56.58 (53.45-59.82)	20.91 (18.09-23.64)	5.32 (4.00-6.82)
## Valencia	Corr&IRE	17.41 (15.73-19.09)	57.31 (54.82-59.64)	20.50 (18.45-22.55)	4.78 (3.91-5.73)
## Valencia	Observed	16.91	58.55	20.18	4.27
## Elche	Indep	15.44 (12.03-18.81)	54.87 (50.34-59.15)	23.25 (19.49-27.29)	6.43 (4.41-8.81)
## Elche	Corr	15.36 (12.37-18.64)	55.48 (51.36-59.83)	23.02 (19.15-26.61)	6.14 (4.24-8.31)
## Elche	Corr&IRE	16.66 (14.41-18.98)	53.71 (50.68-56.95)	21.28 (18.47-24.07)	8.36 (6.95-9.83)
## Elche	Observed	16.95	53.39	20.51	9.15
## Alicante	Indep	14.83 (11.55-18.28)	54.63 (50.51-58.79)	23.85 (20.00-28.11)	6.68 (4.48-8.97)
## Alicante	Corr	14.57 (11.55-18.10)	55.15 (50.86-59.48)	23.84 (19.82-27.76)	6.44 (4.48-8.62)
## Alicante	Corr&IRE	16.62 (14.48-18.79)	51.35 (48.28-54.31)	23.47 (20.69-26.03)	8.57 (7.07-10.17)
## Alicante	Observed	17.93	50.00	22.07	9.48
## Castellón de la Plana	Indep	21.55 (15.77-28.08)	56.64 (50.77-62.69)	17.48 (12.31-23.09)	4.33 (1.92-7.31)
## Castellón de la Plana	Corr	21.38 (15.77-27.31)	57.23 (51.54-63.46)	17.29 (12.31-22.69)	4.10 (1.92-6.92)

```
## Castellón de la Plana Corr&IRE 20.02 (16.15-23.85) 60.72 (55.77-65.38) 16.68 (13.08-20.77) 2.58 (1.54-4.23)
## Castellón de la Plana Observed 20.00 61.54 16.15 2.31
## Orihuela Indep 15.88 (10.42-21.89) 55.09 (48.44-61.98) 22.75 (16.15-29.69) 6.28 (3.12-10.42)
## Orihuela Corr 17.37 (11.98-23.96) 56.96 (50.00-63.55) 20.50 (14.57-26.56) 5.17 (2.08-8.85)
## Orihuela Corr&IRE 19.66 (16.15-23.44) 51.82 (46.88-56.77) 23.82 (19.27-28.12) 4.70 (2.60-7.29)
## Orihuela Observed 18.23 52.08 26.56 2.60
## Torrevieja Indep 15.30 (9.78-21.21) 54.48 (47.28-61.96) 23.56 (16.85-30.45) 6.65 (3.25-10.87)
## Torrevieja Corr 15.49 (10.31-21.74) 55.40 (47.83-63.04) 23.03 (16.29-30.43) 6.09 (2.72-10.33)
## Torrevieja Corr&IRE 19.88 (16.30-23.37) 48.25 (43.48-53.80) 21.89 (17.39-26.63) 9.98 (7.61-12.50)
## Torrevieja Observed 19.02 49.46 21.20 10.33
```

Tables S1-S12: Model assessment

```
old_width <- getOption("width")
options(width = 200)

for (Var in 1:NVars) {

  TableVar <- construct_assessment_table(Var)

  cat("\nTable S", Var, " - Item ", Var, "\n", sep = "")
  print(TableVar, row.names = FALSE, right = FALSE)
}

options(width = old_width)
```

```
##
## Table S1 - Item 1
## Municipality Model 1 2 3 4
## Valencia Indep 4.79 (3.36-6.36) 83.51 (81.09-85.91) 9.54 (7.64-11.64) 2.17 (1.36-3.18)
## Valencia Corr 4.51 (3.27-5.91) 83.74 (81.45-86.00) 9.61 (7.82-11.64) 2.14 (1.27-3.09)
## Valencia Corr&IRE 4.53 (3.36-5.73) 83.50 (81.18-85.55) 9.95 (8.18-11.91) 2.02 (1.27-2.91)
## Valencia Observed 4.36 84.55 9.18 1.91
## Elche Indep 4.05 (2.37-5.93) 82.26 (78.81-85.76) 11.12 (8.14-14.41) 2.57 (1.35-4.07)
## Elche Corr 3.85 (2.37-5.76) 82.45 (78.98-85.76) 11.22 (8.31-14.24) 2.49 (1.19-3.90)
## Elche Corr&IRE 4.23 (2.71-5.76) 81.57 (78.64-84.41) 11.20 (8.64-13.90) 3.00 (1.86-4.41)
## Elche Observed 3.90 81.02 12.20 2.88
## Alicante Indep 5.60 (3.62-7.76) 84.13 (80.69-87.07) 8.36 (5.86-10.86) 1.91 (0.86-3.28)
## Alicante Corr 5.97 (3.97-8.45) 85.00 (81.90-87.94) 7.42 (5.17-9.83) 1.60 (0.69-2.76)
## Alicante Corr&IRE 7.69 (5.69-9.66) 81.42 (78.45-84.31) 8.95 (6.55-11.38) 1.94 (1.03-2.93)
## Alicante Observed 10.00 76.38 10.17 2.93
## Castellón de la Plana Indep 5.27 (2.68-8.46) 84.03 (78.85-88.47) 8.75 (5.00-13.08) 1.96 (0.38-3.85)
## Castellón de la Plana Corr 5.13 (2.69-8.09) 84.26 (79.62-88.85) 8.73 (4.99-12.70) 1.88 (0.38-3.47)
## Castellón de la Plana Corr&IRE 4.22 (1.92-6.92) 86.26 (81.92-90.00) 8.09 (4.99-11.54) 1.43 (0.38-2.70)
## Castellón de la Plana Observed 5.00 85.77 8.85 0.38
## Orihuela Indep 5.55 (2.60-8.85) 84.29 (79.17-89.06) 8.29 (4.17-12.50) 1.87 (0.00-4.17)
## Orihuela Corr 6.64 (3.12-10.42) 85.40 (80.21-90.10) 6.61 (3.12-10.42) 1.35 (0.00-3.12)
## Orihuela Corr&IRE 7.82 (4.69-11.46) 82.41 (77.60-86.98) 8.18 (4.67-11.98) 1.59 (0.52-3.12)
## Orihuela Observed 7.81 83.85 6.77 1.04
## Torrevieja Indep 5.86 (2.72-9.78) 83.97 (78.26-89.13) 8.34 (4.35-12.50) 1.83 (0.00-3.80)
## Torrevieja Corr 6.33 (2.72-10.33) 84.76 (79.35-90.22) 7.30 (3.26-11.96) 1.60 (0.00-3.80)
## Torrevieja Corr&IRE 7.46 (4.35-10.87) 82.21 (77.17-86.97) 8.70 (4.89-13.04) 1.62 (0.54-3.26)
## Torrevieja Observed 8.15 82.07 8.70 1.09
##
## Table S2 - Item 2
## Municipality Model 1 2 3 4
## Valencia Indep 16.93 (14.36-19.73) 66.47 (63.45-69.45) 11.43 (9.36-13.64) 5.17 (3.82-6.73)
## Valencia Corr 17.18 (14.36-19.91) 66.92 (64.00-69.82) 11.05 (9.18-13.18) 4.85 (3.54-6.27)
## Valencia Corr&IRE 17.02 (14.91-19.18) 66.46 (63.55-69.09) 11.73 (9.82-13.82) 4.79 (3.73-6.09)
## Valencia Observed 16.27 68.00 11.27 4.45
## Elche Indep 17.24 (13.73-20.85) 66.28 (62.20-69.83) 11.34 (8.64-14.41) 5.14 (3.39-7.12)
## Elche Corr 16.71 (13.22-20.00) 66.87 (63.05-70.85) 11.35 (8.64-14.41) 5.08 (3.22-7.12)
## Elche Corr&IRE 17.51 (14.58-21.02) 63.87 (60.00-67.63) 12.26 (9.49-14.92) 6.36 (4.75-8.14)
## Elche Observed 21.02 59.49 12.20 7.29
## Alicante Indep 15.85 (12.59-19.31) 66.51 (62.41-70.69) 12.11 (9.14-15.34) 5.53 (3.62-7.76)
## Alicante Corr 15.89 (12.41-19.48) 66.75 (62.76-70.86) 11.92 (8.97-15.00) 5.44 (3.62-7.59)
## Alicante Corr&IRE 17.06 (14.14-20.00) 63.01 (59.66-66.56) 13.22 (10.69-15.87) 6.70 (4.83-8.62)
## Alicante Observed 19.66 60.52 11.38 7.93
## Castellón de la Plana Indep 24.30 (18.08-31.15) 64.55 (58.46-70.38) 7.79 (4.62-11.92) 3.36 (1.53-6.15)
```

```

## Castellón de la Plana Corr      23.85 (18.45-30.00) 64.95 (59.23-70.77) 7.90 (4.23-11.54) 3.30 (1.15-5.77)
## Castellón de la Plana Corr&IRE 23.93 (18.85-29.23) 66.52 (61.15-71.92) 7.27 (4.23-10.38) 2.28 (0.77-3.86)
## Castellón de la Plana Observed 22.31 73.08 3.46 1.15
## Orihuela Indep 16.58 (11.46-22.40) 66.53 (59.90-72.92) 11.64 (6.77-16.67) 5.26 (2.08-8.35)
## Orihuela Corr 17.77 (12.50-23.96) 66.94 (60.42-73.44) 10.60 (6.25-15.10) 4.68 (2.07-8.33)
## Orihuela Corr&IRE 18.51 (13.54-23.44) 64.41 (57.81-70.31) 12.37 (8.33-17.71) 4.71 (2.08-7.81)
## Orihuela Observed 16.15 68.75 11.46 3.12
## Torrevieja Indep 16.80 (10.87-23.37) 66.48 (59.77-73.37) 11.48 (6.52-16.85) 5.24 (2.17-9.24)
## Torrevieja Corr 16.93 (10.87-23.91) 66.83 (60.33-73.37) 11.27 (6.52-16.85) 4.97 (2.16-8.70)
## Torrevieja Corr&IRE 19.26 (14.13-25.00) 60.17 (53.80-66.30) 13.00 (8.70-17.93) 7.58 (4.35-10.87)
## Torrevieja Observed 21.74 57.07 13.04 8.15
##
## Table S3 - Item 3
## Municipality Model 1 2 3 4
## Valencia Indep 7.07 (5.36-9.18) 85.42 (82.91-87.73) 5.72 (4.36-7.36) 1.79 (1.00-2.64)
## Valencia Corr 7.16 (5.45-9.00) 85.88 (83.73-87.91) 5.32 (4.00-6.82) 1.63 (0.82-2.45)
## Valencia Corr&IRE 7.21 (5.91-8.64) 85.56 (83.64-87.18) 5.50 (4.18-6.91) 1.73 (1.09-2.45)
## Valencia Observed 6.73 85.82 5.82 1.45
## Elche Indep 9.56 (6.78-12.54) 84.91 (82.03-87.97) 4.25 (2.54-6.10) 1.29 (0.34-2.37)
## Elche Corr 8.30 (5.76-11.19) 85.64 (82.54-88.64) 4.66 (2.88-6.61) 1.40 (0.51-2.54)
## Elche Corr&IRE 9.90 (7.63-12.37) 83.26 (80.51-85.93) 4.94 (3.39-6.78) 1.90 (1.02-2.88)
## Elche Observed 11.02 82.54 4.41 2.03
## Alicante Indep 11.82 (8.62-15.17) 83.80 (80.69-87.24) 3.38 (1.90-5.00) 1.00 (0.34-2.07)
## Alicante Corr 12.18 (9.31-15.17) 83.81 (80.86-86.73) 3.13 (1.72-4.66) 0.89 (0.17-1.72)
## Alicante Corr&IRE 13.59 (11.55-15.86) 80.79 (78.28-83.28) 4.32 (2.93-6.03) 1.31 (0.52-2.07)
## Alicante Observed 13.10 81.38 3.97 1.03
## Castellón de la Plana Indep 6.94 (3.46-11.15) 85.32 (80.38-89.62) 5.94 (3.08-9.62) 1.81 (0.38-3.47)
## Castellón de la Plana Corr 7.17 (3.85-10.38) 85.86 (81.54-90.00) 5.35 (2.69-8.85) 1.63 (0.38-3.46)
## Castellón de la Plana Corr&IRE 6.06 (3.46-9.23) 87.73 (84.23-91.15) 5.14 (2.69-7.69) 1.08 (0.00-2.31)
## Castellón de la Plana Observed 6.15 85.77 6.54 1.15
## Orihuela Indep 9.15 (4.69-14.08) 84.92 (79.17-90.10) 4.57 (1.56-7.81) 1.35 (0.00-3.12)
## Orihuela Corr 10.79 (6.25-16.15) 84.63 (79.17-89.58) 3.54 (1.04-6.25) 1.03 (0.00-2.60)
## Orihuela Corr&IRE 11.29 (7.81-15.10) 82.79 (78.12-86.98) 4.61 (2.08-7.29) 1.31 (0.00-2.60)
## Orihuela Observed 9.90 84.38 4.17 1.04
## Torrevieja Indep 9.48 (4.89-14.67) 84.69 (78.80-89.67) 4.44 (1.63-7.61) 1.40 (0.00-3.80)
## Torrevieja Corr 10.38 (5.98-15.76) 84.73 (79.35-90.22) 3.75 (1.09-7.07) 1.14 (0.00-2.72)
## Torrevieja Corr&IRE 10.97 (7.61-14.67) 83.64 (79.35-87.50) 4.31 (1.63-7.07) 1.09 (0.00-2.72)
## Torrevieja Observed 10.33 84.78 3.80 1.09
##
## Table S4 - Item 4
## Municipality Model 1 2 3 4
## Valencia Indep 5.65 (4.09-7.36) 87.66 (85.73-89.55) 5.01 (3.55-6.55) 1.69 (0.91-2.55)
## Valencia Corr 5.46 (4.00-7.00) 88.12 (86.09-90.00) 4.80 (3.55-6.27) 1.61 (0.91-2.45)
## Valencia Corr&IRE 5.65 (4.64-6.82) 87.92 (86.27-89.55) 4.86 (3.73-6.09) 1.56 (1.00-2.18)
## Valencia Observed 5.73 87.91 5.18 1.18
## Elche Indep 5.72 (3.56-8.14) 87.68 (85.08-90.17) 4.93 (3.05-7.12) 1.67 (0.68-2.88)
## Elche Corr 5.35 (3.39-7.46) 88.11 (85.42-90.85) 4.91 (3.22-6.78) 1.63 (0.68-2.71)
## Elche Corr&IRE 6.27 (4.58-7.97) 86.37 (84.07-88.48) 5.10 (3.56-6.78) 2.26 (1.36-3.05)
## Elche Observed 6.44 86.10 4.75 2.71
## Alicante Indep 9.19 (6.38-12.24) 86.72 (83.62-89.66) 3.10 (1.72-4.66) 0.99 (0.17-1.90)
## Alicante Corr 9.10 (6.38-11.90) 87.07 (83.97-89.83) 2.91 (1.55-4.48) 0.92 (0.17-1.90)
## Alicante Corr&IRE 10.95 (9.31-12.93) 83.88 (81.55-86.21) 3.79 (2.41-5.35) 1.38 (0.69-2.07)
## Alicante Observed 10.52 84.31 3.10 1.55
## Castellón de la Plana Indep 6.35 (3.08-10.00) 87.69 (83.46-91.54) 4.49 (2.31-7.31) 1.47 (0.00-3.08)
## Castellón de la Plana Corr 5.98 (3.08-9.62) 88.12 (83.85-91.92) 4.46 (1.92-7.31) 1.43 (0.38-3.08)
## Castellón de la Plana Corr&IRE 5.10 (3.08-7.69) 90.23 (87.30-93.08) 3.85 (1.92-6.15) 0.82 (0.00-1.92)
## Castellón de la Plana Observed 6.92 87.31 5.38 0.38
## Orihuela Indep 7.36 (3.65-11.98) 87.52 (82.29-92.19) 3.85 (1.04-6.77) 1.28 (0.00-3.12)
## Orihuela Corr 8.89 (4.69-13.54) 87.19 (82.29-91.67) 2.98 (1.04-5.73) 0.94 (0.00-2.60)
## Orihuela Corr&IRE 10.00 (7.29-13.02) 84.54 (80.73-88.02) 4.01 (1.56-6.77) 1.44 (0.52-2.60)
## Orihuela Observed 9.38 84.90 3.65 1.56
## Torrevieja Indep 8.56 (3.80-13.59) 86.75 (81.52-91.85) 3.50 (1.09-7.07) 1.19 (0.00-2.72)
## Torrevieja Corr 8.44 (4.35-14.13) 87.27 (82.07-91.85) 3.23 (0.54-5.98) 1.06 (0.00-2.72)
## Torrevieja Corr&IRE 9.45 (7.07-12.50) 85.79 (81.52-89.13) 3.80 (1.63-6.52) 0.96 (0.00-2.17)
## Torrevieja Observed 9.24 87.50 2.72 0.54
##
## Table S5 - Item 5
## Municipality Model 1 2 3 4
## Valencia Indep 17.81 (15.18-20.55) 56.02 (52.91-59.27) 20.73 (18.00-23.73) 5.43 (4.00-7.00)
## Valencia Corr 17.18 (14.45-19.91) 56.58 (53.45-59.82) 20.91 (18.09-23.64) 5.32 (4.00-6.82)
## Valencia Corr&IRE 17.41 (15.73-19.09) 57.31 (54.82-59.64) 20.50 (18.45-22.55) 4.78 (3.91-5.73)
## Valencia Observed 16.91 58.55 20.18 4.27
## Elche Indep 15.44 (12.03-18.81) 54.87 (50.34-59.15) 23.25 (19.49-27.29) 6.43 (4.41-8.81)
## Elche Corr 15.36 (12.37-18.64) 55.48 (51.36-59.83) 23.02 (19.15-26.61) 6.14 (4.24-8.31)
## Elche Corr&IRE 16.66 (14.41-18.98) 53.71 (50.68-56.95) 21.28 (18.47-24.07) 8.36 (6.95-9.83)
## Elche Observed 16.95 53.39 20.51 9.15
## Alicante Indep 14.83 (11.55-18.28) 54.63 (50.51-58.79) 23.85 (20.00-28.11) 6.68 (4.48-8.97)
## Alicante Corr 14.57 (11.55-18.10) 55.15 (50.86-59.48) 23.84 (19.82-27.76) 6.44 (4.48-8.62)
## Alicante Corr&IRE 16.62 (14.48-18.79) 51.35 (48.28-54.31) 23.47 (20.69-26.03) 8.57 (7.07-10.17)
## Alicante Observed 17.93 50.00 22.07 9.48
## Castellón de la Plana Indep 21.55 (15.77-28.08) 56.64 (50.77-62.69) 17.48 (12.31-23.09) 4.33 (1.92-7.31)

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## Castellón de la Plana Corr      21.38 (15.77-27.31) 57.23 (51.54-63.46) 17.29 (12.31-22.69) 4.10 (1.92-6.92)
## Castellón de la Plana Corr&IRE 20.02 (16.15-23.85) 60.72 (55.77-65.38) 16.68 (13.08-20.77) 2.58 (1.54-4.23)
## Castellón de la Plana Observed 20.00      61.54      16.15      2.31
## Orihuela Indep      15.88 (10.42-21.89) 55.09 (48.44-61.98) 22.75 (16.15-29.69) 6.28 (3.12-10.42)
## Orihuela Corr      17.37 (11.98-23.96) 56.96 (50.00-63.55) 20.50 (14.57-26.56) 5.17 (2.08-8.85)
## Orihuela Corr&IRE 19.66 (16.15-23.44) 51.82 (46.88-56.77) 23.82 (19.27-28.12) 4.70 (2.60-7.29)
## Orihuela Observed 18.23      52.08      26.56      2.60
## Torrevieja Indep      15.30 (9.78-21.21) 54.48 (47.28-61.96) 23.56 (16.85-30.45) 6.65 (3.25-10.87)
## Torrevieja Corr      15.49 (10.31-21.74) 55.40 (47.83-63.04) 23.03 (16.29-30.43) 6.09 (2.72-10.33)
## Torrevieja Corr&IRE 19.88 (16.30-23.37) 48.25 (43.48-53.80) 21.89 (17.39-26.63) 9.98 (7.61-12.50)
## Torrevieja Observed 19.02      49.46      21.20      10.33
##
## Table S6 - Item 6
## Municipality Model      1      2      3      4
## Valencia Indep      12.18 (9.82-14.55) 59.73 (56.36-62.82) 22.41 (19.55-25.45) 5.68 (4.27-7.27)
## Valencia Corr      12.15 (9.91-14.27) 60.47 (57.27-63.55) 22.03 (19.36-24.82) 5.35 (3.91-6.82)
## Valencia Corr&IRE 11.99 (10.45-13.55) 61.09 (58.55-63.45) 21.80 (19.55-24.00) 5.12 (4.09-6.09)
## Valencia Observed 10.82      62.36      22.27      4.55
## Elche Indep      11.49 (8.64-14.75) 58.79 (54.24-63.22) 23.57 (19.66-27.97) 6.14 (4.07-8.47)
## Elche Corr      10.78 (8.14-13.73) 58.99 (54.74-63.22) 24.13 (20.00-28.31) 6.10 (4.07-8.31)
## Elche Corr&IRE 12.30 (10.17-14.41) 56.58 (53.39-59.66) 22.85 (19.83-25.93) 8.26 (6.78-9.83)
## Elche Observed 15.59      52.71      21.19      10.34
## Alicante Indep      11.02 (8.27-14.14) 58.67 (53.79-63.28) 24.07 (19.83-28.45) 6.25 (4.14-8.45)
## Alicante Corr      10.32 (7.93-13.45) 58.72 (54.31-63.10) 24.65 (20.52-28.80) 6.31 (4.31-8.62)
## Alicante Corr&IRE 12.20 (10.00-14.48) 56.02 (52.93-59.66) 23.78 (21.03-26.90) 8.00 (6.55-9.66)
## Alicante Observed 13.79      55.00      21.03      9.66
## Castellón de la Plana Indep      14.87 (10.00-20.38) 61.38 (55.38-67.69) 19.14 (13.46-25.00) 4.61 (2.31-7.69)
## Castellón de la Plana Corr      15.74 (10.77-21.15) 62.55 (56.53-68.85) 17.74 (12.69-23.46) 3.96 (1.54-6.92)
## Castellón de la Plana Corr&IRE 13.61 (10.00-17.31) 66.41 (61.15-71.16) 17.14 (13.08-21.54) 2.84 (1.54-4.62)
## Castellón de la Plana Observed 11.15      68.46      18.08      1.92
## Orihuela Indep      12.72 (7.29-18.23) 60.05 (53.12-66.67) 21.82 (15.09-28.14) 5.41 (2.08-9.38)
## Orihuela Corr      13.70 (8.33-19.27) 61.31 (54.69-68.75) 20.24 (14.06-26.56) 4.75 (2.08-8.33)
## Orihuela Corr&IRE 16.63 (13.02-20.31) 55.59 (49.99-61.46) 22.94 (18.23-27.60) 4.84 (2.60-7.29)
## Orihuela Observed 17.19      54.69      23.96      3.65
## Torrevieja Indep      12.51 (7.07-18.48) 59.63 (52.72-66.85) 22.24 (15.22-29.90) 5.62 (2.17-9.78)
## Torrevieja Corr      11.77 (7.07-17.39) 59.68 (52.17-66.85) 22.96 (16.83-29.89) 5.60 (2.17-9.24)
## Torrevieja Corr&IRE 16.75 (13.04-20.65) 52.20 (46.73-57.61) 21.38 (16.30-26.63) 9.68 (7.07-12.50)
## Torrevieja Observed 17.93      51.63      21.20      9.24
##
## Table S7 - Item 7
## Municipality Model      1      2      3      4
## Valencia Indep      6.22 (4.64-8.00) 83.61 (81.27-85.91) 8.23 (6.36-10.18) 1.93 (1.09-2.82)
## Valencia Corr      5.84 (4.36-7.46) 84.08 (81.82-86.45) 8.19 (6.36-10.09) 1.88 (1.09-2.73)
## Valencia Corr&IRE 5.96 (5.00-7.09) 84.29 (82.36-86.18) 7.97 (6.55-9.45) 1.78 (1.09-2.45)
## Valencia Observed 5.45      85.55      7.82      1.18
## Elche Indep      5.39 (3.39-7.63) 83.20 (80.00-86.44) 9.24 (6.61-12.03) 2.17 (1.02-3.56)
## Elche Corr      5.02 (3.22-6.95) 83.56 (80.17-86.61) 9.26 (6.78-12.20) 2.17 (1.02-3.39)
## Elche Corr&IRE 5.75 (4.24-7.29) 82.14 (79.66-84.58) 9.37 (7.29-11.53) 2.73 (1.86-3.90)
## Elche Observed 6.27      81.19      9.83      2.54
## Alicante Indep      9.05 (6.55-11.90) 83.89 (80.69-86.90) 5.76 (3.79-8.10) 1.30 (0.52-2.41)
## Alicante Corr      8.90 (6.38-11.72) 84.46 (81.38-87.41) 5.47 (3.62-7.59) 1.17 (0.34-2.24)
## Alicante Corr&IRE 10.73 (8.97-12.59) 80.82 (78.45-83.28) 6.84 (5.00-8.62) 1.61 (0.86-2.42)
## Alicante Observed 10.86      80.86      6.21      1.55
## Castellón de la Plana Indep      6.23 (3.08-9.62) 83.63 (78.85-88.08) 8.20 (4.62-12.31) 1.94 (0.38-3.86)
## Castellón de la Plana Corr      6.03 (3.08-9.23) 84.24 (79.23-88.46) 7.93 (4.23-11.92) 1.80 (0.38-3.85)
## Castellón de la Plana Corr&IRE 4.69 (2.31-7.31) 87.03 (83.46-90.38) 7.12 (4.23-10.00) 1.17 (0.38-2.31)
## Castellón de la Plana Observed 4.23      87.69      7.31      0.77
## Orihuela Indep      7.91 (4.17-12.50) 84.02 (78.65-89.06) 6.61 (3.12-10.94) 1.46 (0.00-3.65)
## Orihuela Corr      9.10 (5.21-14.06) 84.40 (78.65-89.58) 5.35 (2.59-9.38) 1.14 (0.00-2.60)
## Orihuela Corr&IRE 10.76 (7.81-14.06) 81.08 (76.56-85.42) 6.68 (3.65-9.90) 1.49 (0.52-3.12)
## Orihuela Observed 11.46      81.25      5.73      1.04
## Torrevieja Indep      6.41 (2.72-10.87) 83.44 (77.72-88.59) 8.17 (3.80-13.04) 1.98 (0.00-4.35)
## Torrevieja Corr      7.77 (3.80-12.50) 84.31 (78.79-89.67) 6.47 (2.72-10.34) 1.46 (0.00-3.26)
## Torrevieja Corr&IRE 8.81 (6.51-11.41) 81.15 (76.63-85.33) 8.32 (4.89-11.96) 1.72 (0.54-3.27)
## Torrevieja Observed 8.15      80.98      7.61      3.26
##
## Table S8 - Item 8
## Municipality Model      1      2      3      4
## Valencia Indep      4.95 (3.45-6.55) 86.26 (84.09-88.27) 6.70 (5.18-8.37) 2.09 (1.18-3.09)
## Valencia Corr      4.73 (3.27-6.18) 86.82 (84.82-89.00) 6.45 (5.00-8.09) 2.00 (1.18-3.00)
## Valencia Corr&IRE 4.81 (3.91-5.91) 86.84 (85.27-88.45) 6.41 (5.09-7.73) 1.94 (1.36-2.64)
## Valencia Observed 4.82      86.73      6.73      1.73
## Elche Indep      4.32 (2.54-6.27) 85.81 (82.88-88.64) 7.47 (5.08-10.00) 2.41 (1.19-3.73)
## Elche Corr      3.93 (2.37-5.76) 86.14 (83.22-88.98) 7.55 (5.42-10.00) 2.38 (1.19-3.73)
## Elche Corr&IRE 4.43 (3.05-5.93) 85.02 (82.71-87.29) 7.54 (5.76-9.49) 3.01 (2.03-4.24)
## Elche Observed 4.24      85.42      6.95      3.22
## Alicante Indep      6.28 (4.14-8.62) 86.77 (83.79-89.66) 5.31 (3.28-7.41) 1.64 (0.69-2.76)
## Alicante Corr      6.74 (4.48-9.14) 87.22 (84.31-89.83) 4.65 (2.93-6.55) 1.39 (0.52-2.59)
## Alicante Corr&IRE 8.55 (7.06-10.17) 83.51 (81.21-85.69) 5.98 (4.31-7.76) 1.95 (1.21-2.76)
## Alicante Observed 8.45      82.76      6.21      2.07
## Castellón de la Plana Indep      5.56 (2.69-8.85) 86.55 (82.31-90.38) 6.01 (3.08-9.23) 1.88 (0.38-3.85)

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## Castellón de la Plana Corr 5.24 (2.31-8.46) 87.16 (82.69-91.15) 5.82 (3.08-9.23) 1.78 (0.38-3.46)
## Castellón de la Plana Corr&IRE 3.86 (1.92-6.15) 90.12 (86.92-93.08) 5.08 (2.69-7.69) 0.94 (0.00-1.93)
## Castellón de la Plana Observed 3.46 91.54 4.62 0.38
## Orihuela Indep 6.53 (3.12-10.94) 86.75 (82.29-91.15) 5.19 (2.08-8.85) 1.52 (0.00-3.65)
## Orihuela Corr 8.08 (4.17-13.02) 86.95 (81.77-91.67) 3.81 (1.04-6.77) 1.16 (0.00-3.12)
## Orihuela Corr&IRE 10.14 (7.29-13.02) 83.30 (79.69-86.98) 4.93 (2.08-7.81) 1.64 (0.52-3.12)
## Orihuela Observed 10.42 82.81 5.21 1.04
## Torrevieja Indep 7.13 (3.26-11.41) 86.50 (80.98-91.85) 4.85 (1.63-8.70) 1.52 (0.00-3.80)
## Torrevieja Corr 7.15 (3.79-11.96) 87.08 (82.60-91.85) 4.40 (1.63-7.61) 1.37 (0.00-3.26)
## Torrevieja Corr&IRE 8.50 (5.98-11.41) 84.42 (80.43-88.04) 5.72 (2.72-8.70) 1.36 (0.00-2.72)
## Torrevieja Observed 8.15 86.96 4.35 0.54
##
## Table S9 - Item 9
## Municipality Model 1 2 3 4
## Valencia Indep 17.02 (14.36-19.82) 51.56 (48.55-54.73) 21.85 (19.09-24.73) 9.57 (7.64-11.73)
## Valencia Corr 16.56 (14.18-19.09) 52.30 (49.09-55.73) 21.94 (19.36-24.73) 9.20 (7.36-11.09)
## Valencia Corr&IRE 15.39 (13.91-16.91) 54.61 (52.36-56.73) 22.11 (20.27-24.00) 7.89 (6.91-9.00)
## Valencia Observed 13.82 56.64 22.45 7.00
## Elche Indep 11.44 (8.81-14.41) 46.31 (41.52-51.36) 27.79 (23.39-32.04) 14.46 (11.02-17.97)
## Elche Corr 10.66 (8.14-13.39) 46.28 (41.53-50.85) 28.53 (24.58-32.54) 14.53 (11.36-17.97)
## Elche Corr&IRE 14.27 (12.37-16.10) 41.85 (39.15-44.58) 24.23 (21.53-26.78) 19.65 (17.46-21.86)
## Elche Observed 16.78 40.51 20.51 22.03
## Alicante Indep 9.26 (6.55-12.07) 43.09 (38.10-48.11) 30.27 (26.03-34.66) 17.38 (13.79-21.38)
## Alicante Corr 9.32 (6.90-11.90) 44.06 (39.66-48.79) 30.30 (26.21-34.31) 16.33 (12.93-19.83)
## Alicante Corr&IRE 11.70 (10.00-13.45) 40.13 (37.59-42.59) 26.90 (24.14-29.66) 21.27 (19.14-23.28)
## Alicante Observed 12.76 40.17 22.41 24.14
## Castellón de la Plana Indep 19.87 (14.23-25.78) 52.78 (46.54-59.23) 19.37 (14.23-25.00) 7.98 (4.62-11.92)
## Castellón de la Plana Corr 20.19 (15.38-25.77) 53.71 (47.31-60.00) 18.75 (14.22-23.85) 7.36 (4.23-10.77)
## Castellón de la Plana Corr&IRE 14.72 (11.54-18.08) 62.34 (57.69-66.54) 19.18 (15.38-22.70) 3.76 (2.31-5.38)
## Castellón de la Plana Observed 14.62 62.31 19.62 3.46
## Orihuela Indep 20.78 (14.06-27.60) 52.84 (45.83-59.38) 18.78 (13.02-25.01) 7.60 (3.65-11.98)
## Orihuela Corr 21.78 (15.10-28.65) 53.65 (46.88-60.42) 17.82 (12.49-23.45) 6.75 (3.65-10.94)
## Orihuela Corr&IRE 25.78 (22.40-29.17) 46.38 (41.67-51.04) 22.14 (18.23-26.04) 5.70 (3.65-7.81)
## Orihuela Observed 23.96 47.92 23.96 3.65
## Torrevieja Indep 17.55 (11.41-25.00) 51.28 (44.02-58.15) 21.64 (14.67-28.80) 9.53 (4.89-14.67)
## Torrevieja Corr 15.98 (10.33-22.28) 51.53 (43.48-58.70) 22.72 (16.30-29.89) 9.77 (5.43-14.67)
## Torrevieja Corr&IRE 21.27 (17.93-24.46) 44.18 (39.67-48.91) 21.34 (17.39-25.01) 13.21 (10.87-15.76)
## Torrevieja Observed 20.65 45.65 21.74 11.96
##
## Table S10 - Item 10
## Municipality Model 1 2 3 4
## Valencia Indep 15.45 (12.82-18.09) 53.71 (50.45-56.73) 21.53 (18.82-24.37) 9.32 (7.36-11.27)
## Valencia Corr 14.68 (12.54-17.09) 54.05 (50.91-57.27) 21.95 (19.27-24.73) 9.32 (7.45-11.27)
## Valencia Corr&IRE 13.32 (12.09-14.55) 56.76 (54.82-58.64) 22.12 (20.45-23.73) 7.80 (6.91-8.73)
## Valencia Observed 12.36 58.82 21.82 7.00
## Elche Indep 9.31 (6.95-11.87) 46.02 (41.19-50.68) 28.83 (24.75-33.05) 15.85 (12.54-19.66)
## Elche Corr 8.60 (6.10-11.02) 45.78 (40.85-50.34) 29.76 (25.59-34.07) 15.86 (12.54-19.49)
## Elche Corr&IRE 12.58 (11.19-14.24) 42.57 (40.17-44.92) 23.11 (20.84-25.76) 21.74 (20.00-23.39)
## Elche Observed 13.90 41.86 20.51 23.56
## Alicante Indep 7.84 (5.51-10.52) 43.49 (38.45-48.28) 30.65 (26.90-34.66) 18.01 (14.31-21.90)
## Alicante Corr 7.65 (5.34-10.17) 43.81 (39.31-48.45) 30.99 (26.90-35.00) 17.54 (13.97-21.21)
## Alicante Corr&IRE 10.35 (8.97-11.72) 41.21 (38.96-43.45) 25.14 (22.59-27.93) 23.30 (21.38-25.17)
## Alicante Observed 11.03 41.21 22.76 24.48
## Castellón de la Plana Indep 18.64 (13.08-24.24) 55.14 (48.85-61.15) 18.58 (13.46-24.23) 7.63 (4.23-11.15)
## Castellón de la Plana Corr 18.18 (13.08-23.47) 55.98 (50.00-61.92) 18.56 (13.85-23.46) 7.28 (4.23-10.77)
## Castellón de la Plana Corr&IRE 12.36 (9.62-15.00) 66.08 (62.31-70.38) 18.01 (14.62-21.92) 3.55 (2.31-5.00)
## Castellón de la Plana Observed 9.62 71.92 15.00 3.46
## Orihuela Indep 19.04 (12.50-25.53) 55.14 (47.92-62.50) 18.38 (12.50-25.00) 7.44 (3.65-11.98)
## Orihuela Corr 19.73 (13.02-26.04) 56.30 (48.96-63.54) 17.41 (11.98-23.44) 6.57 (3.12-10.42)
## Orihuela Corr&IRE 23.65 (20.83-27.08) 48.79 (44.27-53.12) 22.37 (18.23-26.04) 5.19 (3.65-7.29)
## Orihuela Observed 23.44 48.44 23.96 3.65
## Torrevieja Indep 14.85 (9.24-21.20) 52.99 (45.65-60.33) 22.10 (15.22-29.35) 10.06 (5.43-15.22)
## Torrevieja Corr 14.14 (8.70-20.11) 53.26 (46.20-60.87) 22.73 (16.30-28.80) 9.87 (5.98-14.67)
## Torrevieja Corr&IRE 18.97 (16.30-21.74) 45.54 (41.83-50.00) 22.66 (19.01-26.63) 12.83 (10.87-15.22)
## Torrevieja Observed 18.48 46.20 23.91 11.41
##
## Table S11 - Item 11
## Municipality Model 1 2 3 4
## Valencia Indep 11.54 (9.36-13.73) 62.73 (59.73-65.73) 15.04 (12.55-17.64) 10.69 (8.63-13.00)
## Valencia Corr 11.82 (9.73-14.27) 63.48 (60.45-66.46) 14.59 (12.45-16.82) 10.11 (8.18-12.27)
## Valencia Corr&IRE 10.66 (9.36-12.09) 64.87 (62.73-67.00) 15.60 (13.82-17.55) 8.87 (7.73-10.09)
## Valencia Observed 9.64 65.45 16.64 8.27
## Elche Indep 6.04 (4.07-8.14) 52.14 (46.78-57.46) 21.89 (18.31-25.76) 19.94 (15.93-24.24)
## Elche Corr 5.91 (3.90-7.97) 52.72 (47.79-56.95) 21.80 (18.47-25.42) 19.56 (15.76-23.56)
## Elche Corr&IRE 9.06 (7.46-10.85) 49.19 (46.44-51.86) 17.53 (14.92-20.34) 24.23 (22.37-26.27)
## Elche Observed 11.19 44.92 17.80 25.93
## Alicante Indep 5.84 (3.79-8.10) 52.01 (47.07-57.24) 22.05 (18.28-25.69) 20.10 (16.03-24.14)
## Alicante Corr 5.42 (3.45-7.59) 51.62 (46.55-56.38) 22.59 (19.14-26.21) 20.37 (16.38-24.31)
## Alicante Corr&IRE 8.32 (6.90-9.83) 49.12 (46.38-51.90) 17.72 (15.17-20.52) 24.83 (22.76-26.90)
## Alicante Observed 10.00 47.24 16.38 25.86
## Castellón de la Plana Indep 16.51 (11.15-22.31) 64.87 (58.85-70.38) 11.27 (7.30-15.77) 7.35 (4.23-11.16)

```

```
## Castellón de la Plana Corr      16.46 (11.54-21.92) 65.28 (59.23-71.15) 11.06 (7.31-15.00) 7.20 (4.23-10.77)
## Castellón de la Plana Corr&IRE 11.11 (8.08-14.62) 75.36 (71.15-80.00) 9.92 (6.54-13.46) 3.61 (2.31-5.00)
## Castellón de la Plana Observed 7.69      82.31      5.77      4.23
## Orihuela      Indep      18.17 (11.98-25.00) 64.92 (58.85-71.35) 10.39 (6.25-15.10) 6.51 (3.12-10.42)
## Orihuela      Corr      19.27 (13.02-25.53) 65.12 (58.85-72.40) 9.74 (5.73-14.06) 5.88 (2.60-9.90)
## Orihuela      Corr&IRE 21.73 (17.71-25.53) 60.97 (55.73-65.62) 12.56 (8.33-16.67) 4.74 (3.12-6.77)
## Orihuela      Observed 20.83      61.46      13.54      3.65
## Torrevieja      Indep      11.90 (7.07-17.40) 62.88 (55.43-70.12) 14.72 (9.24-20.67) 10.49 (5.98-16.30)
## Torrevieja      Corr      12.24 (7.60-17.93) 63.60 (56.51-70.65) 14.27 (9.24-20.65) 9.89 (5.43-14.67)
## Torrevieja      Corr&IRE 16.72 (13.04-20.11) 54.72 (50.00-59.78) 16.03 (11.96-20.65) 12.53 (9.78-15.76)
## Torrevieja      Observed 17.39      50.54      20.65      11.41
##
## Table S12 - Item 12
## Municipality      Model      1      2      3      4
## Valencia      Indep      7.23 (5.55-9.00) 85.42 (83.27-87.45) 6.11 (4.73-7.82) 1.24 (0.64-2.00)
## Valencia      Corr      7.16 (5.36-9.09) 85.80 (83.55-88.00) 5.87 (4.45-7.36) 1.17 (0.55-1.91)
## Valencia      Corr&IRE 7.10 (5.82-8.45) 85.96 (84.18-87.73) 5.76 (4.45-7.18) 1.18 (0.64-1.82)
## Valencia      Observed 6.64      86.73      5.73      0.91
## Elche      Indep      6.56 (4.24-9.15) 85.36 (82.37-88.31) 6.69 (4.41-9.15) 1.38 (0.51-2.54)
## Elche      Corr      6.34 (4.24-8.64) 85.80 (82.88-88.48) 6.53 (4.24-8.81) 1.34 (0.51-2.37)
## Elche      Corr&IRE 7.17 (5.25-9.33) 84.25 (81.69-86.78) 6.91 (5.08-8.98) 1.66 (0.85-2.71)
## Elche      Observed 6.61      84.58      7.29      1.36
## Alicante      Indep      9.94 (7.07-12.93) 84.75 (81.72-87.76) 4.44 (2.75-6.55) 0.87 (0.17-1.72)
## Alicante      Corr      10.31 (7.58-13.62) 84.89 (81.55-87.76) 4.02 (2.59-5.69) 0.77 (0.17-1.55)
## Alicante      Corr&IRE 11.67 (9.48-13.79) 82.06 (79.31-84.66) 5.20 (3.62-7.07) 1.08 (0.34-1.90)
## Alicante      Observed 12.59      80.00      5.52      1.38
## Castellón de la Plana      Indep      6.76 (3.46-10.77) 85.34 (80.77-89.62) 6.59 (3.45-10.38) 1.31 (0.00-2.69)
## Castellón de la Plana      Corr      6.84 (3.85-10.38) 85.79 (81.15-90.00) 6.19 (3.08-9.62) 1.19 (0.00-2.69)
## Castellón de la Plana      Corr&IRE 5.28 (2.69-8.08) 88.07 (84.23-91.54) 5.85 (3.46-8.85) 0.80 (0.00-1.92)
## Castellón de la Plana      Observed 4.23      90.00      4.23      1.54
## Orihuela      Indep      8.71 (4.17-13.54) 85.17 (79.69-90.10) 5.18 (2.08-8.85) 0.95 (0.00-2.60)
## Orihuela      Corr      10.18 (5.73-15.10) 84.89 (79.69-89.58) 4.17 (1.56-7.29) 0.76 (0.00-2.08)
## Orihuela      Corr&IRE 11.10 (7.29-14.58) 82.75 (78.12-86.99) 5.25 (2.60-8.33) 0.91 (0.00-2.60)
## Orihuela      Observed 12.50      80.73      5.73      0.52
## Torrevieja      Indep      8.31 (3.80-13.59) 84.98 (79.35-90.22) 5.52 (2.17-9.24) 1.19 (0.00-2.72)
## Torrevieja      Corr      9.00 (4.89-14.13) 85.24 (79.89-90.23) 4.75 (1.63-8.15) 1.00 (0.00-2.72)
## Torrevieja      Corr&IRE 9.71 (6.52-13.04) 83.35 (79.35-87.50) 5.83 (2.72-9.24) 1.11 (0.00-2.19)
## Torrevieja      Observed 9.24      84.24      4.89      1.63
```

WAIC computation

```
### Model-Indep ###

source(file = file.path("../", "models", "model-indep.R"))

modelData <- list(y = as.matrix(y), zero.theta.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars,
  sex = sex, age = age, muni = muni, NSex = NSex,
  NAges = NAges, NMuni = NMuni, ones = ones,
  NDist = NDist, Lambda = Lambda, from.to = from.to)

# Create the NIMBLE model and initialize its nodes
modelWAIC <- nimbleModel(code = modelCode,
  constants = modelConstants,
  data = modelData,
  inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
    dim = c(NSex, NAges, NVars, NCats)),
    rho = runif(NVars),
    theta = matrix(rnorm(NMuni * NVars, sd = 0.1), nrow =
      NMuni, ncol = NVars),
    sd.theta = runif(NVars)),
  calculate = FALSE)

CmodelWAIC <- compileNimble(modelWAIC) # calculateWAIC needs compiled model to exist
samples <- do.call(rbind, salnimble1) # single matrix of samples
waic1 <- calculateWAIC(samples, modelWAIC)
```



```

### Model-Corr ###

source(file = file.path("../models", "model-corr.R"))

modelData <- list(y = as.matrix(y), zero.sub.Muni.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars,
  sex = sex, age = age, muni = muni, NSex = NSex,
  NAges = NAges, NMuni = NMuni, ones = ones,
  NDist = NDist, Lambda = Lambda, from.to = from.to)

# Create the NIMBLE model and initialize its nodes
modelWAIC <- nimbleModel(code = modelCode,
  constants = modelConstants,
  data = modelData,
  inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
    dim = c(NSex, NAges, NVars, NCats)),
    rho = runif(NVars),
    sub.Muni = matrix(rnorm(NMuni * NVars, sd = 0.01), nrow =
      ↪ NMuni, ncol = NVars),
    M.Muni = matrix(rnorm(NVars * NVars, sd = 0.5), ncol =
      ↪ NVars, nrow = NVars),
    sd.M.Muni = runif(1, min = 0.2, max = 0.8)),
  calculate = FALSE)
CmodelWAIC <- compileNimble(modelWAIC) # calculateWAIC needs compiled model to exist
samples <- do.call(rbind, salnimble2) # single matrix of samples
waic2 <- calculateWAIC(samples, modelWAIC)

### Model-Corr&IRE ###

source(file = file.path("../models", "model-corr&ire.R"))

modelData <- list(y = as.matrix(y), zero.sub.Muni.resp = rep(0, NVars))

modelConstants <- list(NResp = NResp, NCats = NCats, NVars = NVars, sex = sex,
  age = age, muni = muni, NSex = NSex, NAges = NAges,
  NMuni = NMuni, ones = ones, NDist = NDist, Lambda = Lambda,
  from.to = from.to)

# Create the NIMBLE model and initialize its nodes
modelWAIC <- nimbleModel(code = modelCode,
  constants = modelConstants,
  data = modelData,
  inits = list(delta = array(rdirichlet(NSex * NAges * NVars, ones),
    dim = c(NSex, NAges, NVars, NCats)),
    rho = runif(NVars),
    sub.Resp = matrix(rnorm(NResp * NVars, sd = 0.01), nrow =
      ↪ NResp, ncol = NVars),
    M.Resp = matrix(rnorm(NVars * NVars, sd = 0.5), ncol =
      ↪ NVars, nrow = NVars),
    sd.M.Resp = runif(1, min = 0.2, max = 0.8),
    sub.Muni = matrix(rnorm(NMuni * NVars, sd = 0.01), nrow =
      ↪ NMuni, ncol = NVars),
    M.Muni = matrix(rnorm(NVars * NVars, sd = 0.5), ncol =
      ↪ NVars, nrow = NVars),
    sd.M.Muni = runif(1, min = 0.2, max = 0.8)),
  calculate = FALSE)
CmodelWAIC <- compileNimble(modelWAIC) # calculateWAIC needs compiled model to exist

```

```

samples <- do.call(rbind, salnimble3)           # single matrix of samples
waic3 <- calculateWAIC(samples, modelWAIC)

```

```
waic1
```

```

## nimbleList object of type waicNimbleList
## Field "WAIC":
## [1] 192525.4
## Field "lppd":
## [1] -94783.11
## Field "pWAIC":
## [1] 1479.595

```

```
waic2
```

```

## nimbleList object of type waicNimbleList
## Field "WAIC":
## [1] 190447.6
## Field "lppd":
## [1] -93940.28
## Field "pWAIC":
## [1] 1283.548

```

```
waic3
```

```

## nimbleList object of type waicNimbleList
## Field "WAIC":
## [1] 103170.1
## Field "lppd":
## [1] -36079.51
## Field "pWAIC":
## [1] 15505.53

```

Convergence assessment

```

### Model-Indep ###
which((salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "kappa"), 6] > 1.1) |
  → (salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "kappa"), 7] < 100))

```

```
## integer(0)
```

```

which((salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "theta"), 6] > 1.1) |
  → (salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "theta"), 7] < 100))

```

```
## integer(0)
```

```
which((salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "sd.theta"), 6] > 1.1) |  
→ (salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "sd.theta"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "rho"), 6] > 1.1) |  
→ (salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "rho"), 7] < 100))
```

```
## [1] 1 12
```

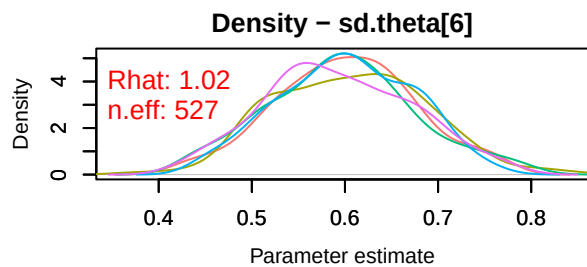
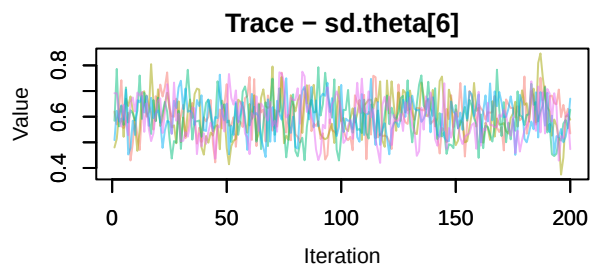
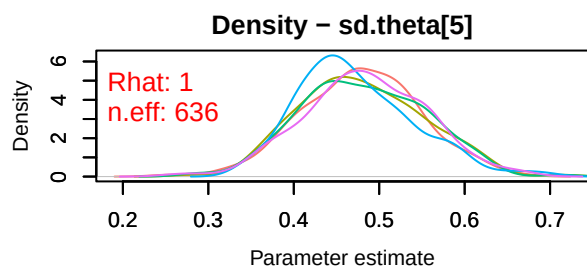
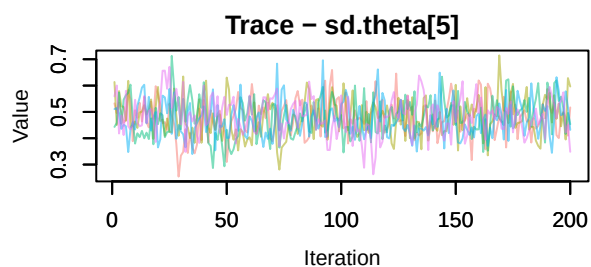
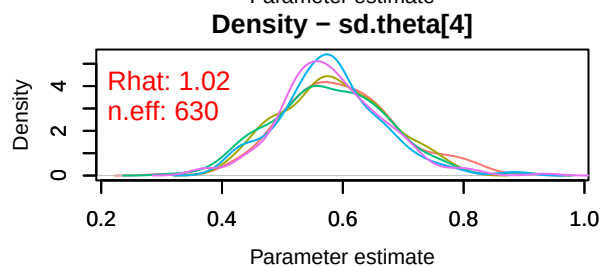
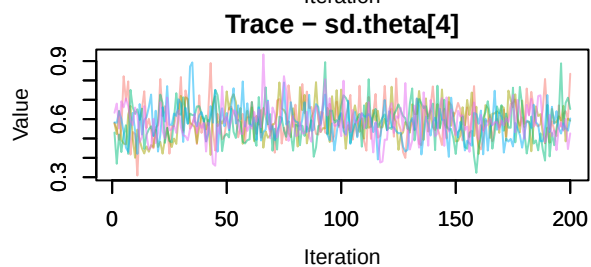
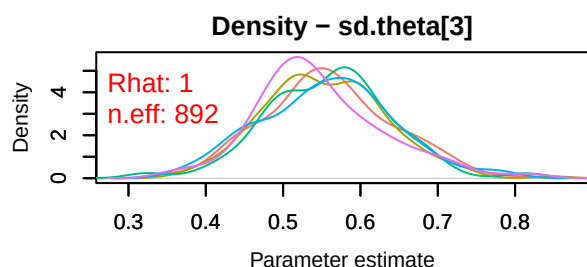
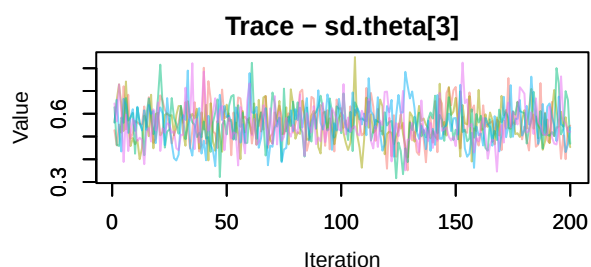
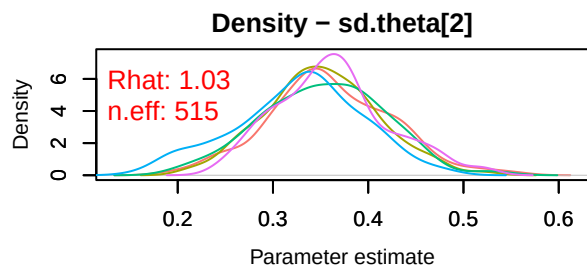
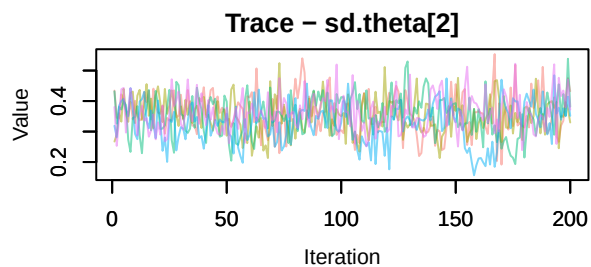
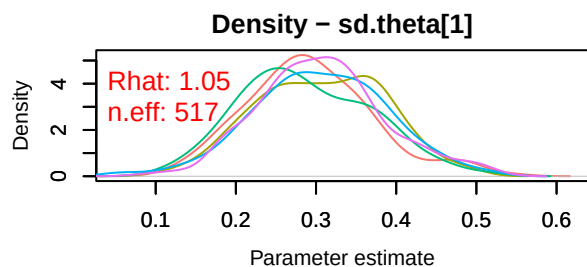
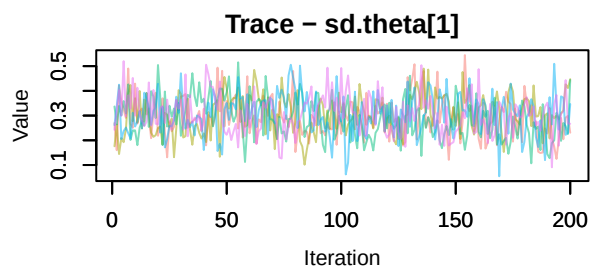
```
salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "kappa"), ][1:10, ]
```

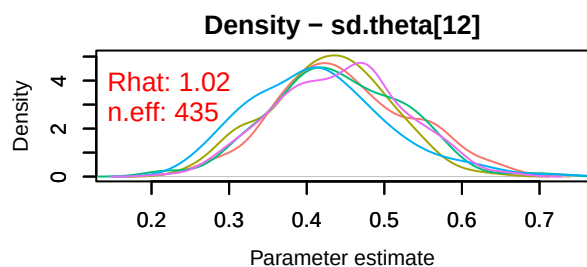
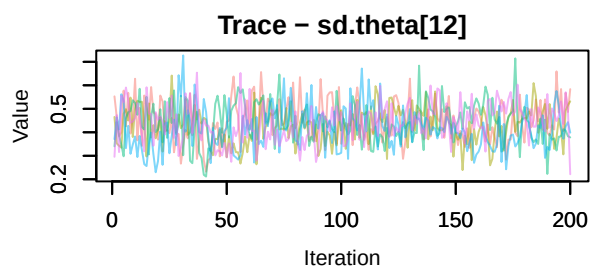
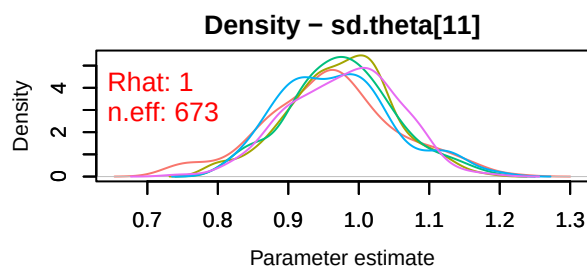
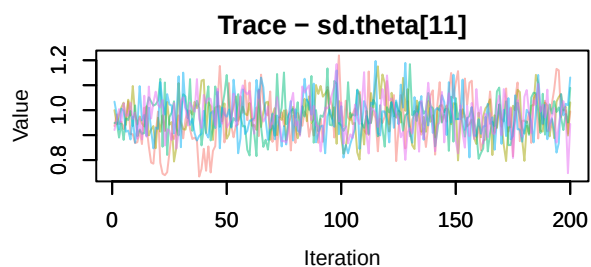
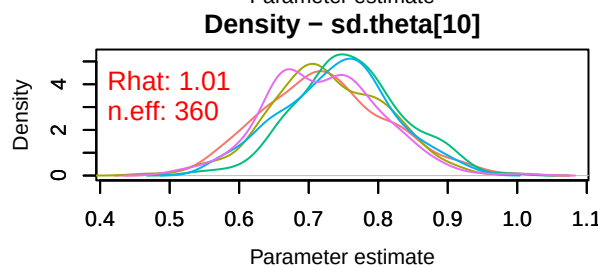
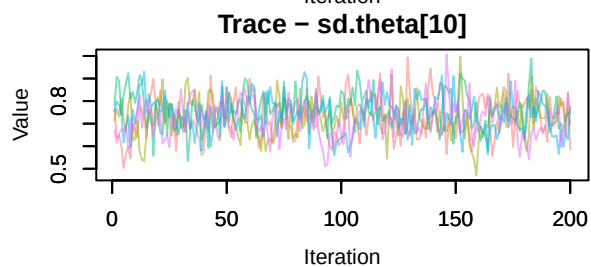
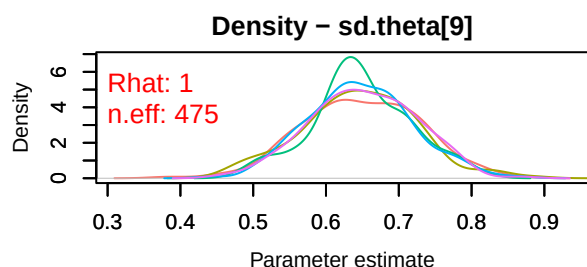
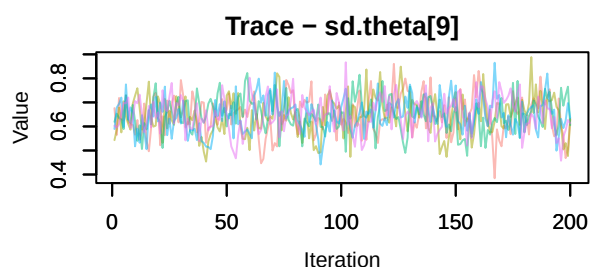
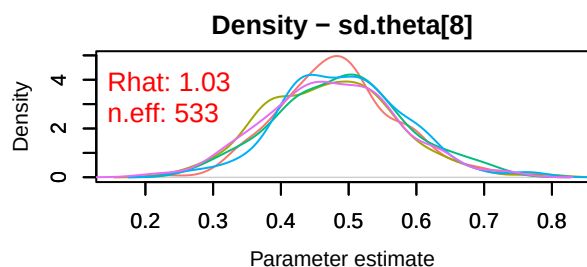
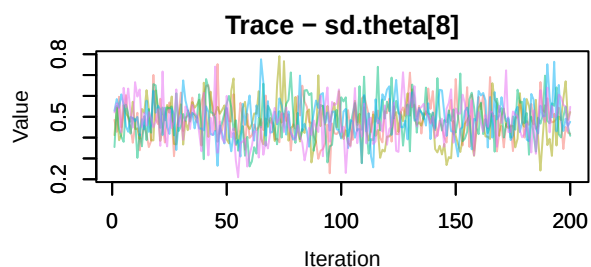
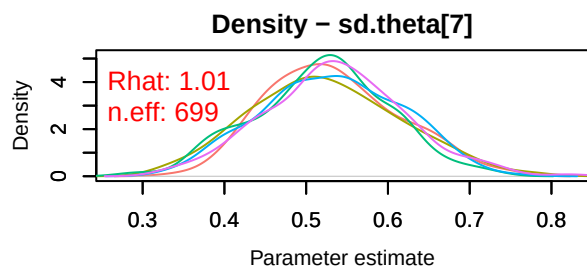
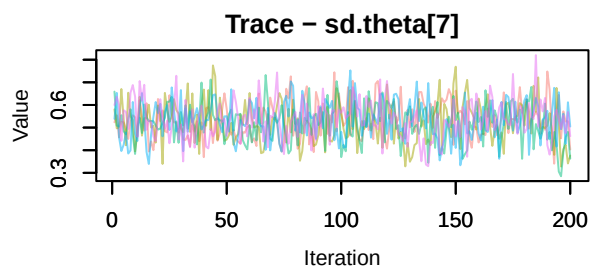
##		mean	sd	2.5%	50%	97.5%	Rhat	n.eff
##	kappa[1, 1, 1, 1]	-2.5754	0.1739	-2.9327	-2.5699	-2.2494	1.00	1174
##	kappa[2, 1, 1, 1]	-2.3844	0.1686	-2.7289	-2.3794	-2.0781	1.00	1208
##	kappa[1, 2, 1, 1]	-2.4628	0.1665	-2.7966	-2.4550	-2.1498	1.00	989
##	kappa[2, 2, 1, 1]	-2.5266	0.1589	-2.8632	-2.5188	-2.2301	1.00	1000
##	kappa[1, 3, 1, 1]	-2.7292	0.1533	-3.0197	-2.7199	-2.4244	1.01	1000
##	kappa[2, 3, 1, 1]	-2.4652	0.1396	-2.7382	-2.4634	-2.1960	1.01	960
##	kappa[1, 4, 1, 1]	-3.2979	0.1852	-3.6819	-3.2894	-2.9725	1.00	1000
##	kappa[2, 4, 1, 1]	-2.8549	0.1541	-3.1922	-2.8493	-2.5666	1.00	780
##	kappa[1, 5, 1, 1]	-3.1442	0.1983	-3.5294	-3.1397	-2.7843	1.01	892
##	kappa[2, 5, 1, 1]	-3.1827	0.1937	-3.5774	-3.1806	-2.8277	1.01	2756

```
salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], "theta"), ][1:10, ]
```

##		mean	sd	2.5%	50%	97.5%	Rhat	n.eff
##	theta[1, 1]	-0.0439	0.6895	-1.3864	-0.0467	1.3720	1.01	1075
##	theta[2, 1]	0.0441	0.6638	-1.2400	0.0196	1.4005	1.00	1138
##	theta[3, 1]	-0.1012	0.6461	-1.2719	-0.1251	1.1998	1.01	1003
##	theta[4, 1]	0.3868	0.6811	-1.0585	0.4176	1.5716	1.01	953
##	theta[5, 1]	-0.2265	0.5726	-1.3945	-0.2280	0.9430	1.02	1077
##	theta[6, 1]	0.0715	0.5737	-1.0723	0.0555	1.1740	1.00	1000
##	theta[7, 1]	-0.0266	0.7226	-1.4569	-0.0339	1.3666	1.00	953
##	theta[8, 1]	0.0445	0.7412	-1.4635	0.0331	1.5323	1.02	1055
##	theta[9, 1]	0.3294	0.4231	-0.4975	0.3266	1.1545	1.02	1012
##	theta[10, 1]	-0.1043	0.7396	-1.5696	-0.1314	1.3917	1.00	956

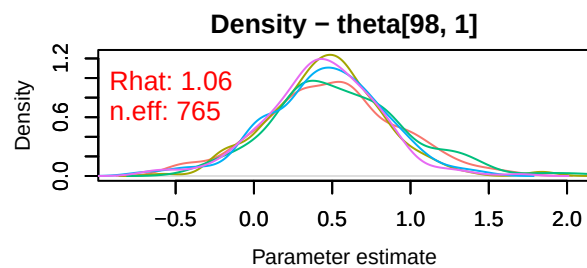
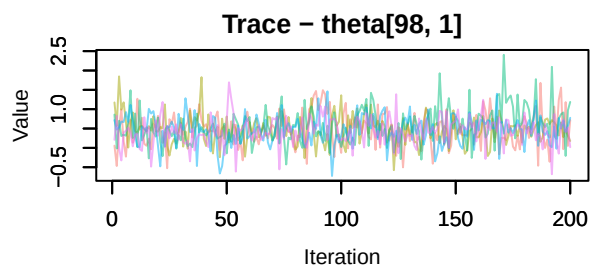
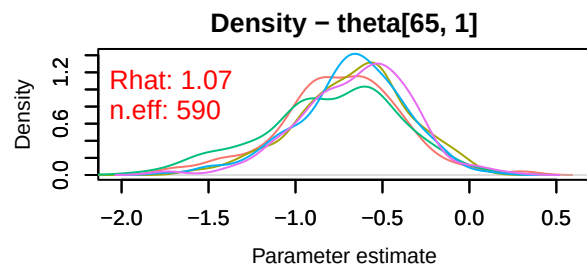
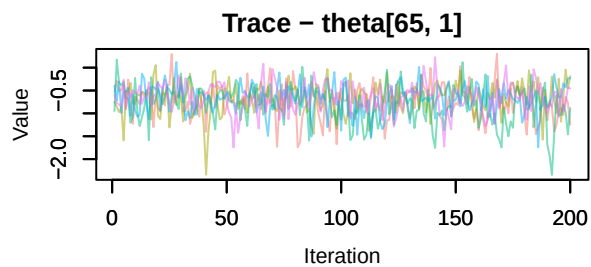
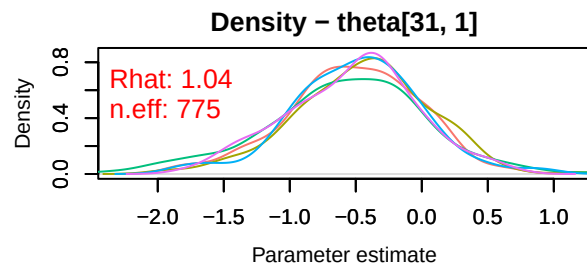
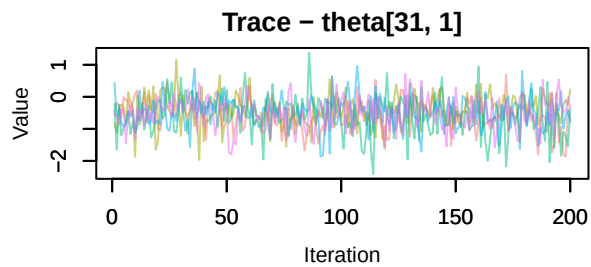
```
MCMCtrace(object = salnimble1,  
  pdf = FALSE, # no export to PDF  
  ind = TRUE, # separate density lines per chain  
  Rhat = TRUE,  
  n.eff = TRUE,  
  params = "sd.theta")
```

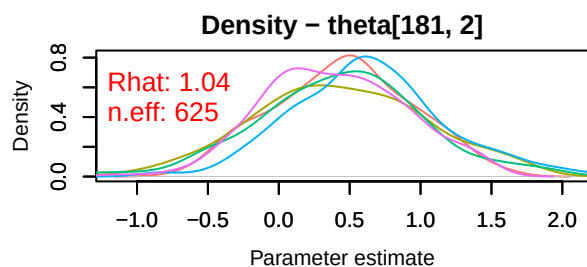
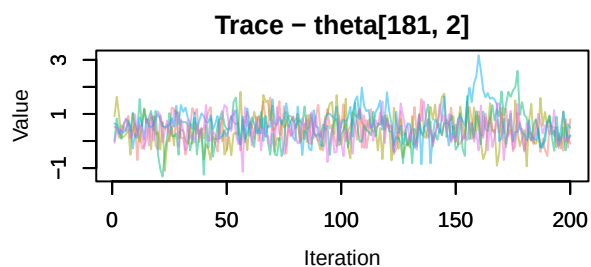
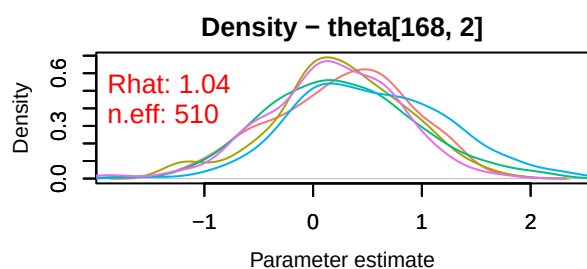
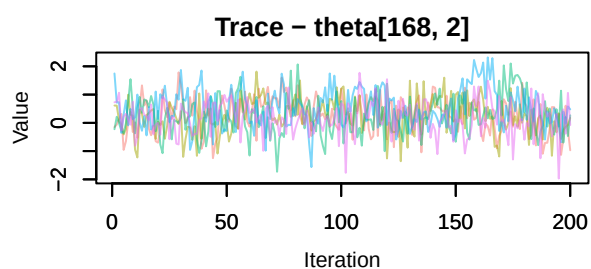
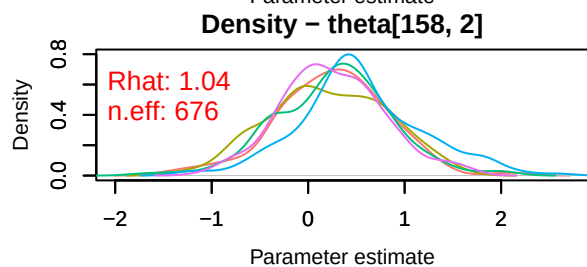
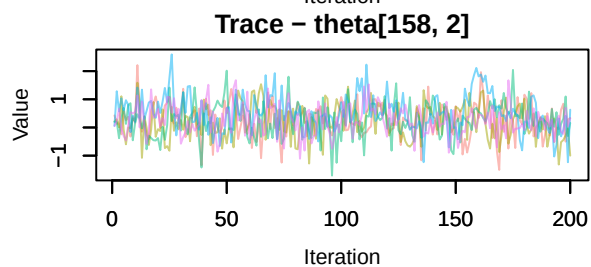
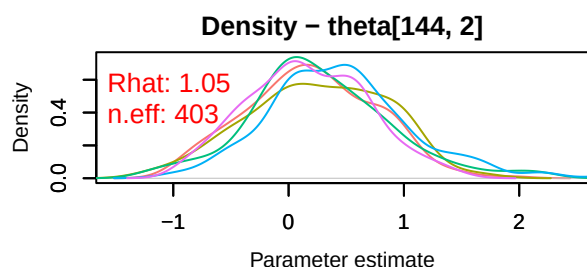
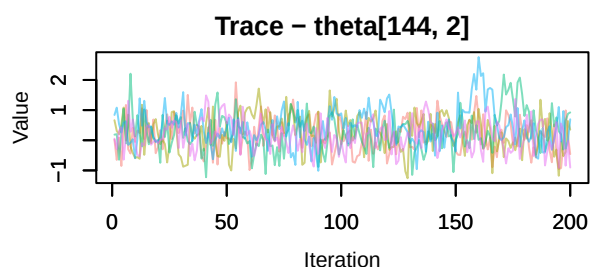
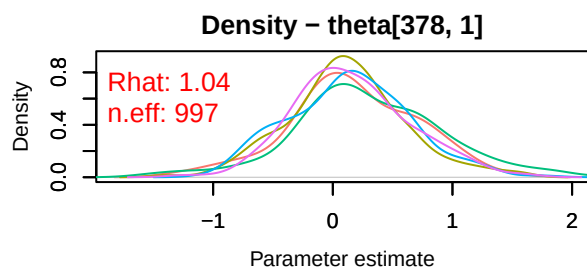
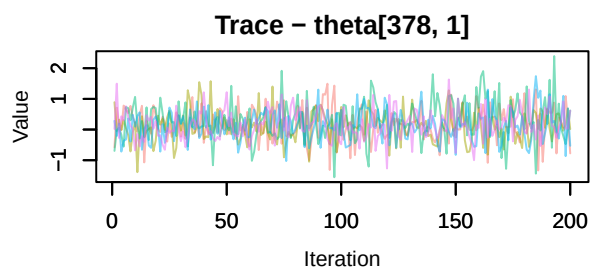
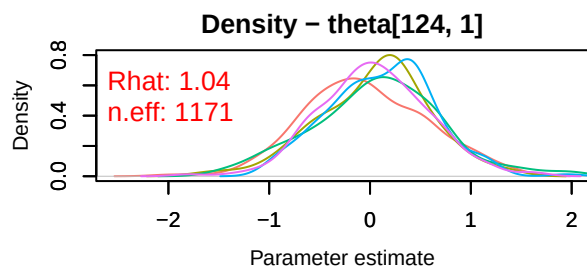
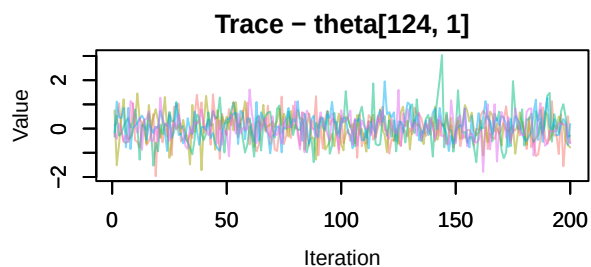


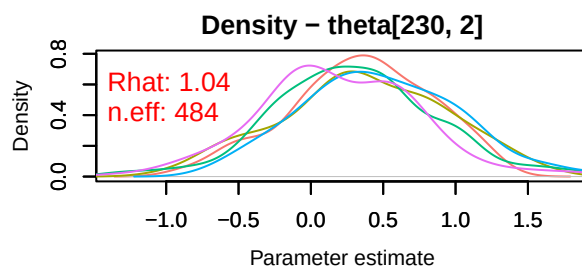
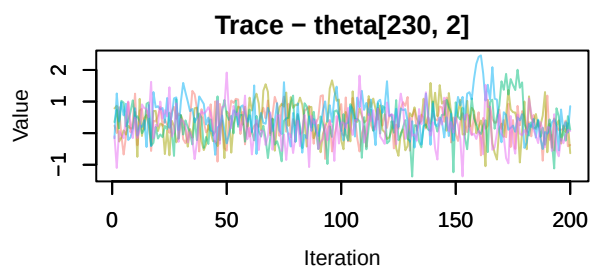
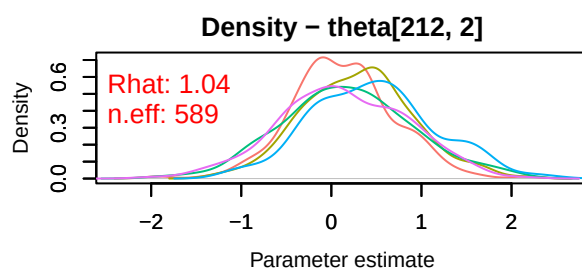
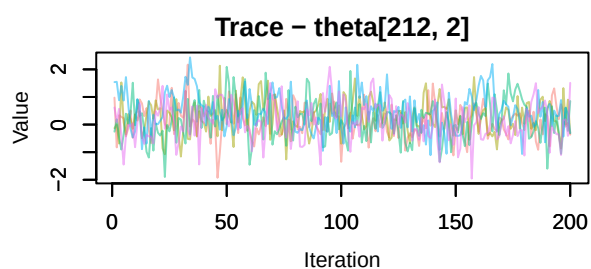
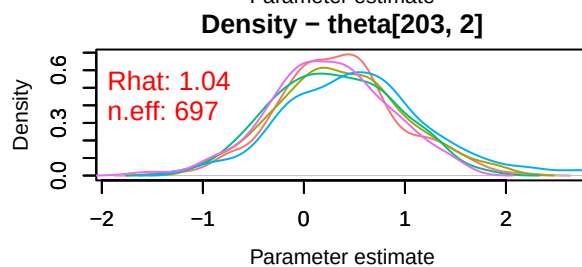
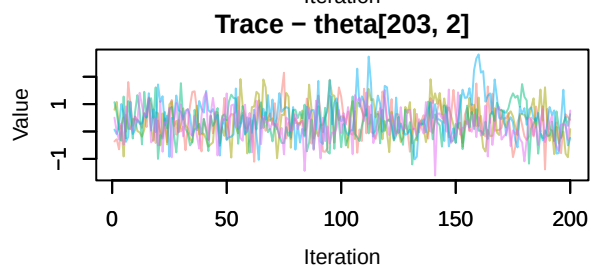
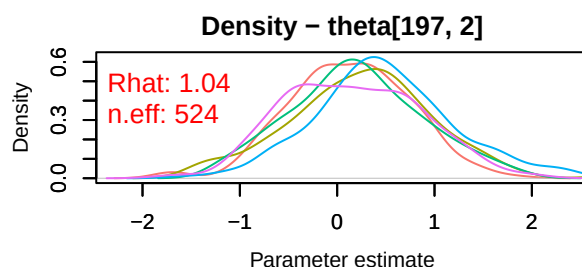
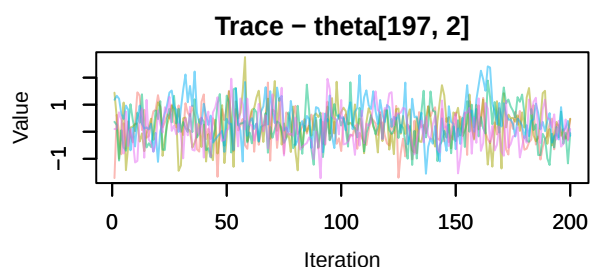
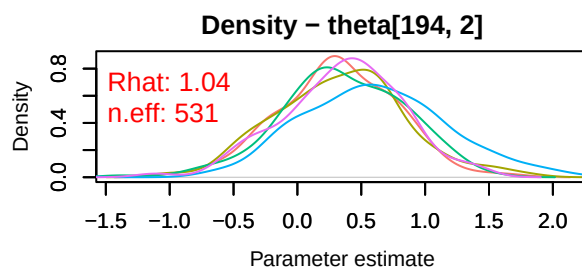
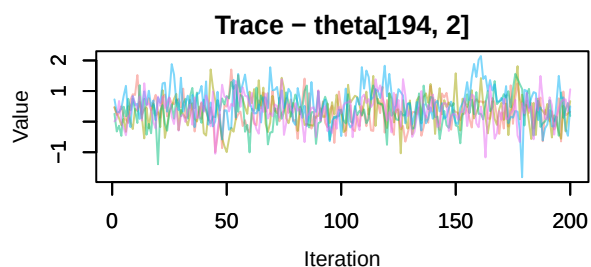
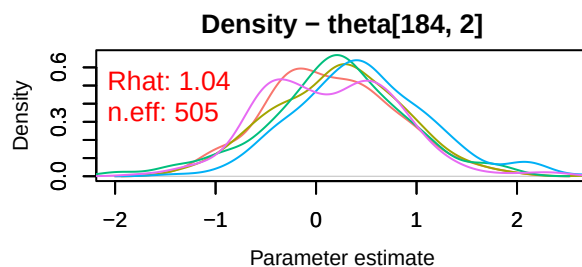
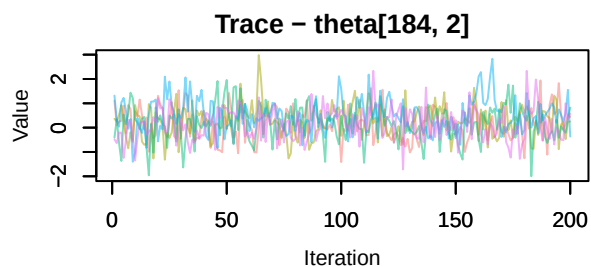


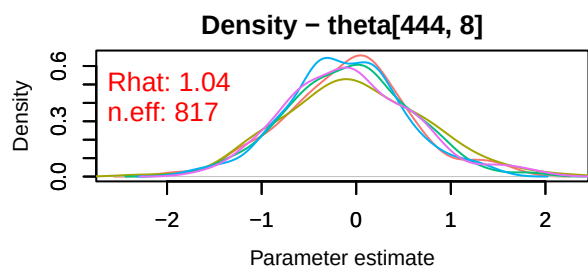
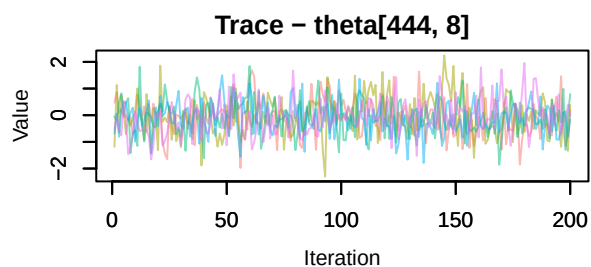
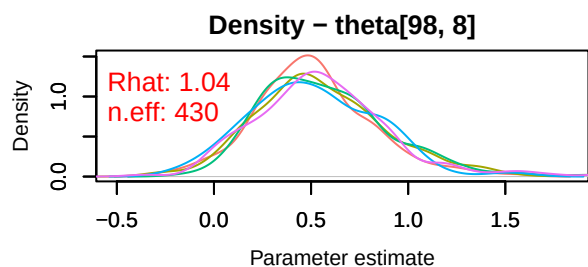
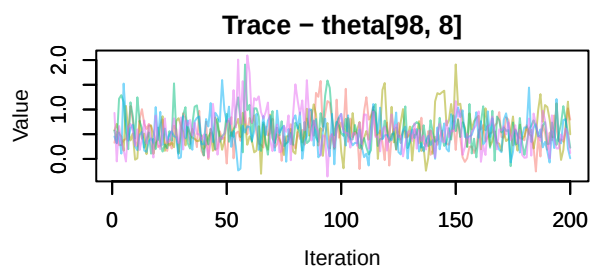
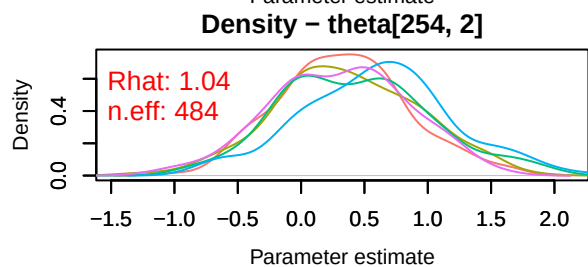
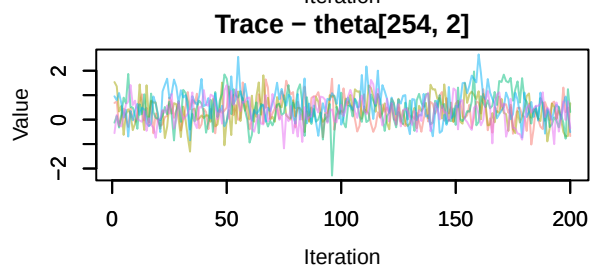
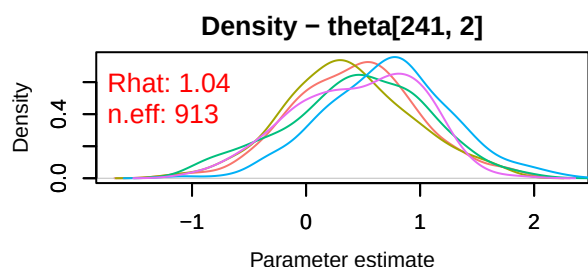
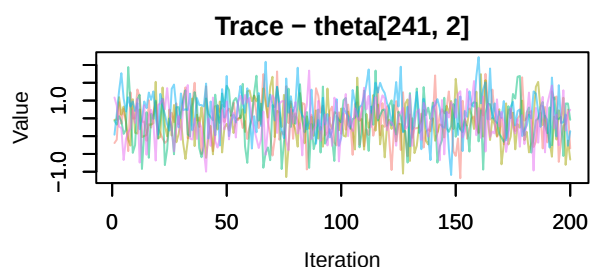
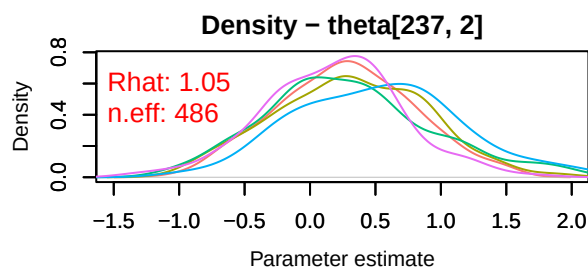
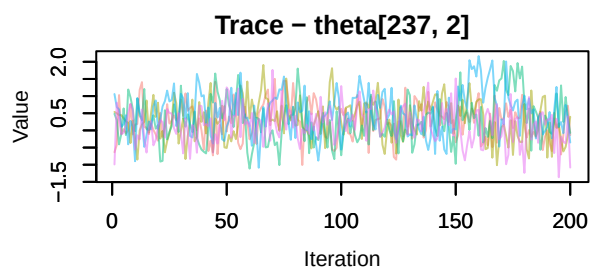
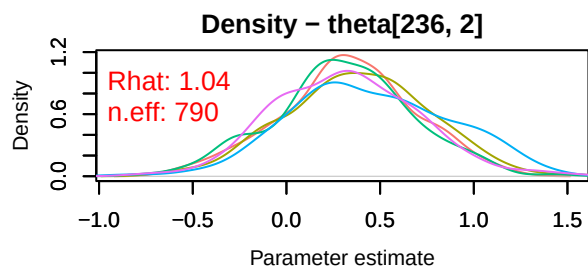
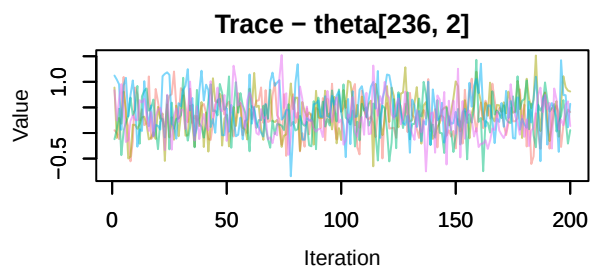
```
test <- "theta"

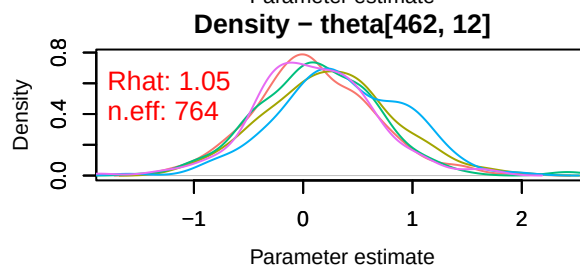
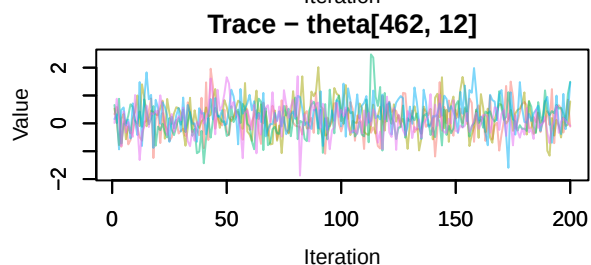
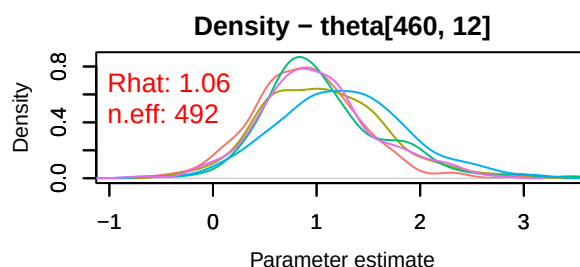
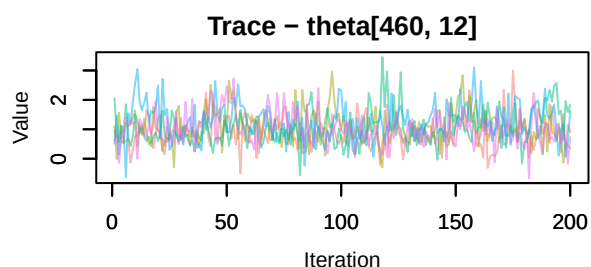
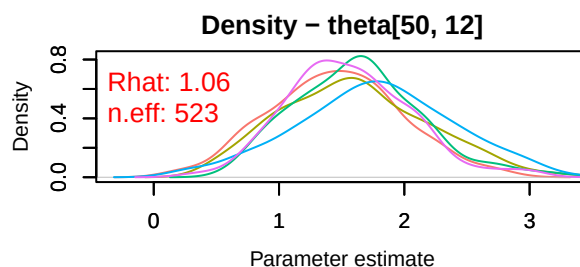
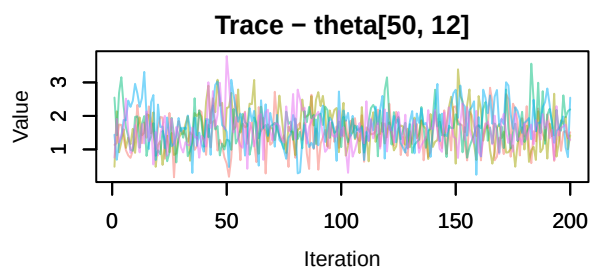
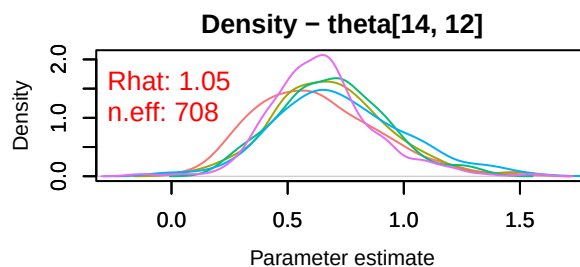
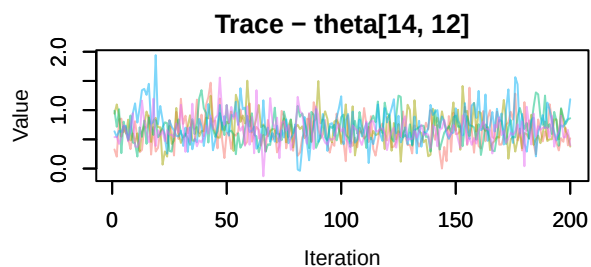
MCMCtrace(object = salnimble1,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  exact = TRUE,
  ISB = FALSE,
  params = rownames(salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]],
    ↪ test), ])[(salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]], test),
    ↪ 6] > 1.03) | (salwinbugs1$summary[startsWith(labels(salwinbugs1$summary)[[1]],
    ↪ test), 7] < 200)])
```











```
### Model-Corr ###
```

```
which((salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "kappa"), 6] > 1.1) |
  ↪ (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "kappa"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "sd.M.Muni"), 6] > 1.1) |
  ↪ (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "sd.M.Muni"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "theta"), 6] > 1.1) |
  ↪ (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "theta"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "rho"), 6] > 1.1) |
→ (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "rho"), 7] < 100))
```

```
## [1] 2 4 5 6 8 9 10 11 12
```

```
which((salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "M.Muni"), 6] > 1.1) |
→ (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "M.Muni"), 7] < 100))
```

```
## [1] 2 4 5 8 9 10 11 12 13 14 16 17 18 19 20 21 22 23
## [19] 24 26 28 29 32 33 34 35 36 38 40 41 44 45 46 47 48 49
## [37] 50 51 52 53 54 55 56 57 58 59 60 61 62 64 65 66 67 68
## [55] 69 70 71 72 74 76 77 80 81 82 83 84 86 88 89 92 93 94
## [73] 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112
## [91] 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130
## [109] 131 132 134 136 137 140 141 142 143 144
```

```
salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "kappa"), ][1:10, ]
```

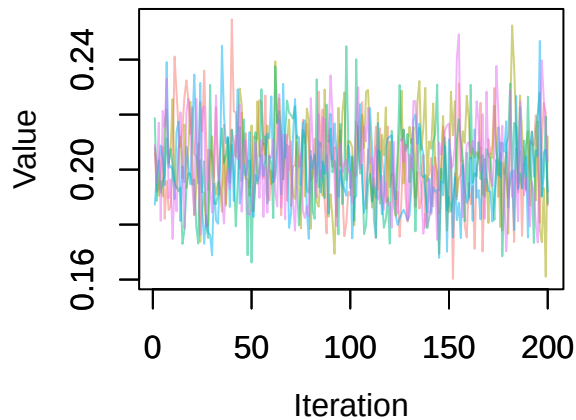
	mean	sd	2.5%	50%	97.5%	Rhat	n.eff
kappa[1, 1, 1, 1]	-2.6112	0.1731	-2.9459	-2.6110	-2.2738	1.01	1058
kappa[2, 1, 1, 1]	-2.3826	0.1647	-2.7222	-2.3787	-2.0768	1.01	861
kappa[1, 2, 1, 1]	-2.4966	0.1690	-2.8469	-2.4918	-2.1651	1.00	982
kappa[2, 2, 1, 1]	-2.5389	0.1684	-2.8639	-2.5331	-2.2225	1.00	1046
kappa[1, 3, 1, 1]	-2.7670	0.1606	-3.0860	-2.7713	-2.4757	1.01	1020
kappa[2, 3, 1, 1]	-2.4918	0.1441	-2.7835	-2.4894	-2.2361	1.00	1000
kappa[1, 4, 1, 1]	-3.3326	0.1989	-3.7429	-3.3224	-2.9798	1.01	960
kappa[2, 4, 1, 1]	-2.9040	0.1578	-3.2336	-2.9044	-2.6051	1.01	1000
kappa[1, 5, 1, 1]	-3.1721	0.1934	-3.5596	-3.1598	-2.8165	1.01	926
kappa[2, 5, 1, 1]	-3.2198	0.1860	-3.5956	-3.2112	-2.8751	1.01	1046

```
salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], "theta"), ][1:10, ]
```

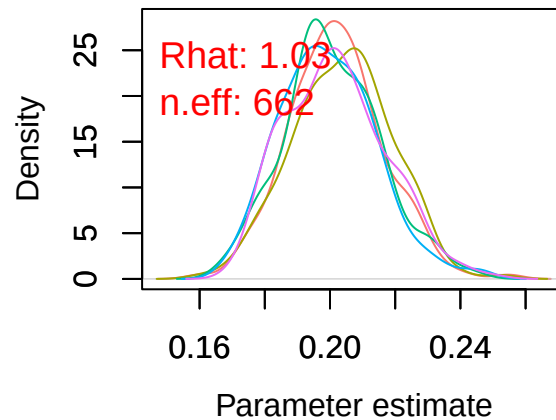
	mean	sd	2.5%	50%	97.5%	Rhat	n.eff
theta[1, 1]	0.0150	0.4553	-0.8817	0.0201	0.9334	1.01	1000
theta[2, 1]	-0.3297	0.2982	-0.9045	-0.3354	0.2538	1.01	945
theta[3, 1]	-0.0762	0.4359	-0.8942	-0.0900	0.7585	1.00	953
theta[4, 1]	-0.1383	0.3822	-0.9428	-0.1223	0.5782	1.00	1000
theta[5, 1]	-0.1882	0.1914	-0.5586	-0.1828	0.1905	1.00	1088
theta[6, 1]	0.0630	0.3232	-0.5488	0.0655	0.6981	1.00	1011
theta[7, 1]	0.0570	0.4576	-0.8418	0.0560	0.9570	1.00	1138
theta[8, 1]	-0.0498	0.4168	-0.9187	-0.0486	0.7417	1.00	955
theta[9, 1]	0.0888	0.1247	-0.1581	0.0906	0.3315	1.01	1123
theta[10, 1]	-0.0414	0.4793	-0.9397	-0.0442	0.9253	1.00	962

```
MCMCtrace(object = salnimble2,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  params = "sd.M.Muni")
```

Trace – sd.M.Muni



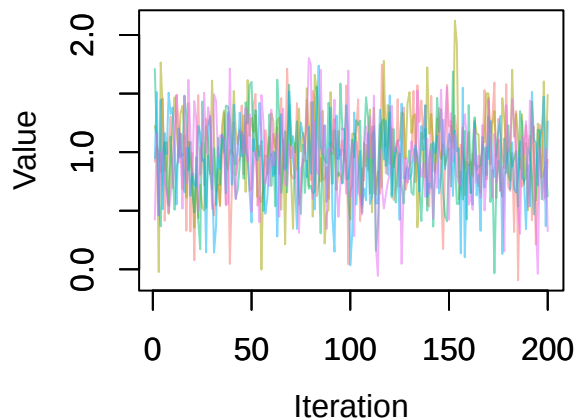
Density – sd.M.Muni



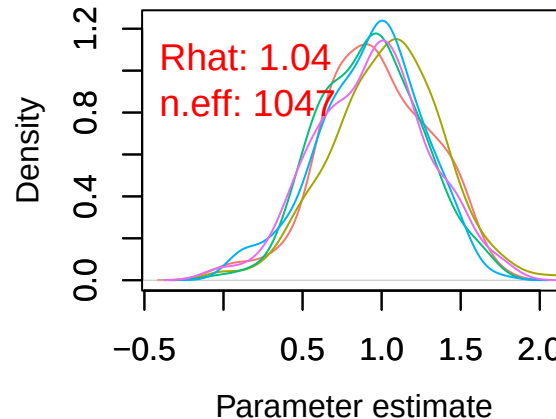
```
test <- "theta"

MCMCtrace(object = salnimble2,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  exact = TRUE,
  ISB = FALSE,
  params = rownames(salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]],
    ↪ test), ])[(salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]], test),
    ↪ 6] > 1.03) | (salwinbugs2$summary[startsWith(labels(salwinbugs2$summary)[[1]],
    ↪ test), 7] < 200)])
```

Trace – theta[467, 9]



Density – theta[467, 9]



```
### Model-Corr&IRE ###

which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "kappa"), 6] > 1.1) |
  ↪ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "kappa"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "sd.M.Muni"), 6] > 1.1) |
→ | (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "sd.M.Muni"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "theta"), 6] > 1.1) |
→ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "theta"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "rho"), 6] > 1.1) |
→ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "rho"), 7] < 100))
```

```
## [1] 1 2 3 4 5 6 8 9 10 12
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "M.Muni"), 6] > 1.1) |
→ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "M.Muni"), 7] < 100))
```

```
## [1] 14 16 20 21 22 23 24 26 32 33 36 37 38 44 48 49 50 52 53
## [20] 54 55 56 57 58 59 60 61 62 64 66 68 69 70 71 72 74 80 82
## [39] 84 86 88 92 93 96 97 98 100 101 102 103 104 105 106 107 108 109 110
## [58] 112 113 114 115 116 117 118 119 120 121 122 124 125 126 127 128 129 130 131
## [77] 132 134 138 140 141 144
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "sd.M.Resp"), 6] > 1.1) |
→ | (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "sd.M.Resp"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "psi"), 6] > 1.1) |
→ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "psi"), 7] < 100))
```

```
## integer(0)
```

```
which((salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "M.Resp"), 6] > 1.1) |
→ (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "M.Resp"), 7] < 100))
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
## [19] 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36
## [37] 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54
## [55] 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72
## [73] 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
## [91] 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144
```

```
salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "kappa"), ][1:10, ]
```

```
##               mean      sd    2.5%    50%   97.5% Rhat n.eff
## kappa[1, 1, 1, 1] -4.0030 0.2768 -4.5731 -4.0036 -3.4708 1.00 1003
## kappa[2, 1, 1, 1] -3.7967 0.2473 -4.2924 -3.7924 -3.3261 1.00  970
## kappa[1, 2, 1, 1] -3.9218 0.2430 -4.4046 -3.9187 -3.4581 1.01 1445
## kappa[2, 2, 1, 1] -3.8904 0.2363 -4.3534 -3.8956 -3.4216 1.00  988
## kappa[1, 3, 1, 1] -4.2577 0.2305 -4.6876 -4.2617 -3.8036 1.01 1111
## kappa[2, 3, 1, 1] -3.8801 0.2070 -4.2995 -3.8756 -3.4905 1.00  965
## kappa[1, 4, 1, 1] -4.9890 0.2604 -5.5495 -4.9876 -4.5058 1.00  944
## kappa[2, 4, 1, 1] -4.2869 0.2238 -4.7612 -4.2826 -3.8899 1.00  934
## kappa[1, 5, 1, 1] -4.7905 0.2562 -5.2968 -4.7829 -4.3253 1.00 1097
## kappa[2, 5, 1, 1] -4.7581 0.2590 -5.2821 -4.7522 -4.2430 1.00 1000
```

```
salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "theta"), ][1:10, ]
```

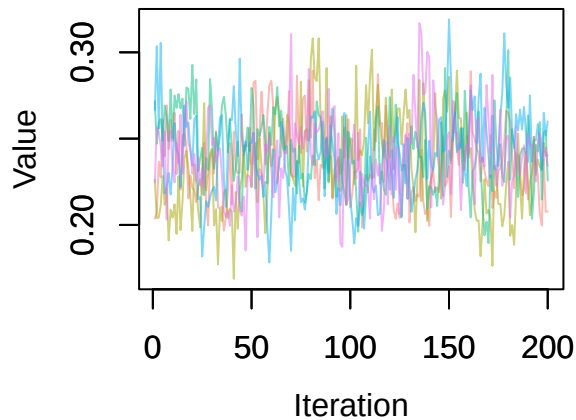
```
##               mean      sd    2.5%    50%   97.5% Rhat n.eff
## theta[1, 1]    -0.0448 0.2911 -0.6366 -0.0442 0.5428 1.00 1084
## theta[2, 1]    -0.0203 0.2791 -0.5612 -0.0281 0.5374 1.00 1443
## theta[3, 1]    -0.0414 0.2641 -0.5677 -0.0434 0.4668 1.00  900
## theta[4, 1]     0.2203 0.2884 -0.3022  0.2122 0.8244 1.01  934
## theta[5, 1]    -0.0733 0.2362 -0.5524 -0.0800 0.3920 1.00  936
## theta[6, 1]     0.0367 0.2505 -0.4401  0.0394 0.5253 1.01  921
## theta[7, 1]    -0.0124 0.3143 -0.6595 -0.0081 0.6258 1.00 1006
## theta[8, 1]     0.0285 0.3105 -0.5866  0.0309 0.6456 1.00 1047
## theta[9, 1]     0.0687 0.1959 -0.3015  0.0717 0.4520 1.00  941
## theta[10, 1]   -0.0266 0.3279 -0.6564 -0.0208 0.5887 1.01 1084
```

```
salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], "psi"), ][1:10, ]
```

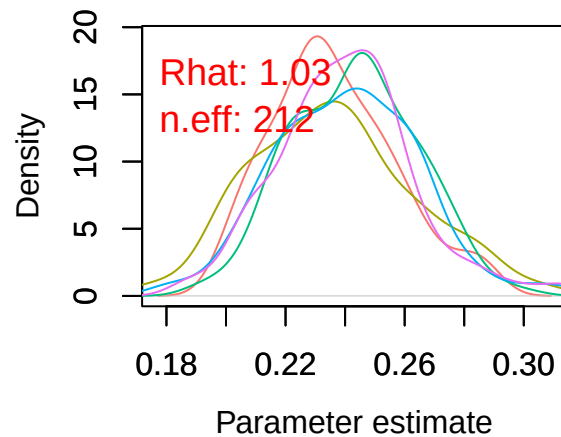
```
##               mean      sd    2.5%    50%   97.5% Rhat n.eff
## psi[1, 1]       4.2731 0.7922  2.8521  4.2324  5.9855 1.00 1000
## psi[2, 1]      -2.9513 0.7271 -4.1727 -2.9746 -1.4593 1.00 1121
## psi[3, 1]       0.2615 1.2542 -2.2734  0.2722  2.6930 1.00  878
## psi[4, 1]      -4.8959 0.6597 -6.1497 -4.9064 -3.4985 1.01  967
## psi[5, 1]       3.1538 0.8661  1.3123  3.1739  4.7894 1.01 1214
## psi[6, 1]      -1.7509 0.6943 -3.0997 -1.7370 -0.3800 1.00 1000
## psi[7, 1]      -0.1897 1.1725 -2.3539 -0.1894  2.0362 1.00 1046
## psi[8, 1]      -0.0807 1.2117 -2.4073 -0.0721  2.1425 1.00 1702
## psi[9, 1]       0.3816 1.1431 -1.8230  0.3410  2.6404 1.01 1253
## psi[10, 1]      1.2283 1.0045 -0.8305  1.2454  3.0538 1.00  956
```

```
MCMCtrace(object = salnimble3,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  params = "sd.M.Muni")
```

Trace – sd.M.Muni

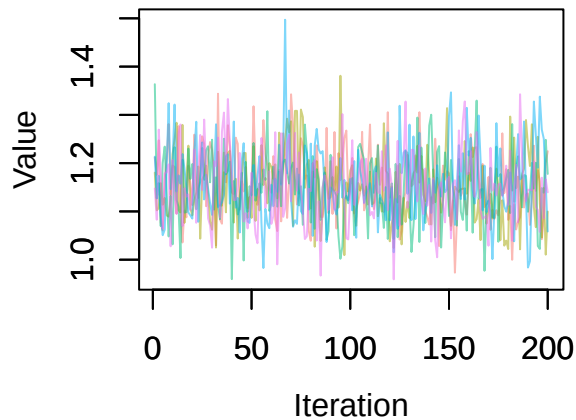


Density – sd.M.Muni

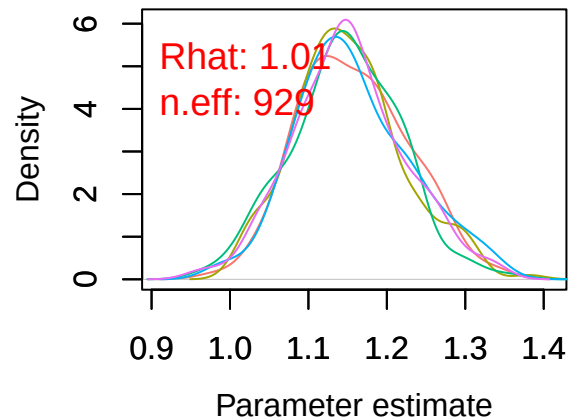


```
MCMCtrace(object = salnimble3,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  params = "sd.M.Resp")
```

Trace – sd.M.Resp

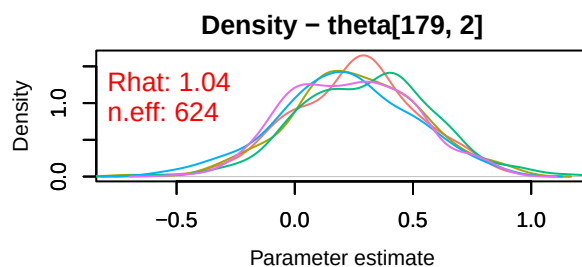
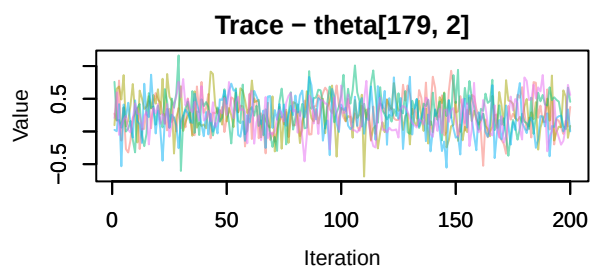
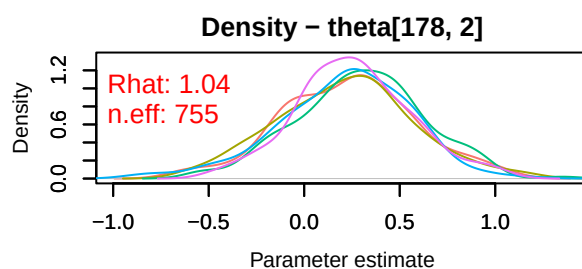
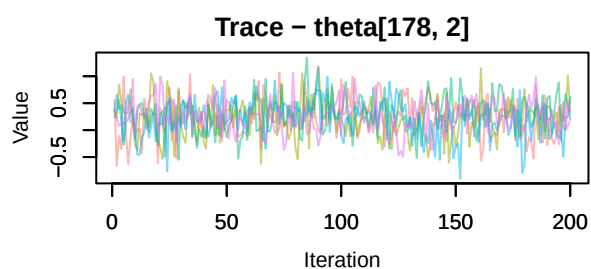
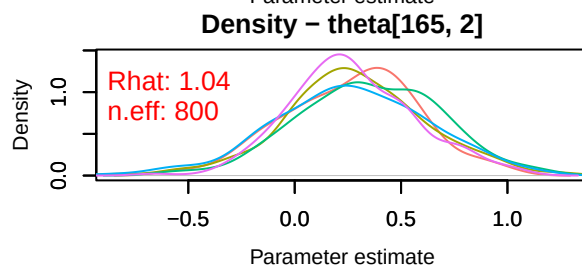
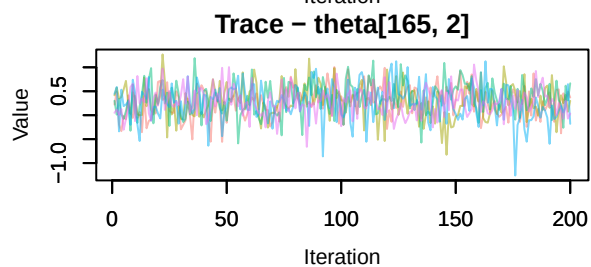
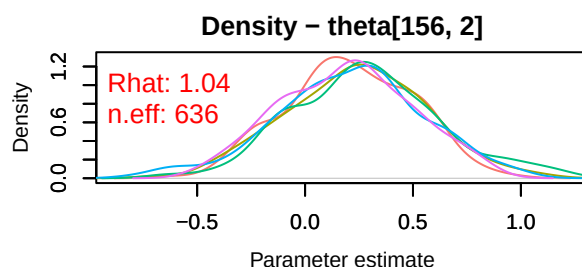
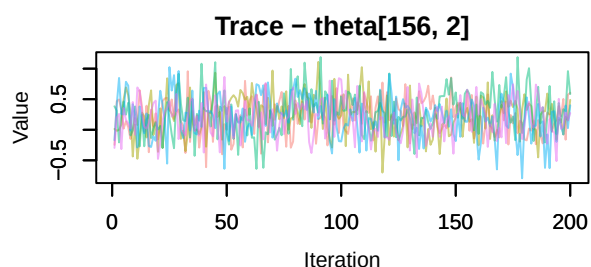
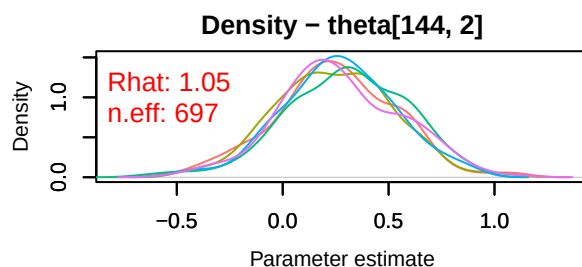
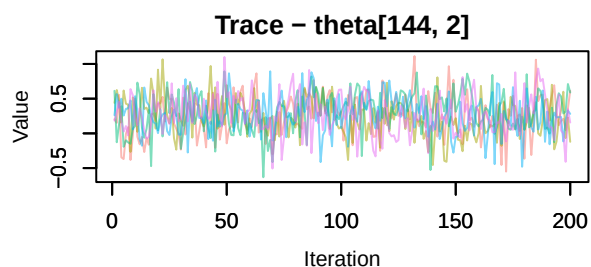
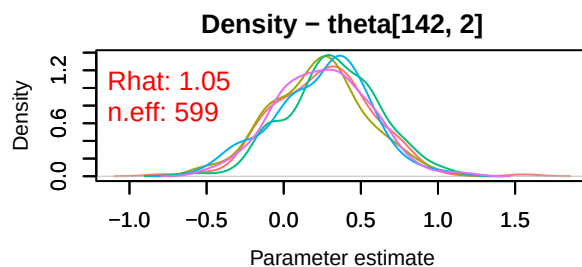
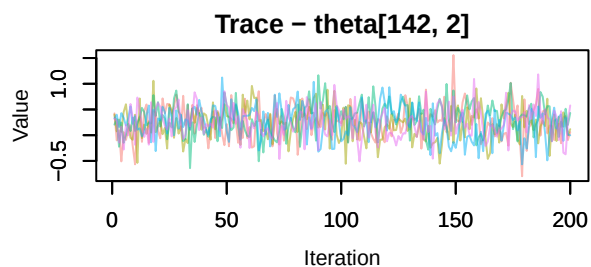


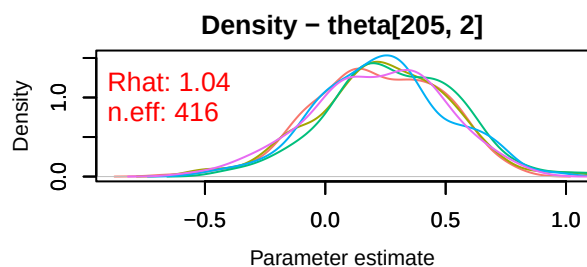
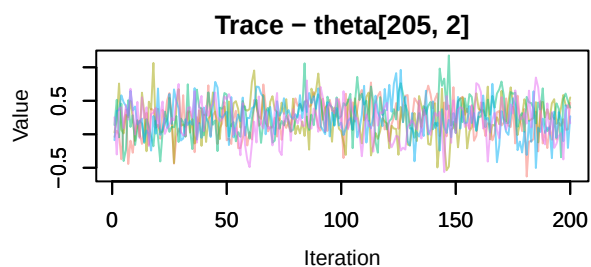
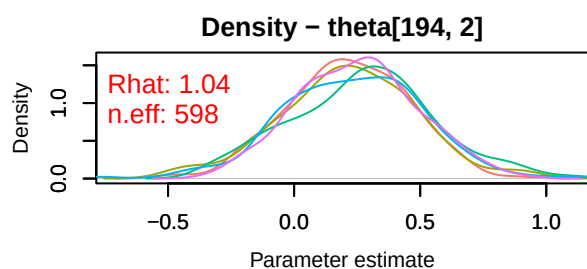
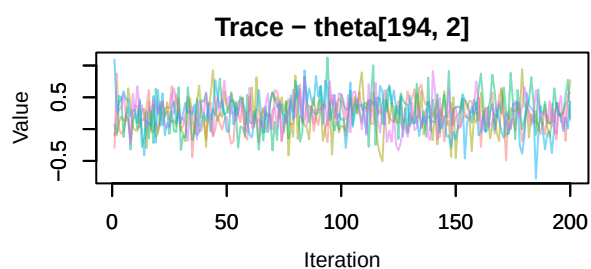
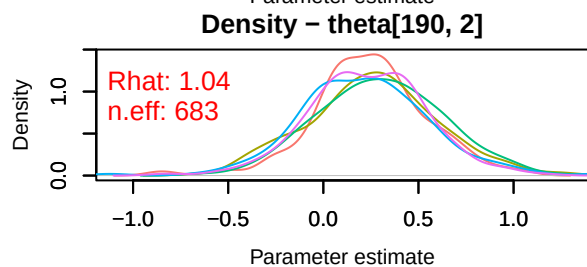
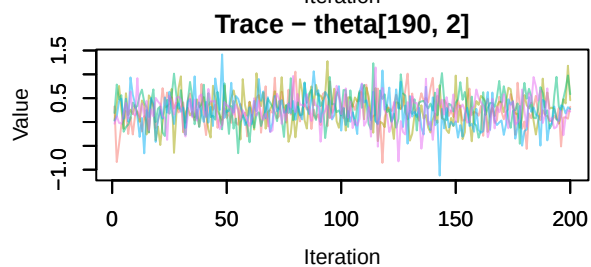
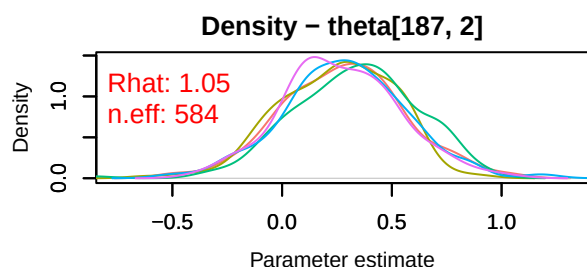
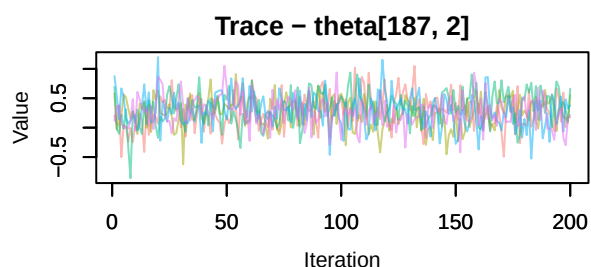
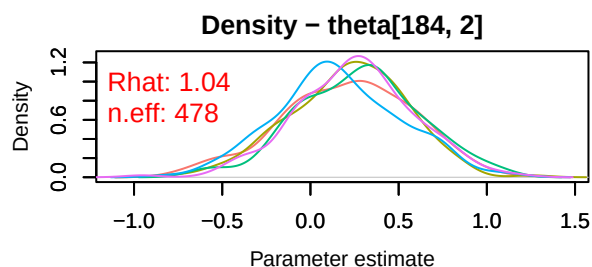
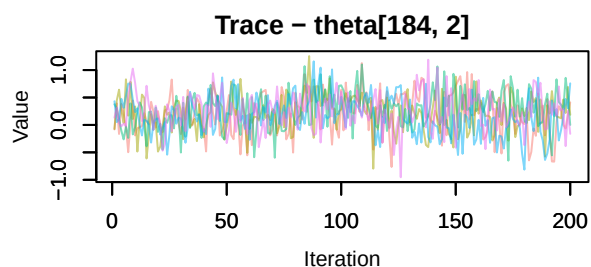
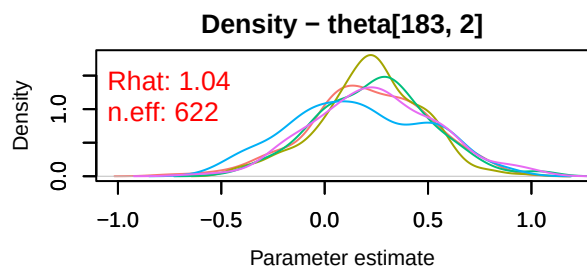
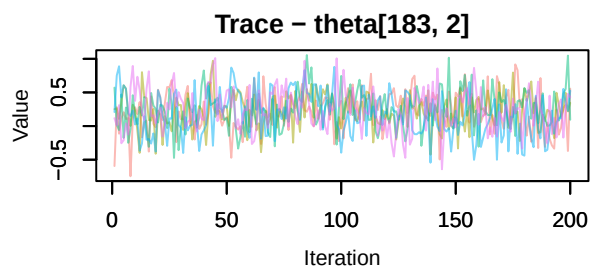
Density – sd.M.Resp

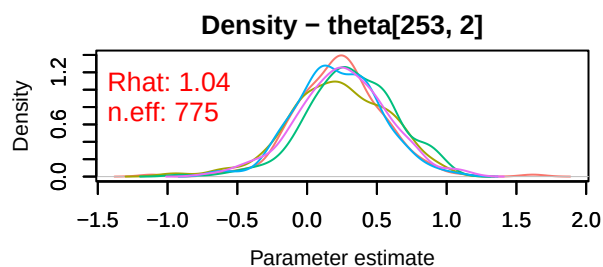
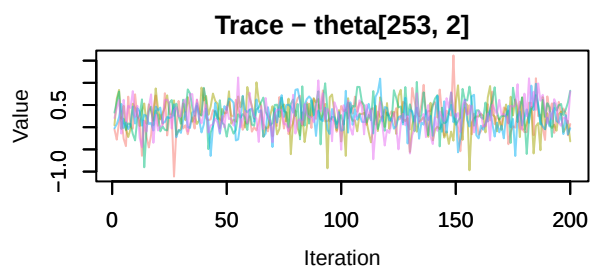
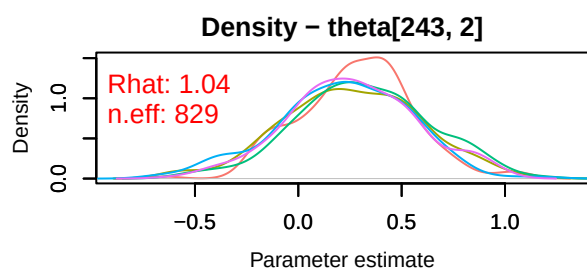
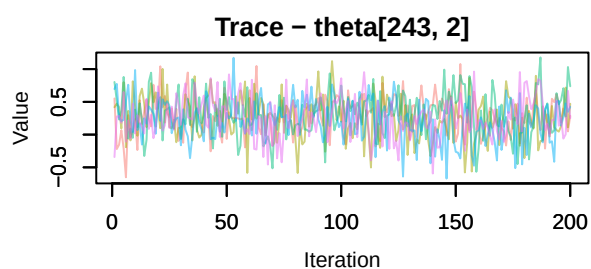
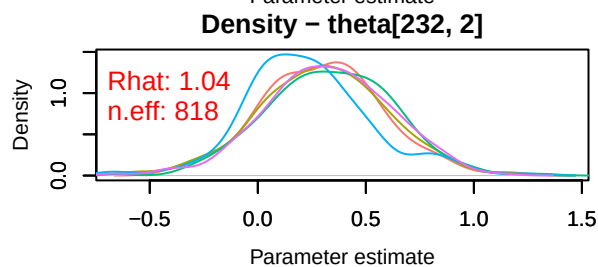
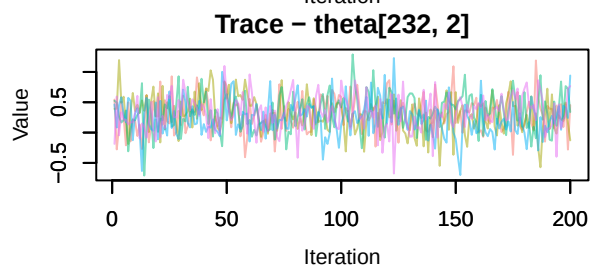
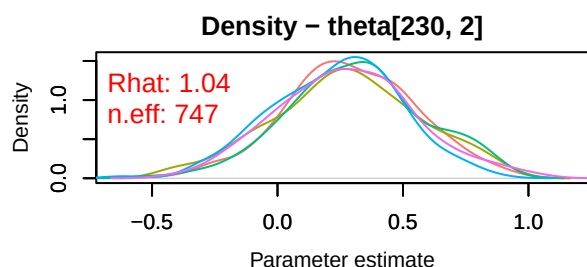
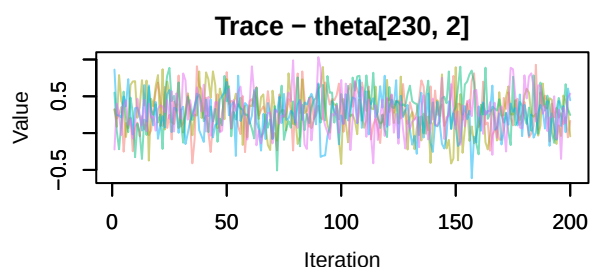
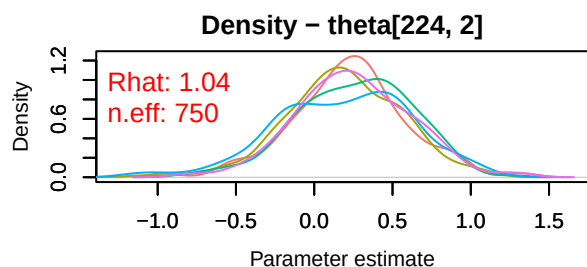
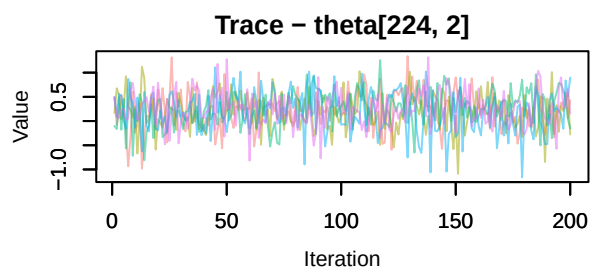
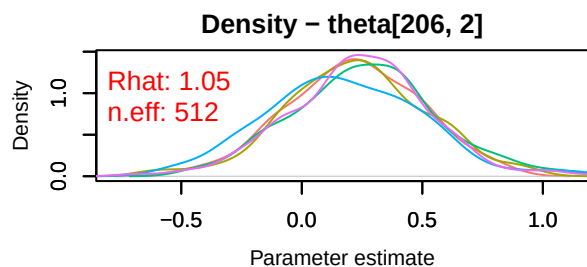
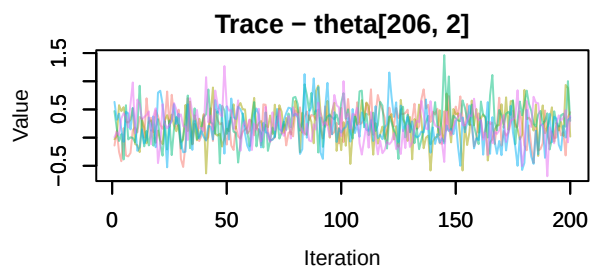


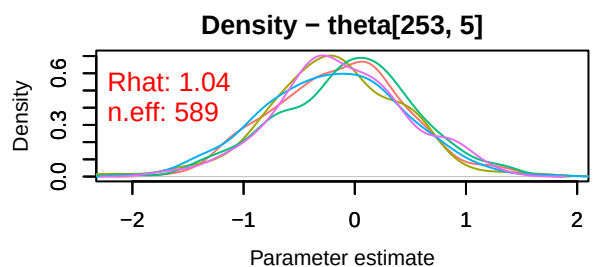
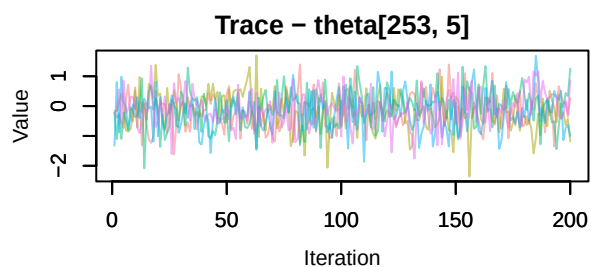
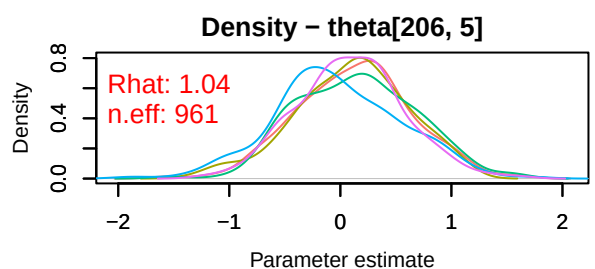
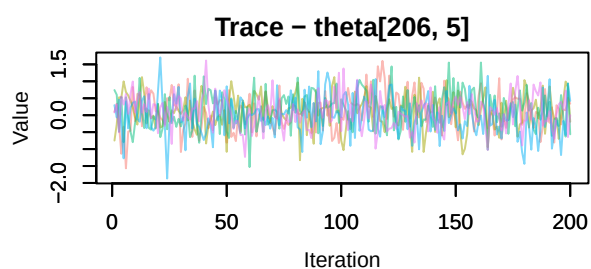
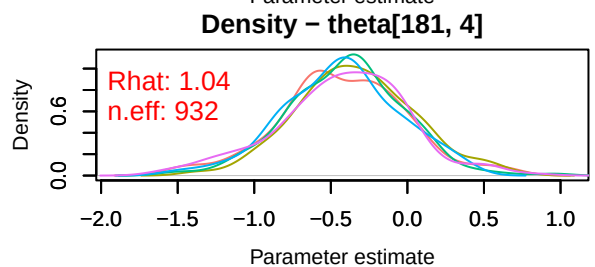
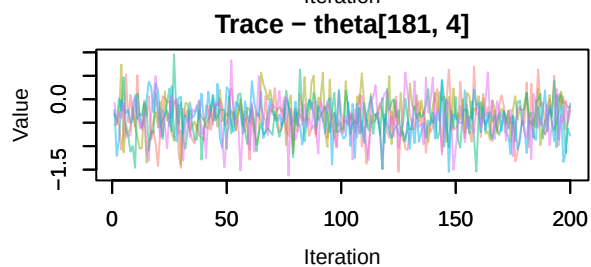
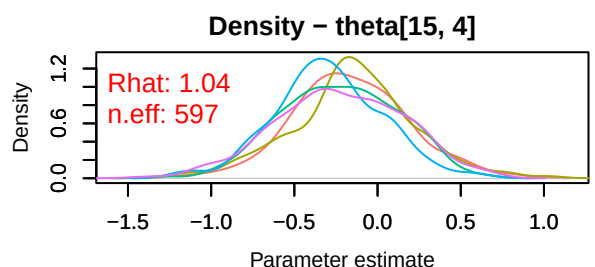
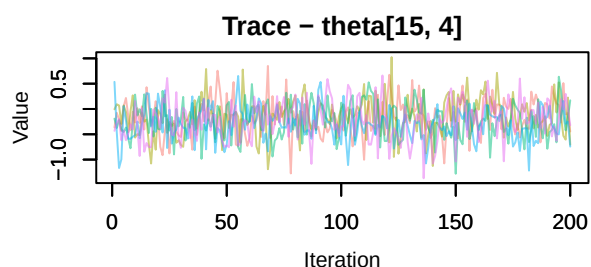
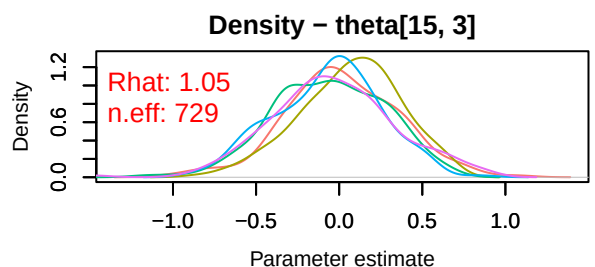
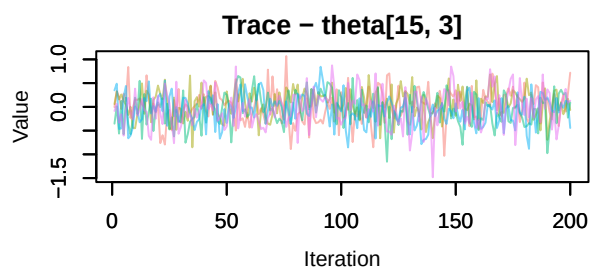
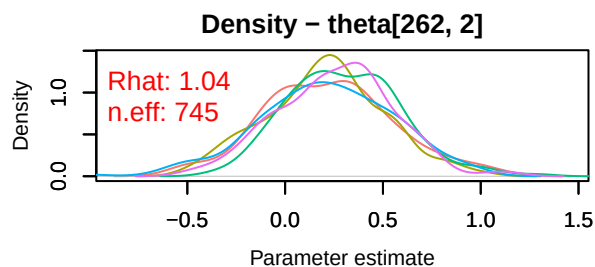
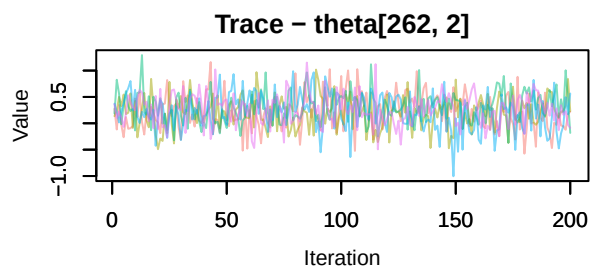
```
test <- "theta"

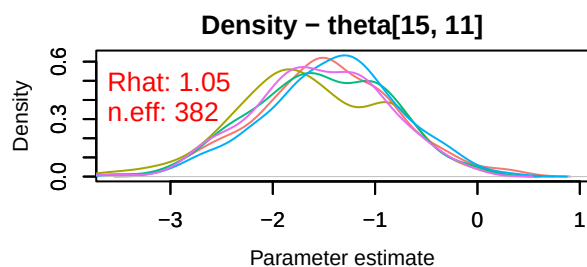
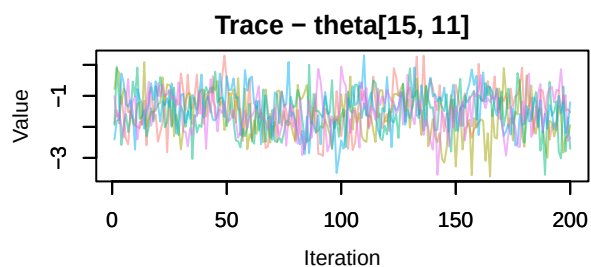
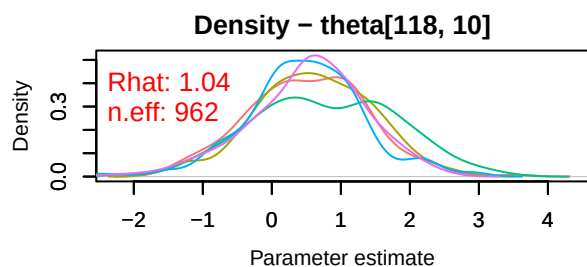
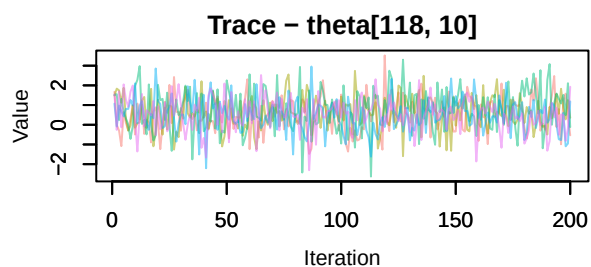
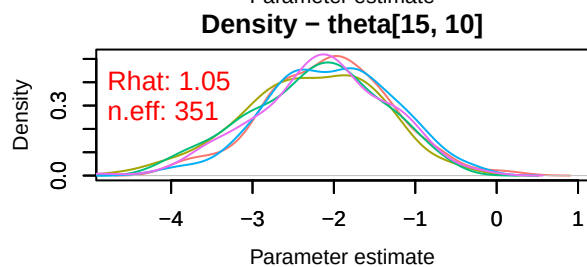
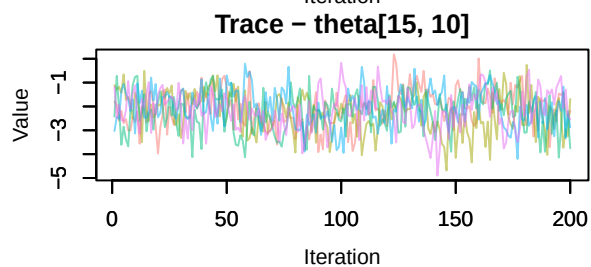
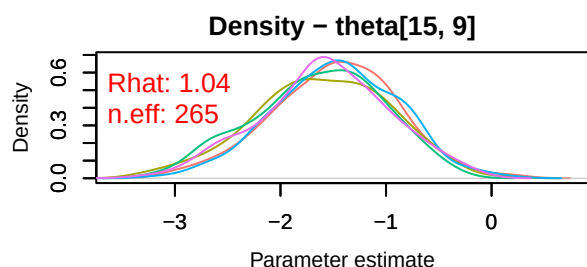
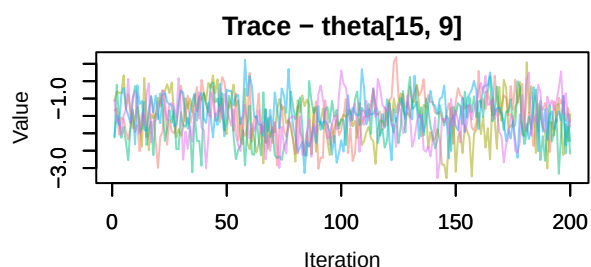
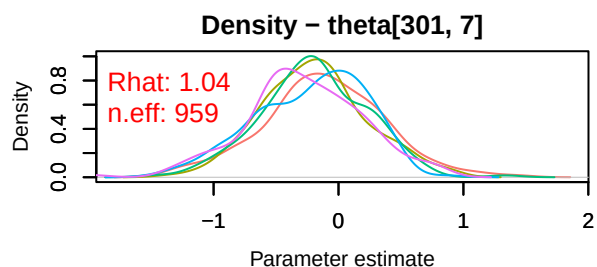
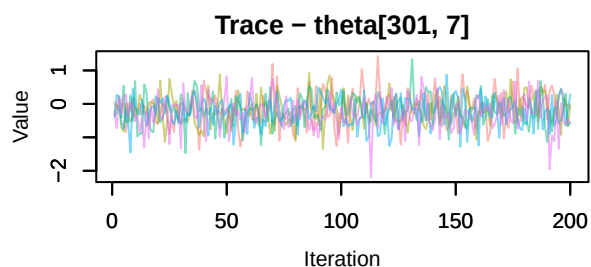
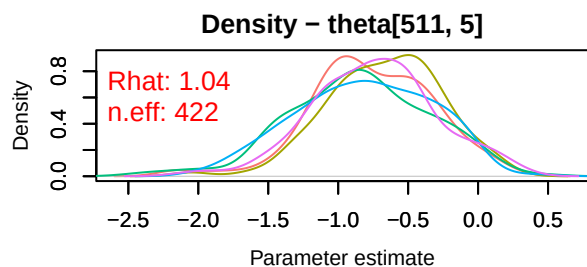
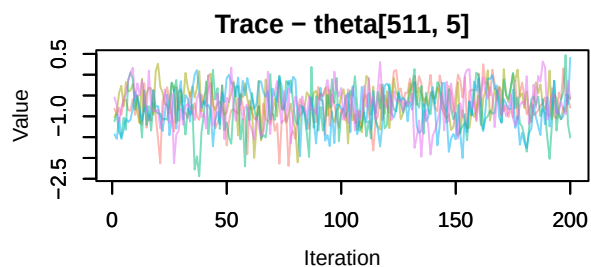
MCMCtrace(object = salnimble3,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  exact = TRUE,
  ISB = FALSE,
  params = rownames(salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]],
    ↪ test), ])[(salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], test),
    ↪ 6] > 1.03) | (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]],
    ↪ test), 7] < 200)])
```

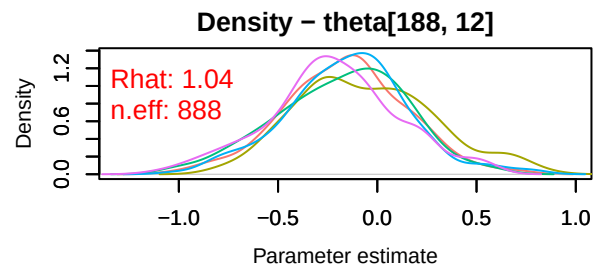
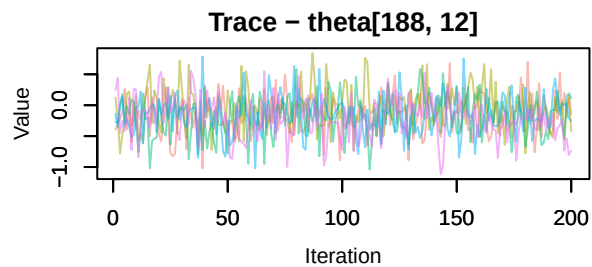
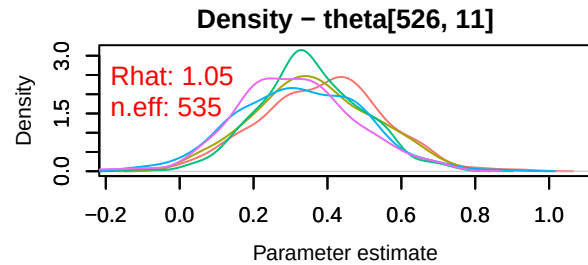
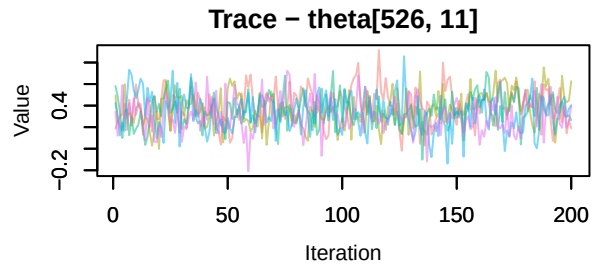
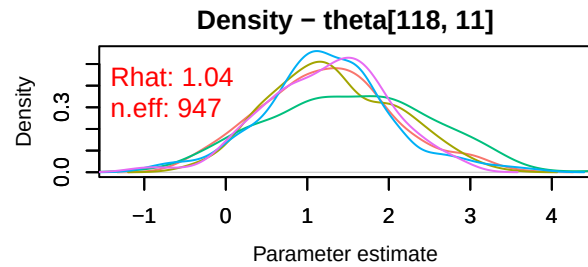
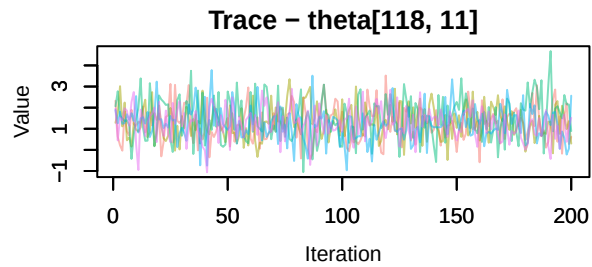













```
test <- "psi"

MCMCtrace(object = salnimble3,
  pdf = FALSE, # no export to PDF
  ind = TRUE, # separate density lines per chain
  Rhat = TRUE,
  n.eff = TRUE,
  exact = TRUE,
  ISB = FALSE,
  params = rownames(salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]],
    ↪ test), ])[(salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]], test),
    ↪ 6] > 1.03) | (salwinbugs3$summary[startsWith(labels(salwinbugs3$summary)[[1]],
    ↪ test), 7] < 200)])
```

